

Understanding the Crystallization Mechanism of Organic-Inorganic Perovskite Films

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Abstract

In recent years, significant progress has been made in the field of organic-inorganic perovskite materials and the accordingly derived photovoltaic technology. With an increasing number of studies on perovskite solar cells, various strategies have been developed for their fabrication, including the use of different crystallizing agents for the formation of the perovskite thin-film layer. These strategies often result in enhanced performances of perovskite solar cells. However, they also increase the complexity of the fabrication process, which in turn narrows the processing window and thus can undermine the economic feasibility of perovskite solar cells. In this work, such crystallizing agents are classified as homogeneous and heterogeneous, where homogeneous is a possible component of the perovskite lattice itself and heterogeneous is not a possible lattice component. Homogeneous crystallizing agents can sometimes be directly involved in the crystallization of perovskite; however, they may also contribute to lattice distortion. Heterogeneous crystallizing agents can directly cause defects in the perovskite layer.

This thesis aims at developing homogeneous and heterogeneous crystallizing agents, aiding in the formation of phase-pure (single-component) perovskites. At the same time, these agents are selected such that they can be removed at relative ease after the film has formed.

In particular, methylammonium chloride (MACl) is identified as a relatively universal crystallizing agent for the preparation of methylammonium lead triiodide (MAPbI₃) and formamidinium lead triiodide (FAPbI₃) films, which are among the most important perovskites studied in literature. In addition, the low boiling point of MACl allows it to be added to the precursor solution directly and also in gaseous form, thus influencing the perovskite crystallization. However, in the case of MAPbI₃, the addition of MACl in the precursor led to the formation of pinholes and defects during the formation of the desired black phase film. In contrast, placing the wet perovskite film in MACl vapor not only avoids pinholes and defects, but also helps to obtain films with larger grains. Similar to MAPbI₃, the perovskite films obtained by adding MACl to the precursor solution of FAPbI₃ show pinholes and defects that make the perovskite films unstable in air. As a modification, methylammonium bromide (MABr) was used as a homogeneous passivation agent, which significantly improved the stability of the perovskite films. The resulting perovskite solar cells achieved state-of-the-art efficiencies of over 23%.

Moreover, the use of MACl, although it allows for higher performances, introduces additional experimental steps, which can decrease the reproducibility of the device. Therefore, a

functionalized, low-boiling and easy-to-remove coordination solvent, dimethyl sulfide (DMS), is proposed for MAPbI₃-based devices. The single-component perovskite films produced with this heterogeneous crystallizing agent showed a reduced number of defects, resulting in a performance of 21.6% for MAPbI₃, among the highest reported for this composition in literature.

Kurzfassung

In den letzten Jahren wurden erhebliche Fortschritte auf dem Gebiet der organisch-anorganischen Perowskit-Materialien und der darauf basierenden Photovoltaik-Technologie erzielt. Mit der zunehmenden Zahl von Studien über Perowskit-Solarzellen wurden verschiedene Strategien für ihre Herstellung entwickelt, einschließlich der Verwendung unterschiedlicher Kristallisationsmittel für die Bildung der Perowskit-Dünnschicht. Diese Strategien führen häufig zu einer verbesserten Leistung von Perowskit-Solarzellen. Sie erhöhen jedoch auch die Komplexität des Herstellungsprozesses, was wiederum das Prozessfenster verkleinert und somit die wirtschaftliche Machbarkeit von Perowskit-Solarzellen untergraben kann. In dieser Arbeit werden solche Kristallisationsmittel als homogen und heterogen klassifiziert, wobei homogen ein möglicher Bestandteil des Perowskit-Gitters selbst ist und heterogen kein möglicher Gitterbestandteil. Homogene Kristallisationsmittel können manchmal direkt an der Kristallisation von Perowskit beteiligt sein, sie können aber auch zur Gitterverzerrung beitragen. Heterogene Kristallisationsmittel können direkt zu Defekten in der Perowskitschicht führen.

Ziel dieser Arbeit ist die Entwicklung von homogenen und heterogenen Kristallisationsmitteln, die die Bildung von phasenreinen (Einkomponenten-) Perowskiten unterstützen. Gleichzeitig sollen diese Mittel so ausgewählt werden, dass sie nach der Filmbildung relativ leicht entfernt werden können.

Insbesondere Methylammoniumchlorid (MACl) wurde als relativ universelles Kristallisationsmittel für die Herstellung von Methylammoniumbleidiiodid (MAPbI₃) und Formamidiniumbleidiiodid (FAPbI₃) identifiziert, die zu den wichtigsten in der Literatur untersuchten Perowskiten gehören. Darüber hinaus kann MACl aufgrund seines niedrigen Siedepunktes sowohl direkt als auch in gasförmiger Form in die Vorläuferlösung gegeben werden, was die Kristallisation des Perowskits beeinflusst. Im Falle von MAPbI₃ führte die Zugabe von MACl in den Präkursor jedoch zur Bildung von Nadellöchern (engl. Pinholes) und Defekten bei der Ausbildung des gewünschten Perowskitfilms in der photoaktiven schwarzen Phase. Wird der nasse Perowskitfilm dagegen MACl-Dampf ausgesetzt, werden nicht nur Pinholes und Defekte vermieden, sondern auch Filme mit größerer Korngröße erhalten. Ähnlich wie bei MAPbI₃ weisen die Perowskitschichten, die durch Zugabe von MACl zur Vorläuferlösung von FAPbI₃ erhalten wurden, Pinholes und Defekte auf, die die

Perowskitschichten an Luft instabil machen. Als Modifikation wurde Methylammoniumbromid (MABr) als homogenes Passivierungsmittel verwendet, wodurch die Stabilität der Perowskitschichten deutlich verbessert wurde. Die resultierenden Perowskit-Solarzellen erreichten Wirkungsgrade von über 23%.

Die Verwendung von MAI ermöglicht zwar höhere Wirkungsgrade, erfordert aber zusätzliche experimentelle Schritte, die die Reproduzierbarkeit der Solarzelle beeinträchtigen können. Daher wird ein funktionalisiertes, niedrigsiedendes und leicht zu entfernendes Koordinationslösungsmittel, Dimethylsulfid (DMS), für MAPbI₃-Solarzellen vorgeschlagen. Die Einkomponenten-Perowskitschichten, die mit diesem heterogenen Kristallisationslösungsmittel hergestellt wurden, wiesen eine geringere Anzahl von Defekten auf, wodurch ein Wirkungsgrad von 21,6 % für MAPbI₃ erzielt werden konnte, einem der höchsten in der Literatur für diese Zusammensetzung.

List of abbreviations

OIHP	Organic-inorganic halide perovskite	MAPbI ₃	Methylammonium lead tri-iodide
PSC	Perovskite solar cell	FAPbI ₃	Formamidinium lead tri-iodide
PCE	Power conversion efficiency	MA	Methylammonium
V _{oc}	Open-circuit voltage	FA	Formamidinium
J _{sc}	Short-circuit current	MAI	Methylammonium iodide
FF	Fill factor	MABr	Methylammonium bromide
VBM	Valence band maximum	MACl	Methylammonium chloride
CBM	Conduction band minimum	FAI	Formamidinium iodide
EQE	External quantum efficiency	FACl	Formamidinium chloride
P _m	Maximum output power	PbI ₂	Lead iodide
t	Tolerance factor	PbBr ₂	Lead bromide
μ	Octahedral factor	PbCl ₂	Lead chloride
ΔE	Defect levels	DMSO	Dimethyl sulfoxide
k _B	Boltzmann constant	DMF	N, N-Dimethylformamide
T	Temperature	DMS	Dimethyl sulfide
M	Defective species	DES	Diethyl sulfide
S	Lattice site	DPS	Dipropyl sulfide
n _e	Electron carrier density	DE	Diethyl ether
n _h	Hole carrier density	Tol	Toluene
Q _{sc}	Space charge region	CB	Chlorobenzene
Q _{surf}	Surface charge	GBL	γ-butyrolactone
GB	Grain boundary		

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1 Introduction

Research on perovskite solar cells (PSCs) has been gaining increasing attention since Miyasaka¹ and co-workers reported the first PSC, then based on the dye-sensitized solar cell architecture known as the Grätzel cell.² They reported a solar cell efficiency of 2.2%³ and 3.8%,¹ in 2006 and 2009, respectively, which was a significant achievement despite the instability against the (typical) liquid electrolyte. In just over a decade, PSCs have reached a certified efficiency of 26.1%, which is close to monocrystalline silicon solar cells at 27.3%.⁴ This demonstrates the tremendous potential of perovskite photovoltaics.

However, while the efficiency of PSCs has skyrocketed, their long-term stability has become a growing concern. Therefore, to understand the relationship between efficiency and long-term stability of PSCs, it is necessary to start from the physical and chemical properties of the material itself to gain an in-depth understanding of various aspects, such as the morphology of the film, the carrier transport properties within the film, the bonding mode of the crystal formation, the characters of defects in the film, and the formation of grain boundaries. In addition, the discussion of crystallization mechanisms and the development of subsequent methods for the production of thin films with high-quality crystallization are particularly important.

In **Chapter 2**, the fundamentals of crystals are first studied in this thesis, followed by a study of the crystalline structure, the phase transitions of different crystals, and the reasons for these phase transitions. Then, the effects on the stability of PSCs in terms of their device structure, which includes the causes of grain defects in perovskite films, as well as the bandgap regulation of the material, and the preparation of perovskite films were introduced. Finally, the perovskite crystallization process is reviewed.

After the fundamental study, an overview of the fabrication protocols used in this thesis and analysis methods of perovskite solar cells are presented in **Chapter 3**.

Chapter 4 presents the experiments for the fabrication of low-defect and pure-phase perovskite films. First, MACl was introduced into MAPbI₃ as a homogeneous dopant. Due to its low boiling point, MACl was found to be easily expelled from the perovskite by evaporation during annealing. It was also found that the addition of MACl in perovskite

precursor could cause pinholes due to the evaporation of MACl when annealing, resulting in not fully covered film. The rapid evaporation rate of MACl was thought to cause this problem. In order to test the speculate, the vapor pressure of the MACl was increased to slow down its rate of evaporation by using the vapor effect. The results showed that when the vapor pressure of MACl was sufficient, perovskite films could form high-quality films with full coverage. The experiment in this chapter indicates that treating perovskite films with extrinsic MACl vapor can result in large grain sizes and high-quality films.

This work was published as: *In situ methylammonium chloride-assisted perovskite crystallization strategy for high-performance solar cells (ACS Materials Letters 4 (3), 448-456)* with this author being the first author.

Such MACl vapor treatment can be considered as a homogeneous interfacial crystallization engineering, although further studies are needed to determine the mechanism. In **Chapter 5**, it was observed through in situ optical microscopy that the crystal formation time of the perovskite varies with different compositions. Specifically, a larger A-cation size or smaller X-halide ion size slows down the crystallization process. Thus, the longest crystal formation time can be achieved by adjusting the size and ratio of the appropriate A-cation and X-anion. This chapter focused on single component FAPbI₃, a perovskite with a longer crystal formation time, and improved its stability by modifying the interface using MABr after forming a film of pure phase. Compared to the control group, the modified film achieved higher stability in air, and the subsequent solar cells exhibited a high efficiency of 23%.

This work was published as: *Crystallization dynamics and stabilization of FAPbI₃ single-phase perovskite (Energy & Environmental Science 17 (4), 1407-1415)* with this author being the first author.

In addition to the homogeneous interface treatment technique, the heterogeneous interface treatment technique is also a commonly used method. However, the use of heterogeneous methods may lead to impurities in the pure phase perovskite. There are various assisted crystallization methods, but the anti-solvent method is one of the most common. In **Chapter 6**, a heterogeneous assisted crystallization was achieved using a coordination solvent DMS that has a suitable functional sulfur group. This simplified the experimental process and prevented the introduction of impurities. Here, DMS was used as a green coordination solvent, which has been shown to effectively remove the high boiling point solvent DMSO

commonly used in perovskite, resulting in a pure MAPbI₃ phase. As a result, PSCs were fabricated using all green solvents. This resulted in a high efficiency of 21.6%, among the highest in the literature for MAPbI₃.

This work was published as: *Coordination Chemistry as a Universal Strategy for a Controlled Perovskite Crystallization (Advanced Materials, 2302889)* with this author being the first author.

Chapter 7 provides a future outlook. This thesis posits that developing a fabrication process that is as streamlined as possible using perovskite materials with as few components as possible is a strategic way forward for efficient PSCs due to commercialization and environmental reasons. Especially a focus on eco-friendliness, besides efficiency and stability, will be important in future fabrication processes.

2 Fundamentals of organic-inorganic halide perovskite

This chapter introduces a fundamental basis for organic-inorganic halide perovskite (OIHP) focusing on crystal structures, defects as well as optical and electronic properties. In addition, the different PSC architectures, the fabrication methods used to produce these devices, and crystallization strategies are reviewed. It also discusses the main factors that can affect stability during the process of perovskite fabrication from crystals to solar cells.

2.1 Crystal structure of perovskite

The crystal structure of perovskite is according to the ABX_3 formula, which is usually cubic, tetragonal, or orthorhombic. One particular mineral, based on this structure, is $CaTiO_3$. It was named after the Russian mineralogist L.A. Perovski as “perovskite”, which then became the namesake for all perovskites.⁵ As is shown in **Figure 2.1a**, the perovskite structure can be envisaged with an A-cation in the central position. With A as its center, it forms a coordination cuboctahedron of coordination number 12 with the 12 X-ions surrounding it. On the other hand, B with the 6 X-ions surrounding it forms a coordination octahedron with a coordination number 6.

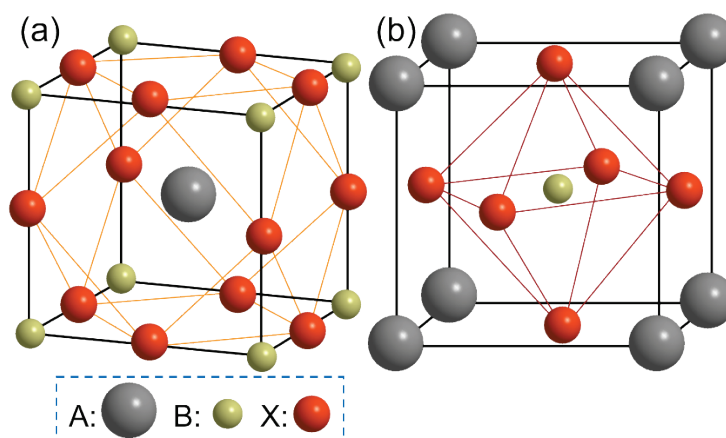


Figure 2.1 Structure of perovskite crystals (a) AX_{12} cubic octahedron; (b) BX_6 octahedron

The A and X ions have similar radii and form dense cubic stacking. In principle, stable perovskite crystals can be formed by any combination of compatible ions. The atomic radii

of the A, B, and X atoms, denoted by R_A , R_B , and R_X , respectively, are utilized in the equations below, where t is the tolerance factor³⁷ and μ is the octahedral factor, to determine the structural parameters and coordination numbers of perovskite crystals.

$$t = (R_A + R_X) / \sqrt{2}(R_B + R_X) \quad (1)$$

$$\mu = R_B / R_X \quad (2)$$

Figure 2.2 shows the t and μ distribution of various perovskites.⁶ Empirically, the crystal structure of ABX_3 is considered stable when the values of t and μ fall within the ranges of 0.81 to 1.11, and 0.44 to 0.90, respectively. Specifically, a cubic lattice with the highest symmetry is formed when t equals 1.0. In contrast, when t falls between 0.89 and 1.0, a rhombohedral structure (tripartite crystal system) is observed. Further, if t is less than 0.96, the symmetry shifts to an orthorhombic structure.⁷⁻⁹ Typically, organic cations are used to fill the A-position in PSCs. Among these, CH_3NH_3^+ (MA^+) ($R_A = 1.8 \text{ \AA}$)¹⁰ and $\text{NH}_2\text{CH}=\text{NH}_2^+$ (FA^+) ($R_A = 2.3 \text{ \AA}$)¹¹ are the most widely used for three-dimensional perovskite fabrication, while other ions such as $\text{CH}_3\text{CH}_2\text{NH}_3^+$ (EA^+) ($R_A = 1.9\text{-}2.2 \text{ \AA}$) also find some applications. On the other hand, metal cations, specifically Pb^{2+} ($R_B = 1.19 \text{ \AA}$) and Sn^{2+} ($R_B = 1.10 \text{ \AA}$), are utilized to represent the B ions. The halogenated anions, such as I^- ($R_X = 2.20 \text{ \AA}$), Cl^- ($R_X = 1.81 \text{ \AA}$), and Br^- ($R_X = 1.96 \text{ \AA}$), are represented by X ions. Unfortunately, even if a reference ionic radius can be stated for the organic cation, defining the actual ionic radius at the A site remains challenging for non-atomic cations. This is especially true since the organic group at the A site has a relatively low decomposition temperature, which can lead to challenges with thermodynamic stability that interfere with obtaining the true ionic radius. Additionally, the lower electronegativity of the iodide anion makes it challenging for the heavy halide ion to be effectively modeled as an unpolarizable hard sphere.

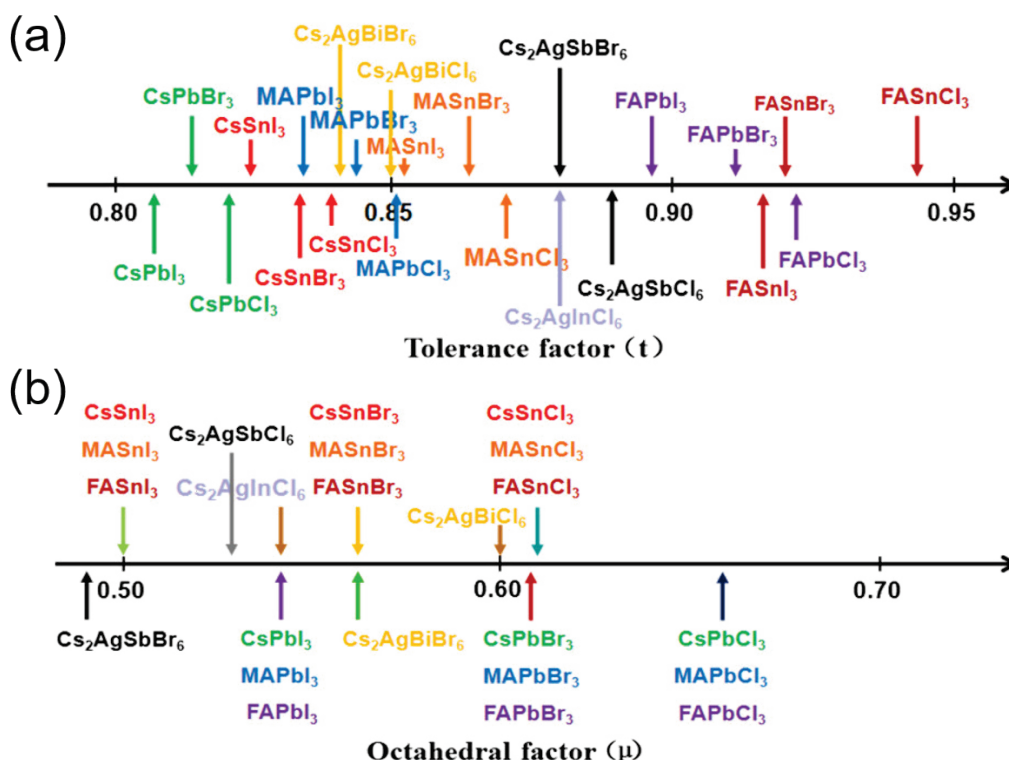


Figure 2.2 (a) The tolerance factor (t) of different perovskites. (b) The octahedral factor (μ) of different perovskites.⁶ Copyright 2019, Wiley-VCH.

Determining the actual ionic radius in perovskite crystals is challenging, but in OIHP, there is an additional difficulty: The A-site component is localized by molecular-type ions that may have intrinsic dipoles and induce octahedral skeleton deformation through interatomic hydrogen bonding. As a result, the common polarization of each unit in OIHP is affected by the molecular orientation of the A site, which gives rise to the apparent polarization.¹² In the case of MAPbI_3 , the orientation of the MA cations, which have an inherent dipole of around 2.3 D, is the primary factor that distinguishes the polar crystalline phase ($I4cm$) from the nonpolar crystalline phase ($I4/mcm$).¹³

Since MAPbI_3 is a classic model for learning OIHPs, the discussion of the MA properties is useful. The use of theoretical calculations is beneficial in studying the effect of MA orientation on nanoscale optoelectronic properties, as well as the interaction between MA and the Pb-I framework. At present, researchers widely accept that the orientation of MA is mainly determined by hydrogen bonding and tends to align with the low-index (Miller index¹⁴) direction. This orientation may deform the Pb-I framework via the H...I hydrogen

bonding, leading to symmetry perturbation and induction of polarization.^{15–18} The bandgap value is relatively unaffected by minor adjustments to the MA orientation. However, altering the orientation can trigger direct-indirect band transitions^{19,20} as well as the Rashba/Dresselhaus effect. This effect arises due to the lack of inversion symmetry in a crystal system, causing the energy bands to split as a result of spin-orbit interaction in solid-state physics.²¹ It can even lead to dynamic Rashba/Dresselhaus effects,^{22,23} which reduce carrier recombination, increase carrier lifetime, and improve optoelectronic performance.²⁴ Apart from the MA orientation, the crystalline phases of perovskite materials have the potential to exert an influence on their bandgap properties. A more comprehensive understanding of the properties of perovskite materials requires a basic understanding of key concepts related to the bandgap and crystalline phases.

2.2 Crystalline phases transition

Inorganic perovskites of the classical type can transition through various phases with multiple applications, including high-temperature cubic and low-temperature polar phases. When organic ligands are present and form OIHPs, the crystallographic structure and bonding of the perovskite lattice are significantly altered, making the prediction of phase transitions much more complicated than for inorganic perovskites. Although there is a possibility that lattice symmetry may be reduced, the perovskites are still distorted while keeping the same chemical formula and coordination numbers as their initial cubic structure. The soft lattice of such ionic materials can lead to highly reconfigurable crystal structures with relatively simple structural rearrangements induced by external stimuli, undergoing a variety of solid-solid phase transitions. Among OIHPs, those categorized according to A cations, i.e., MA lead halide and FA lead halide.

MA-based lead halides belong to the hybrid organic-inorganic MAPbX_3 perovskites class, where X can be I, Br or Cl. Taking MAPbI_3 as an example, the interactions between MA and the inorganic lead iodide host structure are strongly influenced by hydrogen bonding. As the temperature decreases, the cubic structure ($\text{Pm}\bar{3}\text{m}$) appears tilted, similar to a tetragonal to orthorhombic shape, while the $[\text{PbI}_6]^{4-}$ octahedron is distorted (as is shown in **Table 2.1**). In the I4/mcm structure that was formed after the structural change, the orientation of the MA is unevenly distributed among eight spatially symmetric sites.²⁵ While

the orientation inconsistency of the MA cation is eliminated when the phase transfers to *Pnma* at a low temperature.^{26–30} And in the orthorhombic phase, nitrogen atoms in the organic cation can form shorter hydrogen bonds and strengthen the interacting atoms in the inorganic halide framework. In addition, the tetragonal phase transition from tetragonal to cubic phase at room temperature, and to cubic phase at high temperature has received considerable attention in the study of hybrid lead-based halide perovskites. Knop et al.³¹ as well as Onoda-Yamuro et al.³² have already reported this phenomenon during their calorimetric research. The latent heat of this transition process is a first-order thermodynamic transition. **Figure 2.3** illustrates several common deformations of the cubic perovskite structure.³³

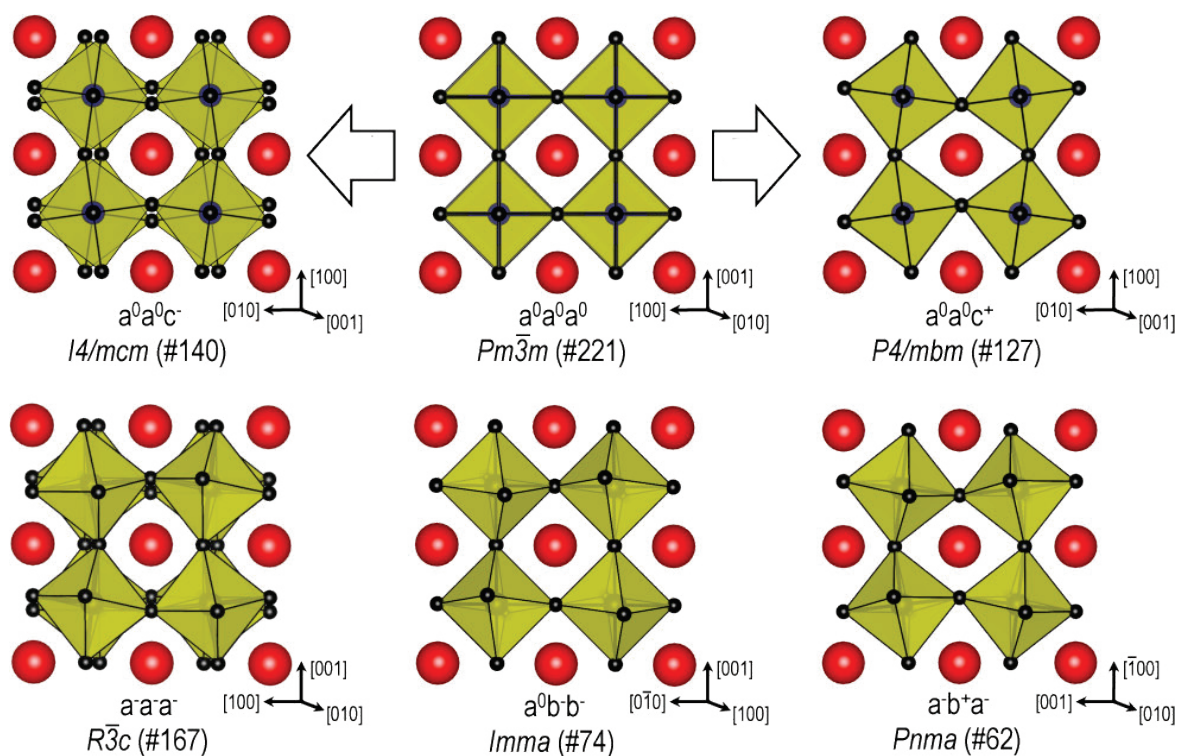


Figure 2.3 Space groups of perovskite crystal and the evolution of cubic structure.³³

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In terms of the structure, Mashiyama et al.³⁴ classified the tetragonal-cubic transition as a substitutional transition which was found in MAPbBr_3 and MAPbI_3 . The tetragonal phase of hybrid lead-halide perovskites demonstrates $[\text{PbI}_6]^{4-}$ octahedra which rotate in an "out-

of-phase" manner along the c-axis, resulting in an $a^0a^0c^-$ structure based on Glazer notation.³⁵ As temperature increases, the extent of octahedral rotation gradually decreases, eventually leading to a cubic structure with an $a^0a^0a^0$ classification, as depicted in **Figure 2.4**. Typically, researchers measure the degree of octahedral rotation in terms of the I-Pb-Pb-I dihedral angle, as shown for α in **Figure 2.4**. The structural analyses by researchers have indicated a progressive decrease in octahedral tilting in MAPbI_3 .^{29,30} In hybrid lead-halide perovskites, the phase that appears at high temperature is characterized by significant structural flexibility due to the movement of organic cations and deformations of the lead-halide framework. This flexibility may explain why there is no strictly pure cubic structure at the higher temperature phase in the previous report, including the observation of residual octahedral rotations in the high-temperature phase of MAPbBr_3 ³⁶ as well as in the high-temperature phase between the phases of MAPbCl_3 exist disordered positioning of Cl anions.³⁷

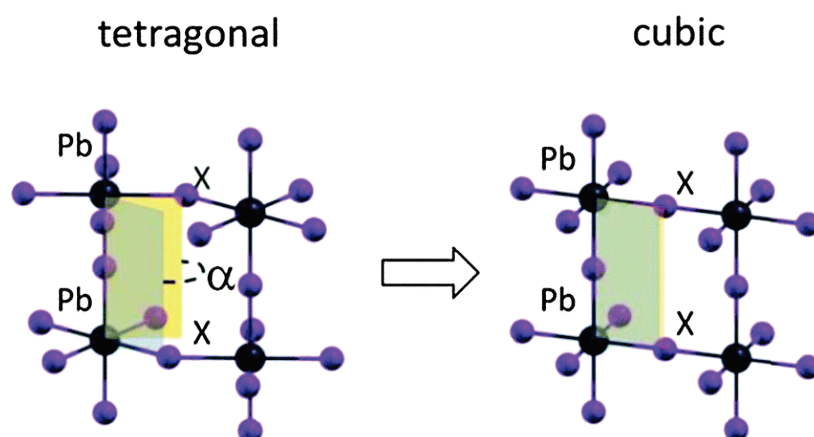


Figure 2.4 MAPbX_3 perovskite from tetragonal to cubic structure.³⁸ Copyright 2016, The Royal Society of Chemistry.

Similar to MAPbI_3 , FAPbI_3 also exists in different perovskite phases due to the more complex hydrogen bonding. For example, the α -phase has a $\text{Pm}\bar{3}\text{m}$ structure and the δ -phase has a P63mc structure at different temperatures³⁹ (as shown in **Table 2.1**). In addition, other phases can occur in FAPbI_3 under special conditions, such as the $\text{Imm}2$ and Immm phases under different pressures.⁴⁰ Extensive research has been conducted on the temperature-dependent polymorphism of FAPbI_3 . At room temperature, it takes the form of a yellow,

non-perovskite δ -phase (P63mc), which is thermally stable. However, its chain-like structure and large bandgap limit electron transport, making it an inefficient solar absorber.⁴¹ At ~ 150 °C, the non-perovskite δ -phase FAPbI₃ (hereafter referred to as δ -FAPbI₃) undergoes a reconstructive phase transformation to the perovskite α -phase FAPbI₃ (hereafter referred to as α -FAPbI₃) which possesses a more favorable bandgap and topology, enabling efficient photovoltaic effects.⁴² In the α -FAPbI₃ phase, [PbI₆]⁴⁻ octahedra are vertex-connected to form a cubic perovskite structure (Pm $\bar{3}$ m).⁴³ Irrespective of the storage environment, the α -FAPbI₃ phase slowly transforms to the thermodynamically stable δ -FAPbI₃ at room temperature.⁴² α -phase FAPbI₃ (trigonal) as one of the two main polymorphs, is of greater interest for optoelectronic applications than δ -phase FAPbI₃ (hexagonal) counterpart, owing to its desirable optical and electronic properties.^{44,45} Although α -phase FAPbI₃ can be obtained directly at 150 °C, it will be more economical but also challenging to obtain stabilized α -phase FAPbI₃ at room temperature.^{46–48} Researchers are presently engaged in efforts to improve understanding of the phase transition relations between the δ and α polymorphs.^{47,49–51} Nonetheless, a comprehensive model that can account for the structural variability of this material during the transient phases of phase stabilization is currently absent.^{40,52,53} Further scientific investigation and technological advances are needed to fully understand the complex and challenging process of producing a stable, black perovskite from simple methods of phase transition between α - and δ -phase FAPbI₃.

Table 2.1 Phase Transitions and Transition Temperatures of MAPbX₃ and FAPbX₃ (X = I, Br, and Cl)

Perovskite	Phase transition	Temperature (K)	Crystal system
MAPbI ₃	$\alpha \rightarrow \beta$	330 ³²	cubic $\rightarrow$ tetragonal
	$\gamma \rightarrow \delta$	161 ³²	tetragonal $\rightarrow$ orthorhombic
	$\alpha \rightarrow \beta$	236 ³²	cubic $\rightarrow$ tetragonal
MAPbBr ₃	$\beta \rightarrow \gamma$	154 ³¹	tetragonal $\rightarrow$ tetragonal
	$\gamma \rightarrow \delta$	149 ³²	tetragonal $\rightarrow$ orthorhombic
MAPbCl ₃	$\alpha \rightarrow \beta$	177 ³²	cubic $\rightarrow$ tetragonal
	$\gamma \rightarrow \delta$	172 ³²	tetragonal $\rightarrow$ orthorhombic
FAPbI ₃	α	423 ³⁹	trigonal
	δ	293 ³⁹	hexagonal

2.3 Band-gap tuning of organic-inorganic halide perovskite

As is known that OIHP, synthesized according to the molecular formula ABX_3 , theoretically has countless types of components, subject to the limitations of the t and μ . Interestingly, the bandgap of OIHP can be tuned by varying the ratio of different A, B, or X ions. The changes in the bandgap, as well as the absolute positions of valence band maximum (VBM) and conduction band maximum (CBM) in perovskites, have been well characterized in numerous experimental^{54–56} and computational^{57–66} studies for various primary ABX_3 compounds and more complex perovskite mixtures. **Figure 2.5**⁶⁷ illustrates the bandgap values for different single-component perovskites.

The interplay of several subtle yet correlated factors,⁶⁷ such as ion type and size, crystal structure, and degree of deviation from the ideal perovskite structure,^{59,64} varies depending on the perovskite composition when different A, B, and X are combined. For instance, Kulkarni et al.⁶⁸ have demonstrated that the bandgap of mixed anion lead halide perovskites ($MAPb(I_{1-x}Br_x)_2$ ($0 \leq x \leq 1$)) can be tuned by adjusting the concentration of halogen precursors in a sequential deposition process. This approach can flexibly modify the optical properties of perovskite hybrids. It is crucial to tailor the bandgap of the light absorber to minimize energy loss in heterojunction and/or tandem solar cells. Therefore, a mixed perovskite with a controllable bandgap is highly advantageous as a light absorber in solar cells.⁵⁴

The dopants strategy is also used to adjust the absorber bandgap of OIHP. Similar to the fabrication of silicon-based semiconductors, at low doping levels the periodic structure of the doped body is perturbed, resulting in an n- or p-type semiconductor. Only when the doping concentration reaches a significant level, or when it is directly alloyed with the perovskite body, does the bandgap change directly, which is commonly referred to as homogeneous doping or hybridized perovskite. The final bandgap of the material is proportional to the quadratic molar ratio of its components,⁶⁹ the nonlinear term being $x(1-x)$ (where the overall equation is $E_g = Cx(1-x)$, with E_g being the bandgap and C being a constant),⁷⁰ which was first introduced by Thompson and Woolley. This was later attributed to crystal potential variations, virtual crystal effects, and aperiodicity by Van Vechten and Bergstresser⁶⁹ in 1969. Saliba et al.⁷¹ introduced Cs cations into (MAFA) perovskites to obtain a triple cation composition. In addition to changing the cation size of the perovskite,

A-site doping can also directly change its energy band structure by forming two-dimensional or one-dimensional crystals, allowing indirect to direct bandgap conversion. Besides chemically changing the bandgap, physical methods like strain can significantly alter the material's band structure by distorting, tilting, or twisting the microscopic lattice and inducing complex phase transitions. These changes alter the boundary conditions of the electron wave function, thereby affecting the bandgap.⁷²

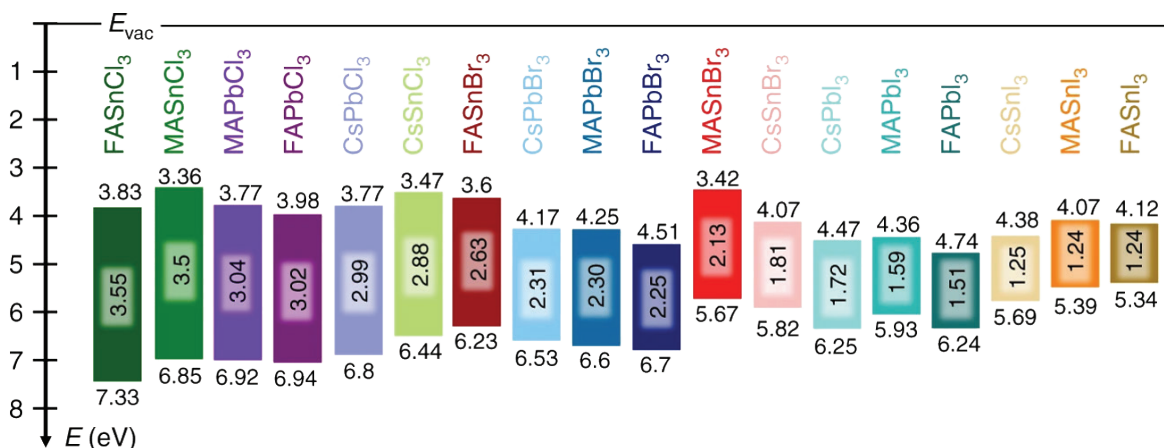


Figure 2.5 Schematic energy levels of the 18 OIHPs.⁶⁷ Copyright 2019, Shuxia Tao et al.

In PSCs, the bandgap is a key parameter with profound implications. It determines the spectral absorption range that allows perovskite cells to efficiently harvest visible and infrared sunlight. Ideally, the material can absorb a broad spectrum of the sun's light if the bandgap is low enough. In practice, however, a low bandgap can easily create hot carriers and dissipate energy (thermalization losses).⁷³ To maximize photovoltaic energy conversion efficiency (η), an optimal bandgap must be calculated according to the Shockley-Queisser (S-Q) detailed-balance model,⁷⁴ achieving wider spectral absorption and lower heat loss. Based on the S-Q detailed-balance model, the maximum η achievable for a single-junction solar cell is 33.7%, and this is possible for an optimum semiconductor bandgap of 1.34 eV.⁷⁴ Scientists have attempted different strategies to overcome the S-Q limit of single-junction solar cells. These methods include wide (1.7 to 1.9 eV) and low (1.1 to 1.3 eV) bandgap combined tandem solar cells,⁷⁵ and gradient bandgap architecture single junction solar cells.⁷⁶

The material selection of the perovskite film based on the bandgaps and crystalline phases is important and efficient, besides this, understanding crystallographic defects in perovskite films is important during film fabrication.

2.4 Fabrication method of perovskite thin-films

Since the reaction precursors of perovskite are generally composed of AX and BX₂, the fabrication of perovskite films is generally divided into processes with or without solvents. The solution route is generally divided into one-step fabrication and two-step fabrication, where the metal-halide and cation components are deposited subsequently and not simultaneously. **Figure 2.6** summarizes some of the processes used to fabricate perovskite thin films.

The one-step method involves depositing a single solution containing all perovskite precursors onto the substrate, which is then followed by annealing on the hotplate.^{77,78} Although this method is more common for certain perovskite compositions,^{79,80} it must be noted that: first, various chemical reactions may occur in the solution of perovskite precursors; second, mixing several types of solvents as well as additives together may also affect the crystallization of perovskite, resulting in various types of defects; the OIHP films and photovoltaic device performance can be influenced by all these factors.⁸¹

In the now less commonly used two-step method, AX and BX₂ were separately prepared in precursor liquids and sequentially spin-coated onto the substrate, which was then heated on a hot plate to form a perovskite film.⁸² Typically, the first precursor (e.g., PbI₂) is deposited by spin-coating and then the film is either immersed in a second precursor solution (e.g., MAI)^{83,84} or further spin-coated with the second precursor⁸⁵. Initially, devices fabricated using the two-step method performed better than those fabricated using the one-step method. However, improvements in solvent engineering have reversed the situation.⁸⁶ This is due to the introduction of an anti-solvent in the one-step method to induce film crystallization, resulting in films with excellent crystallization.⁸⁷ Several other methods are available for depositing perovskite films, including spray coating,⁸⁸ flash evaporation,⁸⁹ inkjet printing,⁹⁰ chemical vapor deposition,⁹¹ and many others.^{92–96}

It should be noted that the typically lower performance of the devices obtained compared to the common techniques of the time makes these alternative methods less present in the

literature. Likewise, various adaptations of thermal evaporation have been demonstrated for depositing perovskite layers. The thermal evaporation process eliminates the need for a solvent and facilitates the production of a uniform, large-area device surface free of pinholes. It also allows easy control of the film thickness, which is difficult to achieve with the spin-coating method. It also prevents the formation of trap/defect sites that can result from the use of solvents during precursor synthesis.^{97–99} The most common technique is based on two-source deposition, where the precursors are co-evaporated to form a perovskite layer and can be extended to three-source deposition or more, depending on the components. Unlike the solution method, the thermal evaporation approach typically results in the formation of pure perovskite crystals without the use of an annealing process.¹⁰⁰ For instance, PSCs with a power conversion efficiency (PCE) of 20.28% for a small-area (0.16 cm²) device and 18.13% for a perovskite mini-module (21 cm²) were obtained by Li et al.¹⁰¹ through the co-evaporation of MAI and PbI₂. Meanwhile, Liu et al.¹⁰² fabricated uniform and high-quality perovskite films on glass (400 cm²) and flexible (300 cm²) substrates by sequentially evaporating PbI₂, FAI, and CsI. The small-area (0.09 cm²) devices on the rigid substrate achieved PCEs ranging from 19.50 to 20.50%.

In addition to this, single-source evaporation,^{103,104} sequential evaporation¹⁰⁵ are also used for thermal evaporation. Despite the remarkable uniformity achieved with evaporation-based techniques, implementing passivation strategies or controlling the perovskite composition remains a significant challenge.¹⁰⁶ Meanwhile, many studies still use annealing to improve the evaporated film properties. These steps are critical to achieving high performance and superior stability.^{107,108} In addition, improved methods of film production have been explored since. However, the goal of easier handling and stability improvements is a powerful motivation when selecting new methods.

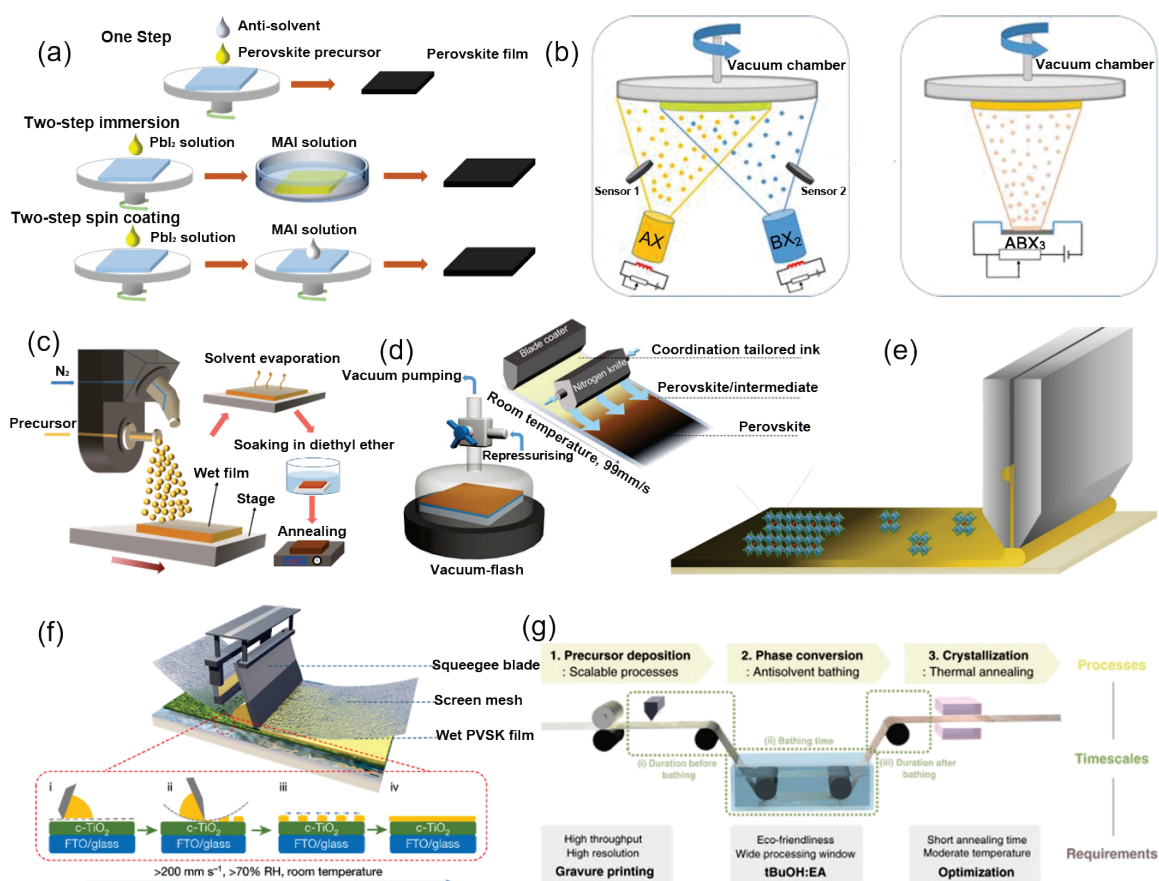


Figure 2.6 Different ways of making perovskite thin film: (a) one-step and two-step spin coating;¹⁰⁹ Copyright 2022, Wiley-VCH GmbH. (b) evaporation methods;¹¹⁰ Copyright 2020, Elsevier B.V. (c) spray coating;¹¹¹ Copyright 2020, American Chemical Society. (d) blade coating and flash evaporation;¹¹² Copyright 2021, Wiley-VCH GmbH. (e) slot-die coating;¹¹³ Copyright 2019, Elsevier Ltd. (f) screen print;¹¹⁴ Copyright 2022, Changshun Chen et al. (g) roll-to-roll print.¹¹⁵ Copyright 2020, Young Yun Kim et al.

2.5 Crystallization of perovskite thin films

The stability problems of perovskite itself limit the large-scale industrial production of PSCs during fabrication.^{116–119} Perovskite chemical compositions are generally water-soluble, meaning they tend to degrade under normal environmental conditions with greater relative humidity than 25%.¹²⁰ To overcome this problem, researchers have proposed numerous approaches that fall into three broad categories: chemical passivation, compositional engineering, and new synthesis methods.¹²¹ The main purpose of these strategies is to eliminate bulk defects as well as surface defects of the crystal. In other words, the study of

the crystallization process of perovskite is the main basis for optimizing the device stability as well as for proposing practical solutions.

Perovskites are similar to other crystals in that their nuclei can all be considered as the maximum number of new phases (solid-liquid phase transfer), i.e., clusters of several atoms or molecules aggregated into one. (See **Figure 2.7a**) The mechanism of the first-stage phase transition is the formation of a sub-stable phase as the concentration of the solution changes, and then the formation of a primary crystal nucleus due to the collision of particles during Brownian motion (It refers to the random motion of particles that are suspended in a medium that is either a liquid or a gas)¹²². The diffusion of the particles determines the rate of cluster growth on the surface after nucleation, and in this process, the clusters are integrated into the lattice to make the crystal larger.^{123,124} The perovskite precursor solution undergoes a change in chemical potential during solvent evaporation and begins to nucleate and grow a new phase when it reaches a supersaturated state. The solution then becomes uniformly dispersed with clusters formed for crystal growth. The size of the critical nucleus determines the Gibbs energy that must be overcome for the nucleation process to occur.¹²⁵ The rate of transformation in perovskite is influenced by various factors, such as the atomic or molecular composition, reaction kinetics, heat flow at the interface, and the initial state of the uniform solution. Typically, an initial metastable nuclei phase is governed by a first-order phase-transition mechanism.¹²⁶ During the perovskite film fabrication, almost all nucleation processes are heterogeneous, which is also related to the nucleating agent used in the crystallization process.¹²⁷ However, homogeneous nucleation processes are most commonly observed in the growth of perovskite single crystals.¹²⁸

As mentioned above, perovskite film fabrication is divided into solid and solution processes. In solution processes, the conversion rate of products and the bond strength of the chemicals involved can be determined by the nature of the reactants, such as acid-base reactions.¹²⁹ In chemical reactions, the conversion pathway of reactants to products can be described kinetically by the reaction rate.^{130,131} Under steady-state conditions, the rate of a reaction will depend on the frequency and specific region of collisions between molecules or ions. To ensure that a kinetic rate expression corresponds to stoichiometry, it must be equilibrated at the molecular level, typically as single or bimolecular reactions. For overall reactions involving multiple steps, including the formation and decomposition of molecular

complexes, the basic reaction can be divided into several steps, with the intermediates acting as reactants in a linkage role.¹²⁶

The prediction of reaction products solely based on the stoichiometry of reactants is often challenging. As a result, intermediates generated during the reaction play a crucial role in understanding complex reaction kinetics. The study of heterogeneous nucleation involves analyzing both primary and secondary nucleation processes. Secondary nucleation happens when crystalline species of the material form nuclei on the surface of larger clusters.¹³²

The fundamental factor that drives the crystallization process is supersaturation, which influences the rate of crystal growth.^{133–135} Rapid nucleation can enhance film coverage while controlling crystal growth can lead to the formation of a highly crystalline material.^{136–138} To promote crystal growth, the newly formed clusters must be energized to transport them to the crystal surface and rearrange their lattice to accommodate crystal growth. Critical clusters, which are analogous to activated complexes in transition state theory, enable an uninterrupted reaction process at the crystal surface and can be continuously formed in solution to achieve dynamic equilibrium. Clusters are generally transported by both diffusions, caused by the thermal motion of the molecules, and convection, caused by the concentration gradient created around the crystal as it grows. Both convection in solution and cluster transport rate can influence the growth rate. The growth rate of crystals can be determined by surface-controlled growth, which can be influenced by nucleation and dislocation growth.^{123,124,139}

For surface crystallization, The Kossel-Stranski model (See **Figure 2.7b**) describes the formation of growth clusters only around the crystal surface and not directly on the surface; these growth clusters are then adsorbed by the crystal and incorporated into the lattice.¹⁴⁰ The lattice sites on the crystal surface may be terrace, ledge, and kink sites, and the number of lattice sites determines the rate of dispersion of the growth clusters onto the crystal surface. At the same time, this rate is different from the rate at which the atoms in the growth cluster diffuse into the lattice, and these two rates together determine the growth rate of the crystal.¹⁴¹ Also the binding energy between the adsorbed atoms and the lattice sites is different, enhancing from the platform to the ledge to the kink site in order. In the Kossel-Stranski theory, the atoms cannot form low-energy nucleation sites by diffusing from the kink ledge to the crystal edge. Nevertheless, screw dislocations can provide a continuous energy ladder for this purpose.^{142,143} Burton, Cabrera, and Frank elucidated the mechanism

of step generation and transport in crystal growth theory (See **Figure 2.7c**).¹⁴⁴ This is because screw dislocations can promote steps on the crystal surface whose magnitude is related to the projection of the lattice distortion vector (Burgers vector) produced by the dislocation.^{145,146} In this theory, additional growth occurs when the steps rotate through the dislocation point.

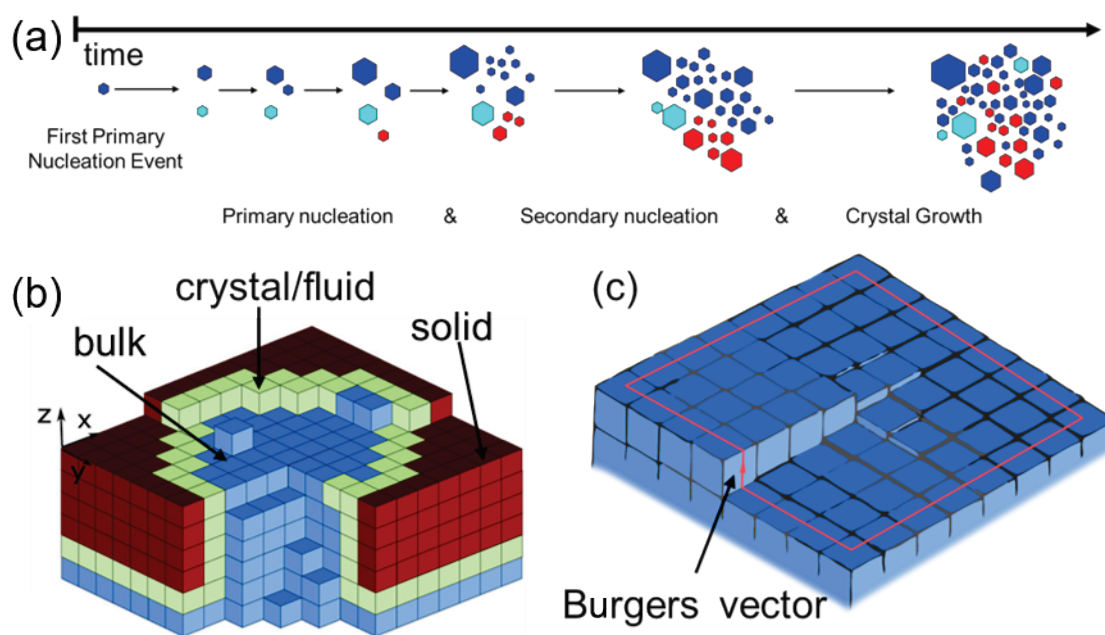


Figure 2.7 (a) Schematic representation of the crystallization of a system with three different polymorphs; Copyright 2019, American Chemical Society (b) A three-dimensional perspective of an ideal pore; Copyright 2015, American Chemical Society (c) A Burgers 'screw' dislocation. Copyright 1998, Springer-Verlag Berlin Heidelberg

Furthermore, Ostwald ripening takes place when smaller particles dissolve or merge with larger particles. In less complex systems, it is commonly believed that the surface energy density remains constant regardless of shape, thus the surface energy is directly proportional to the surface tension.¹⁴⁷ The influence of interfacial curvature on particle size, which impacts energy changes, can result in differences in solubility between larger and smaller growing particles. This leads to the smaller denuded particles contributing to the growth and coarsening of larger particles. The Ostwald ripening phenomenon is known to hinder grain

boundary defects during the growth of perovskite crystals,^{148,149} while the aging effect of perovskite films has been observed to eliminate defects.

A practical experimental framework must complement the model for a comprehensive description to account for the complexities of ripening ratios and coarsening. Typically, a brief high-temperature annealing process can yield an energy process comparable to that of a longer annealing period at a lower temperature. However, achieving the supersaturated state requires distinct activation energies or driving forces in each case, resulting in varying temperature dependencies.¹³⁵

The process of high-temperature annealing generates thermal convection that determines the crystal growth rate, while the temperature gradient of the annealing determines the final crystal morphology. The interfacial energy and the different chemical potentials of the molecules in the process of growing crystals from solution determine the arrangement of the crystal microstructure. Instabilities can cause directional growth, guided by the system's anisotropy, as seen in dendritic growth. In such cases, a metastable phase may grow in a direction that differs from the energetically favorable direction of the final phase.

Another common way to manipulate the crystallization rate of perovskite is through the use of an anti-solvent. Anti-solvent crystallization involves the precipitation of solute particles due to a decrease in solubility. This crystallization achieved by exposing the solution to a nearly insoluble solvent is a solubility strategy of the solute in the solvent, and additionally, the crystallization process can be controlled by introducing airflow to accelerate the evaporation of the solvent.¹⁵⁰⁻¹⁵² The approach employed to achieve supersaturation can influence various crystal properties, such as size, morphology, and coexisting atomic structures.¹⁵³ In addition, methods such as solvent engineering,^{154,155} interface engineering,^{156,157} and defect passivation,¹⁵⁸⁻¹⁶⁰ are used to improve the crystallization of perovskite films for high carrier-transporting ability and low charge recombination.¹⁶¹ Research into the principles of crystallization provides a good basis for the subsequent production of stable solar cells.

2.6 Devices structures of perovskite solar cells

The fundamental architecture of a solar cell incorporates a p-n junction, which combines a p-type semiconductor with an n-type semiconductor. When sunlight strikes a typical semiconductor, it produces electrons and holes, quickly recombined and dissipated in photons or heat. However, at the p-n junction, diffusion due to the difference in charge carrier concentration creates a built-in electric field that points from n to p. Therefore, when the p-n junction absorbs photons, the electrons produced will be moved to the n-type semiconductor by the built-in electric field, and the opposite holes will be transferred to the p-type semiconductor, resulting in the accumulation of charges on both sides of the p-n junction, which will form a current loop when connected by wires. This is the basic principle of solar cell operation.

The structure of the PSCs was first derived from the dye-sensitized solar cell, in which perovskite was used as a dye. And this kind of structure has evolved into the common structures that are still active in the laboratory: FTO /compact-TiO₂ /mesoporous-TiO₂ /perovskite /spiro-OMeTAD /Au (Ag) (See **Figure 2.8**). Since OIHP materials are intrinsic semiconductors, additional additions of p-type and n-type materials to stabilize charge transport must be considered when making stable solar cells. Here, compact-TiO₂ /mesoporous-TiO₂ is considered an n-type material for charge separation and transport, spiro-OMeTAD is considered a p-type material for hole transport. Therefore, this structure can be described as a typical n-i-p structure.

In addition, planar heterojunction as a common structure in optoelectronic devices has also been applied to PSCs. OHIP, on the other hand, is a very high-performance material with reported charge transport down to 1 μm . For example, in a film prepared from the 3 CH₃NH₃I+PbCl₂ mixture,¹⁶² diffusion lengths for electrons can reach ~ 1069 nm and holes ~ 1213 nm.¹⁶² Therefore, the fabrication of planar PSCs does not require many buffer layers to achieve a high-performance device. It is usually sufficient to add a p-type material for hole transport and an n-type material for electron transport to both sides of the perovskite layer. Due to the excellent carrier transport properties, it can be observed that the design of PSC devices is mainly determined by the perovskite layer.

Based on the evolutionary history of PSCs, we have categorized the popular device structures into three categories (See **Figure 2.8**).¹⁶³ As is known, although the perovskite

layer plays a crucial role in the performance of PSCs, the electron transport layer (ETL), and hole transport layer (HTL) also play a very important role in guiding and separating electrons and holes in the structure of PSCs, preventing the recombination of electrons and holes, and improving the efficiency of charge transport to ultimately achieve high-performance devices. Therefore, we categorize them into n-i-p and p-i-n structures according to the deposition order of the ETL, light-absorbing layer, and HTL. While the n-i-p structure in which the ETL is fabricated as a mesoporous structure is also called meso-structure, the other two are called planar heterojunction structures. The selection and optimization of these layers are critical to achieving highly efficient and stable PSCs. In addition, the interface between the layers is also a factor to consider during the fabrication of high-performance solar cells.

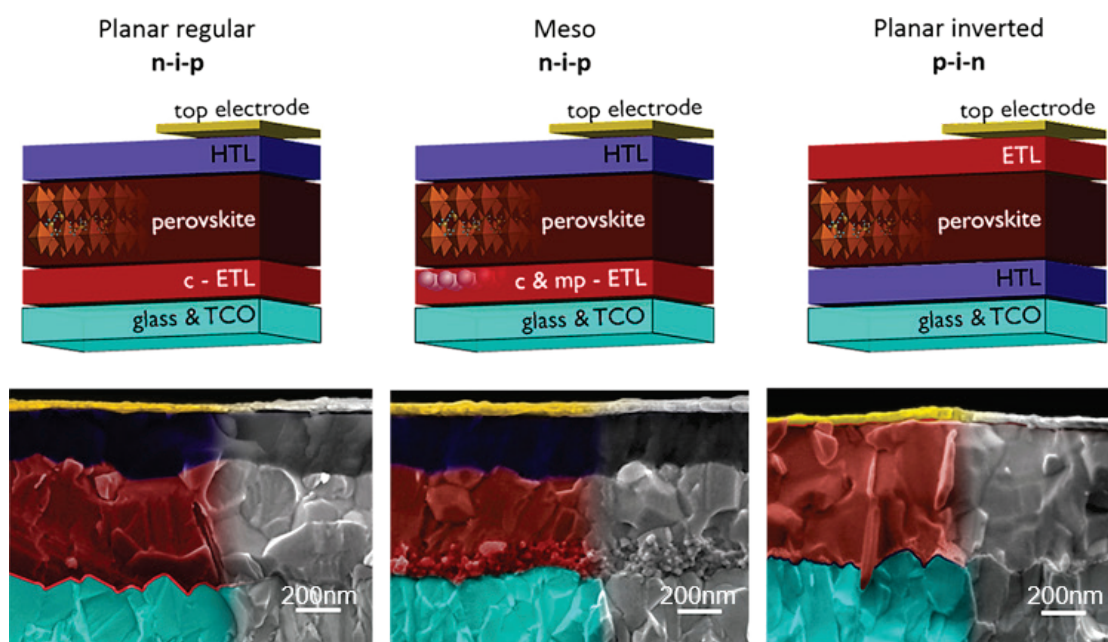


Figure 2.8 Schematics of the three PSC architectures, including planar regular (n-i-p), mesoporous (n-i-p), and planar inverted (p-i-n).¹⁶³ Copyright 2018, American Chemical Society.

2.7 Stability of perovskite solar cells

The instability of PSCs is primarily due to the degradation of OIHP, which is known to be moisture sensitive¹⁶⁴ and has poor long-term thermal stability¹⁶⁵. In a humid environment,

the degradation of OIHP poses a significant challenge due to its hygroscopic nature, which makes moisture one of the major factors contributing to this instability.¹⁶⁶ Additionally, the high hydrophilicity of OIHP (such as MAPbI₃) is easily degraded to hydrates (such as (CH₃NH₃)₄PbI₆·2H₂O) by moisture in the environment.¹⁶⁷ The perovskite's structural integrity depends heavily on hydrogen bonding between its organic and inorganic units. However, this bond is highly sensitive to the polarity of water molecules, which can significantly weaken or break the bond.¹⁶⁸ In addition to moisture-related stability issues, UV exposure is another important factor that can affect the stability of PSCs. Similar to other solar cell technologies, illumination can cause degradation in PSCs.^{169,170} When exposed to air, light catalyzes perovskite decomposition into methylamine, PbI₂, and I₂. This decomposition process is accelerated by the combined effect of light and dry air on the perovskite film.¹⁷¹ In addition, the photogenerated electrons produced by the perovskite convert the oxygen in the air to superoxide, which then reacts with MA. To enhance stability and tolerance to oxygen, replacing the acid-proton-containing MA component with species lacking such protons has been suggested.¹⁷² Aside from these, thermal stability is also a significant concern for PSCs, as exposure to high temperatures can cause the device to degrade.¹⁷³ According to some researchers, the instability of perovskite materials may be attributed to grain boundaries (GBs) or the surface of the material.^{174–178} A promising strategy to enhance the stability of perovskites is to cap their grain boundaries with suitable protective materials.^{179–182} The crystal structure and phase of perovskite materials can be significantly impacted by temperature, with previous research indicating a phase change from tetragonal to cubic at temperatures between 54–56 °C.^{30,179} During operation, solar modules are often exposed to elevated temperatures to meet international standards such as the IEC 61656 climatic chamber tests, which require solar cells to exhibit thermal stability up to 85 °C.¹⁸³ A stabilized crystal structure is important for the fabrication of stable perovskite films. However, the organic components in perovskite are sensitive to moisture, oxygen, light, and heat,^{54,56,184} making the improvement of their stability critical. However, due to the low activation energy required for ionic migration, the ionic nature of OIHP materials may also affect their long-term stability, as ionic migration directly destroys the crystal structure of perovskite.¹⁸⁵ In fact, ions are susceptible to migration by external electrical bias, thermal stress, and light.¹⁸⁶ For example, when external electric bias is applied to PSCs, I ions tend to become carriers and migrate, and some metal atoms of metal electrodes (e.g., Ag) may migrate through the transport layer to the perovskite layer and

combine with I to form defects, resulting in non-radiative recombination of charges, which affects the device performance.¹⁸⁷

For the above reasons, an essential step in advancing perovskite toward commercial production is the careful study of stability issues.

2.8 Summary

The crystalline state of the perovskite film plays a major role in the performance of the device. First, the diverse compositions of perovskite materials allow for a wide range of bandgap choices. In contrast, there are fewer choices of efficient materials for carrier transport layers in solar cells, so the right perovskite components must be chosen to match the energy levels in solar cells. Due to the structural properties of perovskite, for given perovskite, it is also necessary to consider its different crystalline phases formed under different conditions, because the phase change can also affect the matching with the charge carrier transport and also lead to the degradation of the device. Thus, it is necessary to consider how to keep the perovskite in a constant phase. To achieve this goal, researchers have tried many different conditions and methods for preparing perovskite films, as well as investigating the degradation mechanism and the types of defects and their causes. This research process also requires considering the device as a whole and learning about the external environmental conditions that are detrimental to the stability of the device. Most importantly, the process of perovskite crystallization itself must be carefully considered, as the final quality of the grains in the film is the basis for efficient and stable devices.

3 Materials and methods

This chapter will briefly introduce the device structure, fabrication method and characterization tools used in this thesis.

3.1 Samples and device preparation

Materials

*This part follows the work: **Zuo, W.-W.**; Yang, Y.-G.; Fu, W.-F.; Li, M.; Malekshahi Byranvand, M.; Di Girolamo, D.; Pascual, J.; Li, M.-Q.; Li, L. N.; Abate, A. Saliba, M. In Situ Methylammonium Chloride-Assisted Perovskite Crystallization Strategy for High-Performance Solar Cells. *ACS Mater Lett* 4, 448–456 (2022).¹⁸⁸ **Zuo, W.**; Fu, W.; Wang, K.; Das, C.; Byranvand, M. M.; Wang, K. L.; Chaudhary, A.; Lim, J.; Li, M.; Saliba, M. Crystallization dynamics and stabilization of FAPbI₃ single-phase perovskite. *Energy Environ Sci* 2023, 17 (4), 1407–1415.¹⁸⁹ and **Zuo, W.**; Byranvand, M. M.; Kodalle, T.; Zohdi, M.; Lim, J.; Carlsen, B.; Magorian Friedlmeier, T.; Kot, M.; Das, C.; Flege, J. I.; Zong, W.; Abate, A.; Sutter-Fella, C. M.; Li, M.; Saliba, M. Coordination Chemistry as a Universal Strategy for a Controlled Perovskite Crystallization. *Adv. Mater.* 35, 2302889 (2023).¹⁹⁰*

Synthesis of non-stoichiometric NiO_x nanoparticles: 1 mmol nickel nitrate hexahydrate (Aladdin) was dispersed in 200 mL of deionized water to obtain a dark green solution. Then NaOH solution (10 mol L⁻¹) was added to adjust the pH of the solution to 10. After stirring for 5 min, the precipitate was thoroughly washed twice with deionized water and dried at 80 °C for 6 h. The resulting green powder was then calcined at 270 °C for 2 h. After that, a dark black powder is obtained. Finally, the obtained nanoparticles are dispersed in deionized water at 10 mg mL⁻¹.

All of the chemicals and solvents were obtained commercially and were used without further purification. SnO₂ colloid precursor (15% in H₂O colloidal dispersion) was purchased from Alfa Aesar. Clevios VP AI 4083 PEDOT: PSS water dispersion was acquired from Heraeus (Germany). MAI, MACl, MABr and FAI, FAcI were purchased from Greatcell Solar Materials. PbI₂ and PbCl₂ were purchased from TCI EUROPE N.V. PC₆₁BM was purchased from Nichem Fine Technology Co., Ltd. C₆₀ (99.5%), bathocuproine (BCP, 99.9%), Spiro-OMeTAD (99.5%), 4-Tert-butyl pyridine (99.9%), lithium bis(trifluoromethylsulphonyl)

imide (99.95%), and FK209 Co (III) TFSI salt (98%) were purchased from Advanced Election Technology CO., Ltd, TCI, Sigma-Aldrich, and Dyesol, respectively. All solvents, including dimethyl sulfoxide (DMSO), chlorobenzene (CB), toluene (Tol), dimethyl sulfide (DMS), diethyl sulfide (DES), dipropyl sulfide (DPS), anisole (ANS), and acetonitrile (ACN) are anhydrous and were purchased from Sigma-Aldrich.

Devices Preparation

In this thesis, the experiments are designed based on p-i-n and n-i-p structures.

p-i-n structure devices

(ITO/PEDOT:PSS/perovskite/PCBM/BCP/Ag and ITO/NiOx/perovskite/C₆₀/BCP/Ag)¹⁸⁸

The ITO substrates with a sheet resistance of $\sim 15 \Omega \text{ sq}^{-1}$ were thoroughly cleaned with acetone, ethanol, and deionized water for the sequence in an ultrasonic bath for 15 min, respectively. And then treated in UV-Ozone cleaner for 15 min. PEDOT: PSS was obtained by spin-coating on ITO surface under 4500 rpm 40 s^{-1} and then annealing at 140°C for 20 min. For the perovskite layer, the $\text{CH}_3\text{NH}_3\text{PbI}_{3-x}\text{Cl}_x$ solution was spin-coated onto the hole transport layer at 4000 rpm for 40 s in an N_2 -filled glovebox. The annealing of perovskite films was carried out by following a typical gradient increased temperature method that can be seen elsewhere: after it was dried at room temperature for ~ 20 min, the samples were slowly heated from 60 to 100°C at a rate of $10^\circ \text{C } 10 \text{ min}^{-1}$ on a hot plate. Some samples were placed inside a petri dish and some others were placed into the petri dish with MACl powder as the MACl vapor source. When MACl vapor fills the volume within the Petri dish, white crystals of MACl start appearing at the bottom of the Petri dish. After that, the concentration of 20 mg mL^{-1} PC_{61}BM in CB solution was spin-coated onto the perovskite layer at 2000 rpm for 40 s in an N_2 -filled glove-box. Finally, BCP (8 nm) and Ag (100 nm) were thermally deposited onto the PC_{61}BM layer under vacuum at 2×10^{-6} Torr.

For the pristine $\text{CH}_3\text{NH}_3\text{PbI}_3$ devices fabrication, NiOx layer was spin-coated onto the ITO substrates at 5000 rpm for 40 s and dried at 150°C for 30 min. The coated samples were then transferred to an N_2 -filled glove-box. The spin-coating program of MAPbI_3 solution is the same as $\text{CH}_3\text{NH}_3\text{PbI}_{3-x}\text{Cl}_x$, for which $100 \mu\text{L}$ CB was used as the anti-solvent. Then, C_{60} (23 nm)/BCP (8 nm)/Ag (100 nm) were sequentially thermally evaporated onto the perovskite layer.

For perovskite precursor, a stock solution of 1.5 mol L^{-1} of PbI_2 dissolved in 6:1 V/V DMF/DMSO mix solvent was prepared. An approximate amount of MAI or MACl is weighed and the volume of PbI_2 stock solution required to obtain the correct stoichiometry (1:1 molar ratio) is calculated.

n-i-p structure devices

(ITO/SnO₂/perovskite/spiro-OMeTAD/Au)^{189,190}

Patterned ITO substrates ($7 \text{ } \Omega/\text{sq}$, AGC) were ultrasonically cleaned with 2% Hellmanex water solution, deionized water, acetone, and IPA. Before use, the as-cleaned substrates were further cleaned with UV ozone treatment for 15 min. Then the SnO_2 solution (2.67%, diluted by water) was spin-coated onto the substrates at 3000 rpm for 25 s and annealed on a hot plate at $150 \text{ }^\circ\text{C}$ for 30 min under ambient atmosphere. After cooling down to room temperature, the substrates were treated with UV ozone for 15 min. The perovskite precursor solution was prepared with 1.3 M MAPbI_3 in an anhydrous solvent of DMSO. Then the perovskite precursor solution was spin-coated onto SnO_2 with a two-step program at 1000 and 6000 rpm for 5 and 27 s, respectively, in a nitrogen-filled glovebox. During the spin-coating process, anti-solvent was dripped on the spinning substrate slowly for 5 s before the end of the program. The volume of anti-solvent CB, Tol, DPS, DES, DE, and DMS were $100 \text{ } \mu\text{L}$, $200 \text{ } \mu\text{L}$, $200 \text{ } \mu\text{L}$, $200 \text{ } \mu\text{L}$, $300 \text{ } \mu\text{L}$ and $300 \text{ } \mu\text{L}$, respectively. Then the as-fabricated film was annealed on a hot plate at $100 \text{ }^\circ\text{C}$ for 60 min. After cooling down to room temperature, the spiro-OMeTAD solution (73.2 mg in 1 mL ANS) doped with $18 \text{ } \mu\text{L}$ of Li-TFSI (520 mg/mL in ACN), $29 \text{ } \mu\text{L}$ of FK209 Co(III) TFSI salt (300 mg/mL in ACN), and $29 \text{ } \mu\text{L}$ tBP was coated at 4000 rpm for 30 s. Finally, 100 nm of gold was thermally evaporated under a pressure of 10^{-6} mbar as the back electrode.

For FAPbI_3 precursor, 1.8 M FAPbI_3 : MACl (1:0.35) nominal stock solution in 1 mL DMF: DMSO 8:1, calculate the actual molar concentration from the density of the solution. Make sure that the stock solution is clear with no residual powder in the solution and under the lid. The stock solutions of FAPbI_3 need to be warmed overnight at $25 \text{ }^\circ\text{C}$ in the shaker and then left to cool to room temperature for 30 mins before using. After spin-coating, the film needs to be annealed at $150 \text{ }^\circ\text{C}$ for 10 min.

For MABr passivation, 5 mg/mL MABr dissolved in IPA and spin coat 6000 rpm on FAPbI₃ film at room temperature, the volume of MABr dropping was 200 μ L, followed by annealing at 100 °C for 10 min.

3.2 Device characterization

In this thesis, current density-voltage characteristics of PSCs under 1 sun illumination were performed using a programmable Keithley 2400 source meter under AM 1.5G solar irradiation at 100 mW cm⁻² (Newport, Class AAA solar simulator, 94023A-U). The incident-photon-to-current efficiency (IPCE) measurement was performed using a system combining a xenon lamp, a monochromator, a chopper, and a lock-in amplifier with a calibrated silicon photodetector.

Solar cell performance parameters

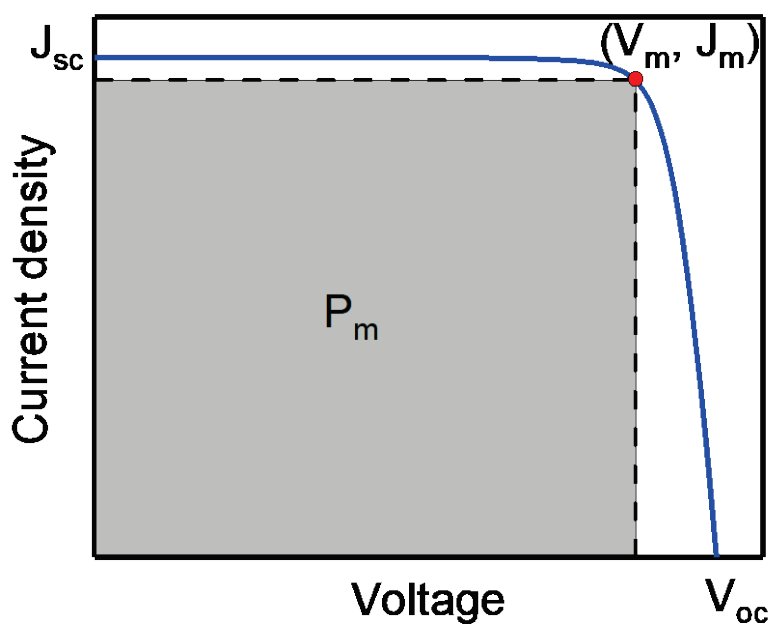


Figure 3.1 Generic J-V curve of solar cells

Open circuit voltage (V_{oc})

The open-circuit voltage corresponds to the voltage when the external circuit of the cell has been disconnected or the current is zero, i.e. the maximum output voltage (See in **Figure 3.1**). Under sunlight, the basic physical process of photovoltaic technology involves electrons being excited from the valence band (VB) of a semiconductor to its conduction

band (CB), leaving holes in the VB. The difference between the quasi-Fermi levels for electrons and holes determines the achievable V_{oc} of a solar cell.¹⁹¹ In monojunction devices, V_{oc} consists of a value close to the theoretical maximum, while in heterojunction devices, V_{oc} depends mainly on the bandgap of the perovskite.

Short circuit current density (J_{sc})

Short-circuit current is the current density on the external circuit when the solar cell is placed under AM1.5 spectral conditions and an irradiation intensity of 100 mW/cm². The voltage on the solar cell is zero, and it is the value corresponding to the vertical coordinate when the horizontal coordinate in the J-V curve is zero (See in **Figure 3.1**). The J_{sc} is related to the device preparation process and perovskite material types, such as perovskite components, additive types, annealing, and interfacial modification.¹⁹²

Maximum output power (P_m)

Since the solar cell operating voltage and current change with the load resistance, the operating voltage and current values corresponding to different resistance values are plotted to obtain the solar cell voltammetric characteristic curve.¹⁹³ When the selected value of load resistance can make the maximum product of output voltage and output current, the maximum output power can be obtained, which is expressed by the symbol P_m . At this time, the operating voltage and current are called the best operating voltage and current with the symbols V_m and I_m , respectively.

Fill factor (FF)

The fill factor, which is the ratio of the product of V_m and I_m and the product of open-circuit voltage and short-circuit current, is another important parameter in solar cells. Thus, the FF value is always less than 1. In a solar cell, small series resistance (R_s) and large shunt resistance (R_{sh}) are generally required for high FF.^{194,195} R_s is related to the resistance of device materials and resistive contacts, which affects FF, especially at high current densities, while R_{sh} results from current leakage through the device.

$$FF = (V_m \times J_m) / (V_{oc} \times J_{sc}) = P_m / (V_{oc} \times J_{sc}) \quad (3)$$

Power conversion efficiency (PCE)

The conversion efficiency of solar cells is the ratio of P_m (maximum output power) to P_{in} (sum of incident light power), which refers to how much of the incident light energy can be converted into effective electrical energy. The photoelectric conversion efficiency of solar cells, which depends on the device structure, junction characteristics, material properties, operating temperature, radiation damage from radioactive particles, and environmental changes, is an important parameter to measure the quality and technical level of the device.¹⁹⁶

$$PCE = P_m/P_{in} = (V_{oc} \times J_{sc} \times FF)/P_{in} \quad (4)$$

External Quantum Efficiency (EQE)

EQE is the ratio of the number of electrons collected by the cell to the number of photons absorbed by the instrument. EQE reflects the energy conversion efficiency of solar cell devices for different wavelengths of monochromatic light.¹⁹⁷ Typically, a higher EQE corresponds to a higher PCE.

$$EQE = 1240 \times J_{sc} / (\lambda \times P_{in})$$

3.3 Perovskite film characterization

Ultraviolet-visible spectrophotometer (UV-Vis)

The UV-vis spectrophotometer is a kind of analytical instrument based on the principle of UV-vis spectrophotometry, which analyzes materials by using a map of the absorption of radiation by their molecules in the ultraviolet-visible region of the spectrum. It consists of a light source, monochromator, absorption cell, detector, and signal processor. The light source must provide a stable, continuous spectrum with adequate brightness. For ultraviolet, a hydrogen or deuterium lamp is usually used. For the visible, tungsten or tungsten halogen lamps are used. The function of the monochromator is to break up the composite light emitted by the light source and separate the monochromatic light of the wavelength of interest. The dispersion elements are prisms and gratings. A glass absorber is used for measurements in the visible range and a quartz absorber is used for measurements in the ultraviolet range. The function of the detector is to detect the intensity of the light

transmitted through the photoconversion element and to convert the optical signal into an electrical signal. The most commonly used photoconversion elements are photocells, photomultiplier tubes, and photodiode array detectors.¹⁹⁸

In this thesis, the absorbance of the HTLs films was measured with a UV-Vis spectrophotometer (PerkinElmer Lambda 750).

Field-emission scanning electron microscope (SEM)

SEM is an observation instrument that uses a high-energy focused electron beam to scan samples, through interaction between beam and material to excite various physical information, collect this information, amplify, and then image. The resolution of the new SEM reaches 1 nm, the magnification reaches 300,000 times and is more continuously variable, and the depth of field, large field of view, imaging stereo. The combination of scanning electron microscopes and other analytical instruments enables the simultaneous observation of micro-morphologies to analyze the micro-regional composition of materials.¹⁹⁹

In this thesis, the SEM images were obtained from a Quanta 200 FEG; The surface morphology of the electrode was characterized by Scanning electron microscopy (SEM) operated at 20 kV 3.2 nA in a Thermo Fischer SCIOS Dual-beam microscope.

Electrical impedance spectroscopy (EIS)

Electrochemical impedance spectroscopy is an electrochemical measurement method that uses a small-amplitude sinusoidal potential (or current) as the perturbing signal. Because the system is perturbed by a small-amplitude electrical signal, it can avoid a large effect on the system, and it also makes the relationship between perturbation and system response approximately linear, making mathematical processing of measurement results easy. The AC impedance method sends small amplitude sinusoidal disturbance signals of different frequencies to the electrode system, the electrode impedance is obtained from the relationship between the response of the electrode system and the disturbance signals, and the equivalent circuit of the electrode can be derived. The electrode impedance is obtained from the relationship between the response of the electrode system and the disturbance signals, the equivalent circuit of the electrode, and the parameter values of the relevant components in the equivalent circuit, such as the electrode double layer capacitance, the

resistance of the charge transfer process, and the parameters of the diffusion mass transfer process.²⁰⁰

Each kinetic process can be represented by a linear element in electricity or a combination of several linear elements when an electrode system is subjected to a small amplitude disturbance signal, the response of various kinetic processes and the disturbance signal is a linear relationship between the kinetic processes. For example, the charge transfer process can be represented by a resistor, and the bilayer charging and discharging process can be represented by the charging and discharging process of a capacitor. Thus, the electrochemical process is described by an equivalent circuit, and the values of the components of the equivalent circuit can be obtained by the perturbation response of the electrode system, and the reaction mechanism of the electrode system can be deduced.²⁰⁰

Electrochemical impedance spectroscopy is also a frequency domain measurement technique that studies the electrode system by measuring the impedance spectrum over a wide range of frequencies.

In this thesis, the EIS was obtained using an IM6e Electrochemical Workstation (ZAHNER, Germany) with an alternating current amplitude of 10 mV. All the EIS measurements were carried out under dark conditions and recorded at 1 V voltage.

Grazing incidence X-ray diffraction (GIXRD)

Grazing incidence X-ray diffraction is a technique that allows X-rays to pass over the sample surface. The absolute peak intensity of the spectrum for grazing incidence XRD is relatively weak because the probe light is parallel light, which is less energetic than conventional diffracted light. However, grazing incidence XRD is used exclusively for testing thin film samples because parallel light allows better focus on the film surface and is less likely to measure the substrate.²⁰¹

In this thesis, the GIXRD measurements were performed at the BL14B1 beamline of the Shanghai Synchrotron Radiation Facility (SSRF) using an X-ray with a wavelength of 1.24 Å. Two-dimensional (2D) GIXRD patterns were acquired by a MarCCD mounted vertically at a distance of ~223 mm from the sample with a grazing incidence angle of 0.2° and an exposure time of 20 s. The 2D GIXRD patterns were analyzed using the FIT2D software and displayed in scattering vector q coordinates.

X-ray photoelectron spectroscopy (XPS) and Ultraviolet Photoelectron Spectroscopy (UPS)

XPS works by irradiating a sample with X-rays, causing the excitation and emission of the inner electrons or valence electrons of an atom or molecule. The electrons that are excited by the photons are called photoelectrons. The kinetic energy/binding energy of the photons (binding energy, $E_b = h\nu$ (photon energy) - E_k (kinetic energy) - w (work function)) as the horizontal coordinate, the relative intensity (impulse/s) as the vertical coordinate is a photon energy spectrum. Thus, it is possible to obtain the relevant information from the sample.²⁰²

The basic principle used to measure UPS is the same as that used for XPS and is based on Einstein's Law of Photoelectricity. For free molecules and atoms, $E_K = h\nu - E_b - \Phi_{sp}$, where $h\nu$ is the incident photon energy (known value), E_K is the kinetic energy of the emitted electrons (measured value), E_b is the binding energy of the bound electrons in the inner or valence levels (calculated value), and Φ_{sp} is the energy loss of the spectrometer (known value, usually about 4 eV). However, the excitation sources used are much lower in energy than X-rays, so the photo-excited electrons only come from very shallow samples (~ 10 Å) and reflect information about the interaction of electrons near the Fermi energy level of the atom, the valence electrons.²⁰³

In this thesis, the perovskite layer prepared with various solvent treatments is analyzed with X-ray photoelectron spectroscopy using Al k-Alpha source (1486.5 eV) non-monochromitized X-ray. The photoelectrons are collected with the hemispherical analyzer from PHOIBOS HSA3500. The photo-emitted electrons are collected at a right angle to the surface of the sample. The detailed spectra of each element present on the samples are collected with the pass energy of 20 eV with and scan step of 0.1 eV. The collected spectra were analyzed with the Casa XPS software and the peaks were fitted with mixed Gaussian-Lorentzian line shape with Shirley background. The binding energy of the measured spectra is corrected concerning the metallic Pb (137.00 eV) and advantageous carbon (285.00 eV).

Fourier-transform infrared spectroscopy (FTIR)

FTIR uses the Fourier transform principle of interfered infrared light. The light from the light source is split by a beam splitter (similar to a semi-transparent half reflector) into two beams, one transmitted to the moving mirror and the other reflected to the fixed mirror. The two light beams have been reflected by the fixed mirror and the moving mirror and then back to the beam divider, the moving mirror at a constant speed for linear movement, and

thus the beam divider beam after the formation of the two beams of light range difference, leading to interference. Interfering light in the beam splitter converges through the sample cell, through the sample after the interfering light with sample information to the detector, and then through the Fourier transform signal processing, and eventually get the transmission or absorption with the number of waves or wavelength of the infrared absorption spectrum.²⁰⁴

In this thesis, FTIR spectra were recorded from a neat powder of the respective analyte on a BRUKER JASCO FT-IR-4100 spectrometer.

In situ absorbance measurements

An Ocean Optics spectrometer (Flame) coupled with fiber optics was used to acquire transmission measurements with an integration time of about 0.15 s (spin-coating) to 0.4 s (annealing) per transmission spectrum and a wavelength spacing of about 1.3 nm in an N₂-filled glove box. The equation [$A_\lambda = -\log_{10}(T_\lambda)$] was used to calculate the UV-Vis absorption spectra from the transmission spectra, where A_λ is the absorbance at a certain wavelength (λ), and T_λ is the corresponding transmitted radiation.

In situ Photoluminescence Measurements

The in-situ PL measurements were carried out using a home-built setup in an N₂-filled glove box. Excitation was performed using a laser diode emitting at 405 nm and the PL emission was collected using an optical fiber coupled with an Ocean Optics spectrometer (Flame). A Jacobian correction was applied to the data transforming them from wavelength- to energy-space and fitting them using a Gaussian and a constant background. Integration times of 300 ms were used.

In situ GIWAXS

GIWAXS can be used to measure information about the surface structure of films. This information includes structure size, shape, distribution, period, and orientation.

In this thesis, the GIWAXS data were collected at the 12.3.2 microdiffraction beamline of the Advanced Light Source (ALS) in a custom-made analytical spin-coater allowing for simultaneous film-deposition and diffraction measurements.

4 Chloride source in-situ mediated perovskite crystallization dynamics

This work was published as: **Zuo, W.-W.**; Yang, Y.-G.; Fu, W.-F.; Li, M.; Malekshahi Byranvand, M.; Di Girolamo, D.; Pascual, J.; Li, M.-Q.; Li, L. N.; Abate, A. Saliba, M. *In Situ Methylammonium Chloride-Assisted Perovskite Crystallization Strategy for High-Performance Solar Cells. ACS Mater Lett* 4, 448–456 (2022).

Zuo, W.-W. fabricated and measured all devices, analysed the data, wrote the manuscript and contributed to the design and implementation of the research as well as the analysis of the results.

Optimizing the crystallization method is a critical step in learning how to produce high-performance PSCs. Common approaches for optimization include the use of additives, crystallization agents, and defect passivation. Among chloride (Cl)-based additives, methylammonium chloride (MACl) is the most frequently reported in the literature, although its mechanism and process of action remain to be determined. In contrast to the commonly used preparation process and precursor formulation, MACl is used as an external chloride source with a steam treatment method to optimize the perovskite film. To investigate the mechanisms by which chloride improves the quality of perovskite films, in situ 2D-GIXRD and in-situ real-time X-ray diffraction measurements are utilized in a closed environment. The results indicate that the vapor created by the MACl around the film during the annealing process can significantly enhance film quality. MACl vapor reduces the internal defects of films and promotes the formation of perovskite films with high crystallinity and large grain size. Lastly, compared to the internal method described in this article, the external MACl source's auxiliary crystallization approach suggests a better-optimized crystal effect. Moreover, a ~21% efficient PSC is obtained based on the optimized MAPbI₃ film in this study.

4.1 Introduction

Lead halide-based perovskite materials have emerged as successful light absorbers for PSCs owing to a high optical extinction coefficient,²⁰⁵ low exciton binding energies,²⁰⁶ long-range charge carrier diffusion lengths,^{207–209} and tunable bandgaps^{209–211} via simple components engineering. In the past few years, the power conversion efficiencies (PCEs) of PSCs have boosted from an initial 3.8%²¹² to a recent record of more than 26.0%.^{4,213} Like other kinds of thin-film solar cells, the film quality significantly affects the photoelectric characteristics and device performance. Great efforts, including composition engineering, solvent engineering, interfacial modification, additive regulation, and film-casting processing,^{214–218} have been made to achieve high-quality perovskite film layers, which is key to achieving high efficiency and reproducibility. Appropriate precursor combinations have been proven to be an effective method for assisting homogeneous nucleation or regulating crystallization kinetics.^{219,220} Among them, chloride-containing precursors have been demonstrated to be suitable for controlling and improving the film morphology, which is beneficial in reducing bulk defects and thus supports the development of high-performance solar cells.²¹⁵

Chloride-containing precursors, such as MACl and PbCl_2 , have attracted a lot of attention due to the chloride ion, which could retard the crystallization and improve the quality of perovskite films.^{221,222} In 2013, Stranks²²³ et al. showed that the chloride-containing perovskite could extend the carrier diffusion lengths to 1 μm . Yang²²⁴ et al. proposed that chloride treatment was the main reason for the improvement of carrier transport to the heterojunction interfaces. However, the directly produced pristine halide perovskite (MAPbCl_3) may result in impure phases in the iodide-based perovskite film due to its lower solubility.²²⁵ In addition, it is difficult to obtain high-quality films due to the low energy of formation and the low boiling point of about 150 °C of MAPbCl_3 , which can decompose into PbCl_2 and MACl during the annealing process.²²⁶ During chloride removal from the perovskite films as MACl in the thermal annealing process,^{162,227} undesirable pinholes or cracks may occur.²²⁸ Therefore, methods to effectively prevent pinhole formation are necessary to obtain high-quality films. Typically, there are two types of chlorides available to researchers. One type is the internal chloride source, such as perovskite precursor containing PbCl_2 and MACl , which can dissolve in precursor directly and homogenous spread in the films. Another type is the external chloride source, such as MACl ²²⁹ vapor treatment. In this work, a MACl vapor treatment was used as an external chloride source for

controlling the perovskite thin-film growth. The MACl-gas vapor could effectively regulate the dynamics of perovskite crystal growth, hindering the formation of undesirable pinholes, as Khadka et al.²²⁹ described. The perovskite film formation is investigated by in situ two-dimensional grazing incidence X-ray diffraction (2D GIXRD). The data proves that higher MACl vapor pressure can effectively promote the transition of locally enriched PbI_2 with MACl to form perovskite crystals.

4.2 Results and discussion

To study the influence of different MACl vapor concentrations in the atmosphere during the crystallization of perovskite films, different methods of exposing MAPbI_3 films to MACl vapors are compared. Dualeh et al.²³⁰ reported that when chloride is introduced in the MAPbI_3 precursor solution, a large amount of it will leave the films as MACl vapor during the thermal annealing. The MACl released by the drying film is referred to as the "internal chloride" source. An alternative strategy is introducing the MACl powder and the film together inside a petri dish during the annealing, referred to as the "external chloride" source. By combining these two methods, the experiment in **Figure 4.1** is designed to investigate the effect of MACl vapor on a standard $\text{MAPbI}_{3-x}\text{Cl}_x$ perovskite reference film. In the second case, to increase the concentration of MACl vapor, the film is annealed inside a petri dish, defined as "sealed". In the third case, this is coupled together with the "external chloride" source.

Figure 4.1e shows the experimental procedures for the thermal annealing of the perovskite, based on a gradual increase in temperature. In practice, (I) the samples were heated from 60 to 90 °C with a heating rate of 10 °C/10 min on a hot plate, (II) the samples were heated from 90 to 100 °C at the same rate, (III) and, finally, the samples were held at 100 °C until the samples were completely black.

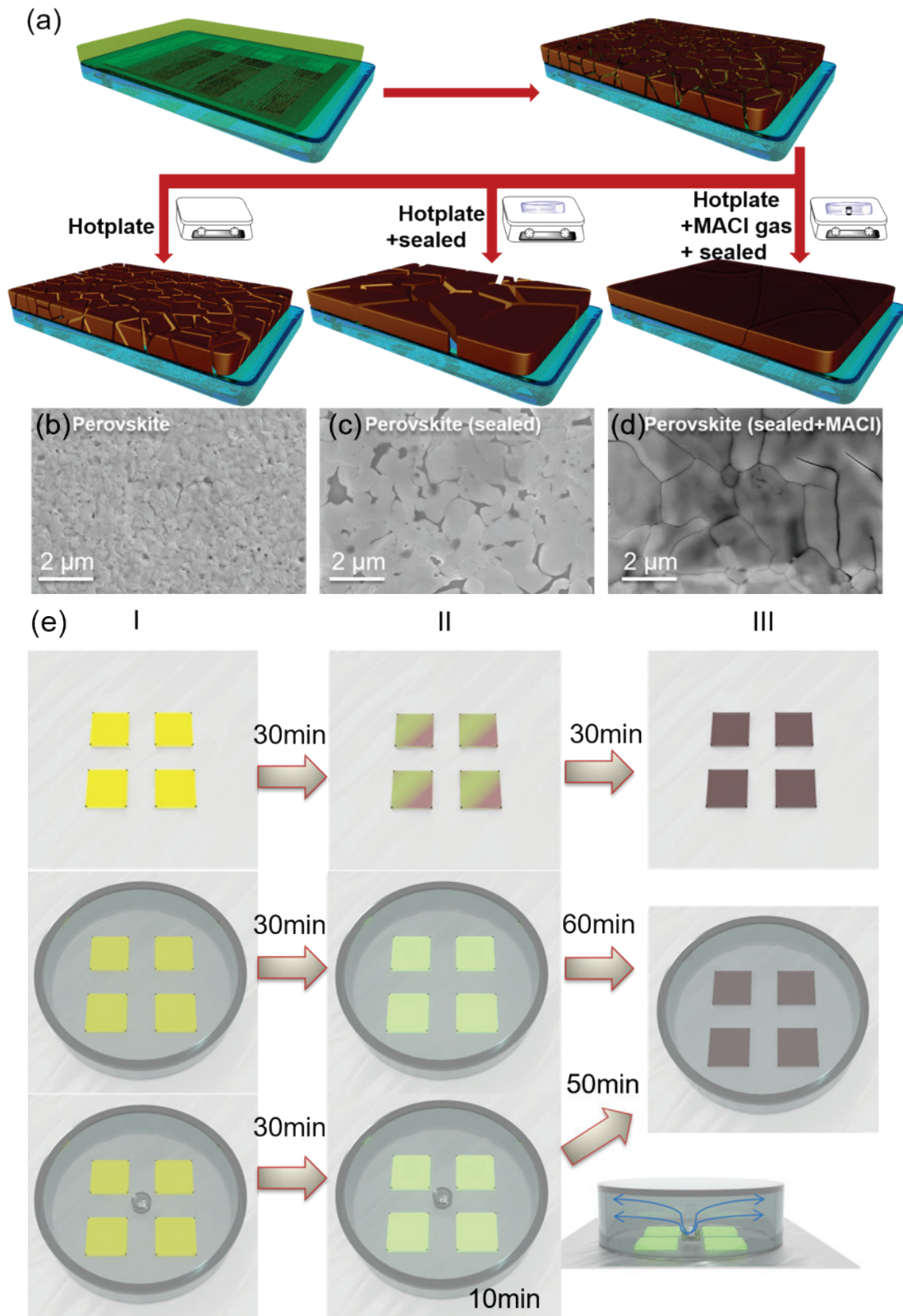


Figure 4.1 (a) Schematic diagram of pristine, sealed and MACl vapor annealing processes for perovskite film. SEM images of (b) pristine $\text{MAPbI}_{3-x}\text{Cl}_x$ thin-film, and the MAPbI_3 -

$x\text{Cl}_x$ thin films treated by (c) sealed (d) MACl vapor annealing processes; (e) Flowchart of annealing process for perovskite film.²³¹ Copyright 2022, American Chemical Society

The morphology of the $\text{MAPbI}_{3-x}\text{Cl}_x$ layers processed with the different methods in **Figure 4.1** is studied by scanning electron microscopy (SEM). It is found that the pristine $\text{MAPbI}_{3-x}\text{Cl}_x$ film exhibits randomly distributed tiny perovskite crystals, with many pinholes on the surface (**Figure 4.1b**). Since the MACl molecules are homogeneously distributed in the film, the escape point during annealing is also a random probability. That can be one of the reasons lead the surface morphology of **Figure 4.1b**. When the sealed annealing process treatment is used for the perovskite film, larger perovskite grains are achieved (**Figure 4.1c**). However, the coverage of the film is significantly reduced, as the surface presents much larger pinholes. As it is a sealed environment, the escaping MACl increased the concentration of MACl-gas and turned into a quasi-external chloride source. However, the vapor pressure is still not sufficient to assist the crystallization process. Due to the low boiling point, using MACl powder in a sealed environment is an excellent way to increase the vapor pressure around perovskite film. In clear contrast, the $\text{MAPbI}_{3-x}\text{Cl}_x$ film obtained from the external chloride source method treatment exhibits larger perovskite crystals and no pinhole on the surface (**Figure 4.1d**).

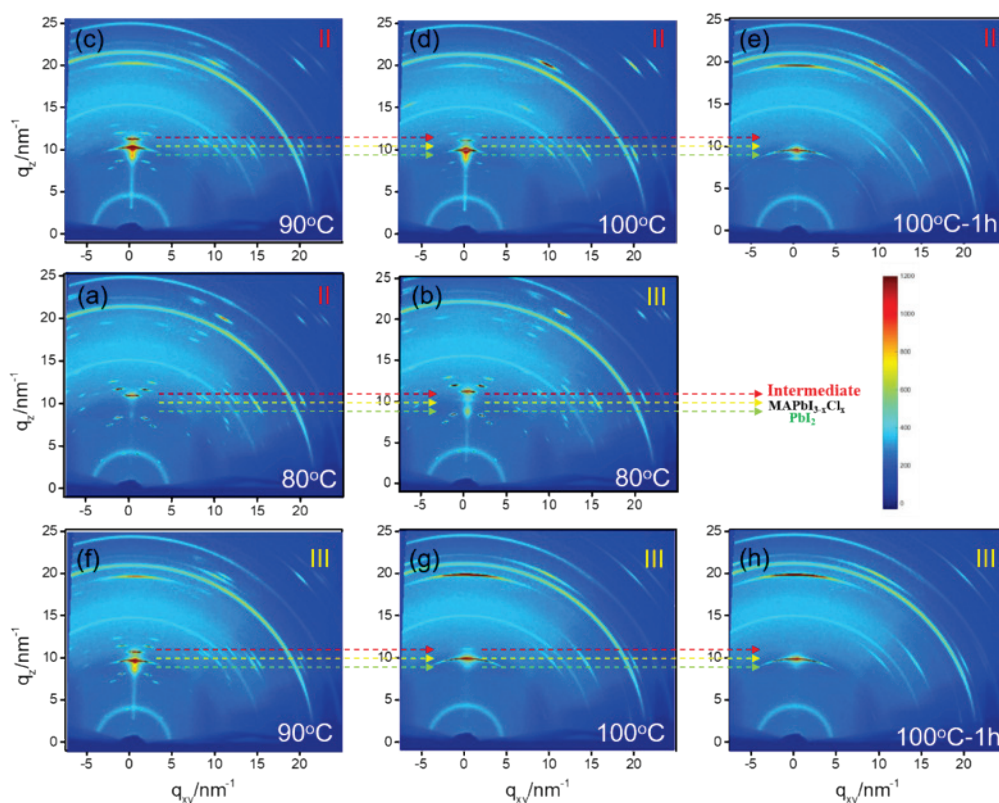


Figure 4.2 2D GIXRD profiles of the in-situ transition process of the films from intermediate compound to perovskite: for the case of sealed annealing at the temperature of (a) 80 °C, (c) 90 °C, (d) 100 °C, (e) 100 °C -1h, for the case of MACl vapor annealing at the temperature of (b) 80 °C, (f) 90 °C, (g) 100 °C, (h) 100 °C -1h.²³¹ Copyright 2022, American Chemical Society

Figure 4.2 shows the in-situ 2D GIXRD patterns performed to investigate the role of excess MACl vapor in the formation of perovskite films. At 80 °C, the signals attributable to intermediate species, PbI_2 and $\text{MAPbI}_{3-x}\text{Cl}_x$ were found for both cases of sealed annealing (**Figure 4.2a**) and MACl vapor annealing processes (**Figure 4.2b**). At the same time, the MAPbI_3 peak is more evident for the sample with MACl vapor. When the annealing temperature increases, the two films show different behavior. In the sealed annealed sample, a continuous decrease of the signals can be observed, which is attributed to the intermediate phase and PbI_2 with increasing temperature. At the same time, the signal of $\text{MAPbI}_{3-x}\text{Cl}_x$ continues to increase. Notably, at the end of the annealing process, a small amount of PbI_2 still exists (**Figure 4.2c-e**). When the excess MACl vapor annealing sample was introduced, the PbI_2 completely disappeared (**Figure 4.2f-h**), confirming the formation of a fully converted perovskite film. The above data shows the MACl vapor can effectively improve

the formation of perovskite and then promote the transition of locally enriched PbI_2 with MACl to form perovskite crystals. It can also explain why the MACl vapor annealing process devices can eventually get better film morphology.

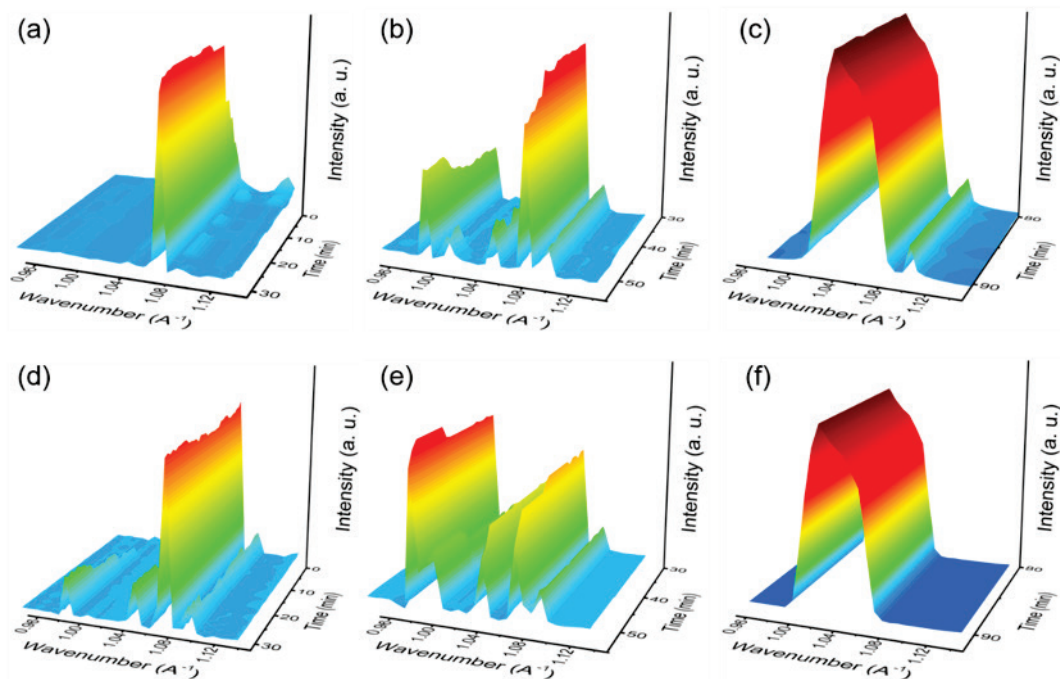


Figure 4.3 Evolution of the angle-dependent diffraction intensity of the in-situ transition process of films from intermediate compound to perovskite: (a) 60-90 °C (b) 90-100 °C, (c) 100 °C for sealed annealing sample, (d) 60-90 °C (e) 90-100 °C, (f) 100 °C for MACl vapor annealing sample.²³¹ Copyright 2022, American Chemical Society

To further understand this process, the evolution of the angle-dependent diffraction was analyzed for the cases of sealed annealing and MACl vapor annealing processes. **Figure 4.3** shows the evolution in intensity during the in-situ transition process for the perovskite films from as-cast films to final perovskite films. In the 60-90 °C step, only the intermediate phase was found for the sealed annealed sample (**Figure 4.3a**), while the intermediate phase, PbI_2 , and $\text{MAPbI}_{3-x}\text{Cl}_x$ were found for the MACl vapor annealed sample (**Figure 4.3b**), with $\text{MAPbI}_{3-x}\text{Cl}_x$ increasing with time. In the 90-100 °C step, $\text{MAPbI}_{3-x}\text{Cl}_x$ increased with time, and the intermediate phase decreased from 100 °C (40 min) for the sample with sealed annealing (**Figure 4.3c**), while for the sample with MACl vapor annealing $\text{MAPbI}_{3-x}\text{Cl}_x$ peak has a much stronger signal than the intermediate phase (**Figure 4.3d**). In the 100 °C step, a small amount of PbI_2 is still present for the sealed annealed sample, while a pure

$\text{MAPbI}_{3-x}\text{Cl}_x$ can be obtained for the MACl vapor annealed sample. These results are consistent with the observations by in situ 2D GIXRD and clearly illustrate the perovskite crystal formation process.

The perovskite crystals were locally restructured during crystallization when the MACl vapor concentration in the environment was increased. But suppose internal chloride is used to provide the MACl vapor since the intermediate phase is limited by the concentration of MACl vapor in the atmosphere. In that case, large crystals can be formed due to the coffee-spot effect²³². During MACl release, pinholes can also be created and even increase the size. It is desired that the concentration of MACl vapor be sufficient to form the new crystal. Hence, compared to the internal chloride source as the reference in this experiment, an external chloride source as MACl vapor in the environment is preferable for fabricating higher quality, pinhole-free perovskite films.

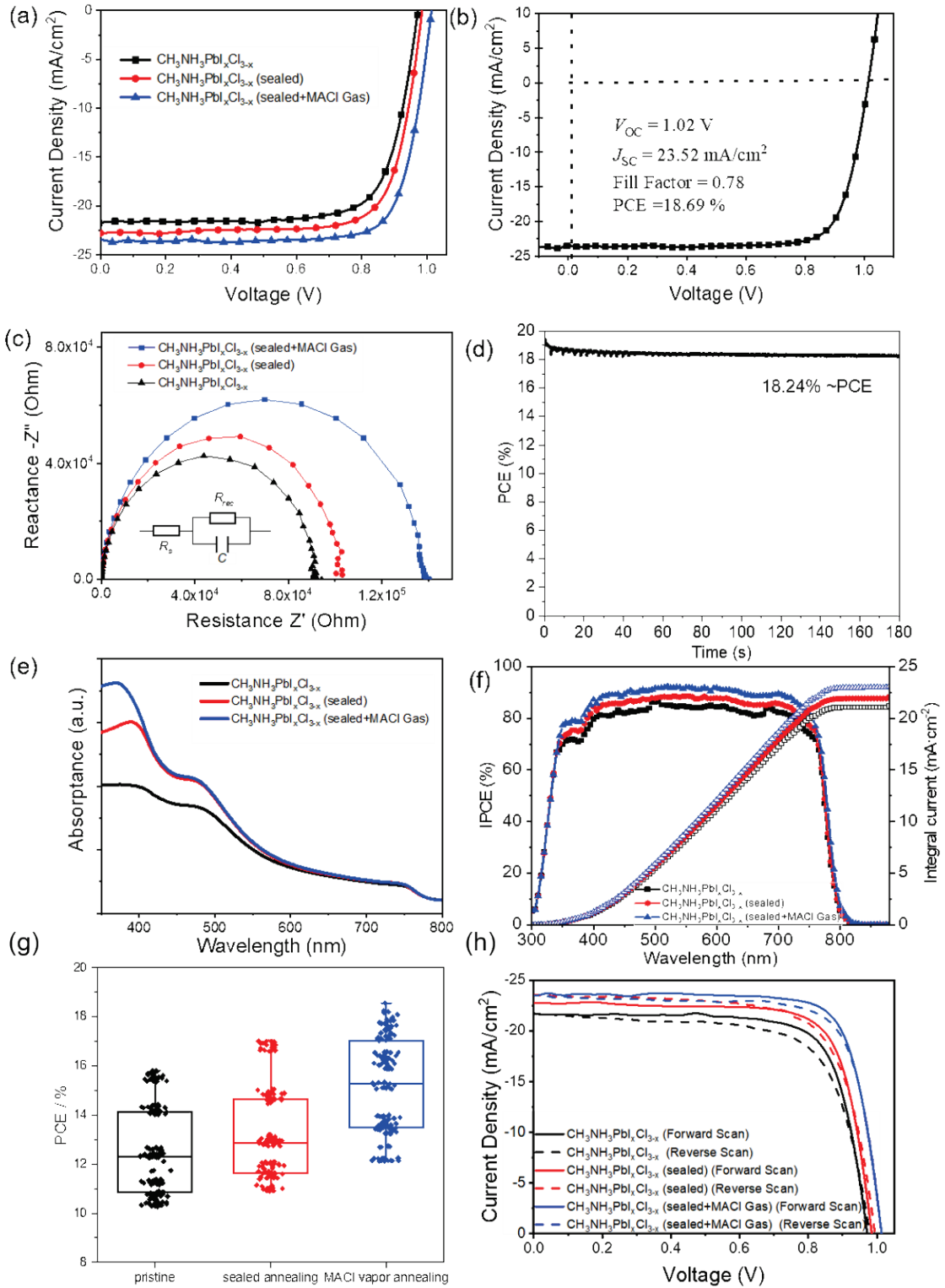


Figure 4.4 (a) $J-V$ curves of PSCs measured under simulated AM 1.5 sunlight of 100 mW cm^{-2} . (b) $J-V$ curves of the champion device under AM 1.5G illumination of 100 mW cm^{-2} . (c) EIS spectra for the PSCs with different concentrations of MACI vapor treatment. (d) The maximum power point for MACI vapor annealing process-based device

at 0.83 V and its corresponding power output. (e) UV-vis absorption spectra of the different annealing processes perovskite films. (f) IPCE spectra of the different annealing processes based on PSCs. (g) A histogram of PCE measured from MACl vapor-annealing process-based PSCs. (h) Measurements of J-V characteristics at forward and reverse scan directions.²³¹ Copyright 2022, American Chemical Society

To investigate the effects of the different annealing processes on device performance, devices are fabricated using an ITO/PEDOT: PSS/Perovskite/PCBM/BCP/Ag stack. **Figure 4.4a** shows the *J-V* characteristics of reference, sealed annealing, MACl vapor annealing-processed PSCs under AM 1.5G illumination with the light intensity of 100 mW cm^{-2} . The champion cell parameters are summarized in **Table 4.1**. The reference device presents a PCE of 15.80% with a J_{sc} of 21.71 mA cm^{-2} , a V_{oc} of 0.97 V, and a *FF* of 0.75. The sealed annealing process-based device shows a notable improvement with a *PCE* of 17.01% with a J_{sc} of 22.79 mA cm^{-2} , a V_{oc} of 0.98 V and a *FF* of 0.76. Finally, the optimized MACl vapor annealing process-based device shows the highest *PCE* of 18.69% with a J_{sc} of 23.52 mA cm^{-2} , a V_{oc} of 1.02 V and an *FF* of 0.78. The *J-V* curve for the best device is shown in **Figure 4.4b**. The MACl vapor annealing processes device performance improvement was attributed to the overall improvement of all parameters. These results confirm the contribution of the MACl vapor-assisted process on perovskite thin-film fabrication for devices and the enhanced beneficial effect of having an external MACl vapor source. The reproducibility of device performance is also studied by fabricating and characterizing over 30 cells. **Figure 4.4g** presents the histogram of the *PCE* of the MACl vapor annealing-processed devices. Electrochemical impedance spectroscopy (EIS) was performed for perovskite films processed with different treatments to comprehend the electron transport process's actual phenomena and recombination resistance. **Figure 4.4c** shows the typical Nyquist plots and equivalent circuits used to fit the experimental data. According to this model²³³, the recombination resistance (R_{rec}) which is attributed to the charge recombination at interfaces, can be obtained from the arc in the low-frequency region. Conversely, it is observed that films fabricated when the MACl vapor increased exhibit a larger arc radius of the Nyquist plot (lower R_{rec}) than devices without MACl vapor treatment. Since the perovskite layers were deposited under the same process except for the different MACl treatment, the increase in R_{rec} suggests the recombination rate decreases after the treatment by MACl vapor, as this parameter is inversely proportional to the recombination

rate.²³⁴ The higher FF value for the device by MACl vapor treatment is partially attributed to the large ratio of R_{sh} to R_s . In short, devices modified by MACl vapor have the highest J_{sc} , V_{oc} , and FF , thus the best performances.

Figure 4.4d shows the steady-state current density and PCE output of the champion device measured at the maximum power point (0.83 V) under AM1.5G 100 mW cm⁻² illumination. The PCE of the device stabilizes at 18.24%. The measurement of J - V characteristics at forward and reverse scan directions reveals that the photocurrent hysteresis could be suppressed by MACl vapor annealing treatment (**Figure 4.4h**). **Figure 4.4e** shows the UV-vis absorption spectra of the MAPbI_{3-x}Cl_x films with a pristine, sealed, MACl vapor annealing process. Similar band edges to the corresponding absorption spectra were obtained, showing a clear increase in intensity for the sealed and MACl vapor annealing-processed film absorption spectra in the range of 450-550 nm. The corresponding incident-photon-to-current efficiency ($IPCE$) spectra of pristine, sealed, MACl vapor annealing processes-based devices were measured (**Figure 4.4f**), showing a good match between the photocurrent densities integrated from $IPCE$ spectra and the J_{sc} obtained in J - V curves.

Table 4.1 Photovoltaic parameters of PSCs with pristine, sealed annealing, and MACl vapor annealing processes.

Device	J_{sc} (mA cm ⁻²)	V_{oc} (V)	FF (-)	PCE (%)	R_s (Ω)
pristine	21.71	0.97	0.75	15.80	41.03
sealed annealing	22.79	0.98	0.76	17.01	31.55
MACl vapor annealing	23.52	1.02	0.78	18.69	17.03

Our investigation explained the critical role of increasing the concentration of MACl vapor in the environment for better fabrication of good-quality MAPbI_{3-x}Cl_x perovskite film. In the following, the films are fabricated with pristine MAPbI₃ precursors without internal chloride, and a clear improvement in perovskite morphology is observed. The MACl influence scheme on MAPbI₃ is shown in **Figure 4.5a** and SEM results in **Figures 4.5b** and **4.5c**. The perovskite film quality is significantly improved, which suggests that the method using external chlorine source MACl vapor treatment can also optimize the film quality of MAPbI₃. Higher MACl vapor pressure is suitable for perovskite crystallization, as is studied

by in situ 2D GIXRD. And make Ostwald ripening process formation mechanism²³⁵ as a reference, in this work, MACl vapor changed the crystal growth kinetic power in the melting pot, which can form a big crystal of the perovskite film. Time-resolved photoluminescence (TRPL) decay plots show that after MACl vapor treatment, the photoluminescence lifetime of MAPbI₃ film is increased (As is shown in **Figure 4.5g**). The enhancement of PL intensity further confirms the synergistic effect of MACl vapor treatment on the suppressed non-radiative recombination associated with a lower trap density (As is shown in **Figure 4.5h**), which is consistent with previous reports.^{236,237} Based on the above work, a batch of devices has been fabricated by pristine MAPbI₃ and MACl-gas treated MAPbI₃ using an ITO/NiO_x/perovskite/C₆₀/BCP/Ag stack. The pristine MAPbI₃ device presents a *PCE* of 19.3% with a *J*_{sc} of 22.3 mA cm⁻², a *V*_{oc} of 1.13 V and a *FF* of 0.77, while the MACl gas-treated device shows a clear improvement with a *PCE* of 20.9% with a *J*_{sc} of 23.0 mA cm⁻², a *V*_{oc} of 1.13 V and a *FF* of 0.80 (As is shown in **Figure 4.5d**). The *PCE* distribution box chart is shown in **Figure 4.5e**.

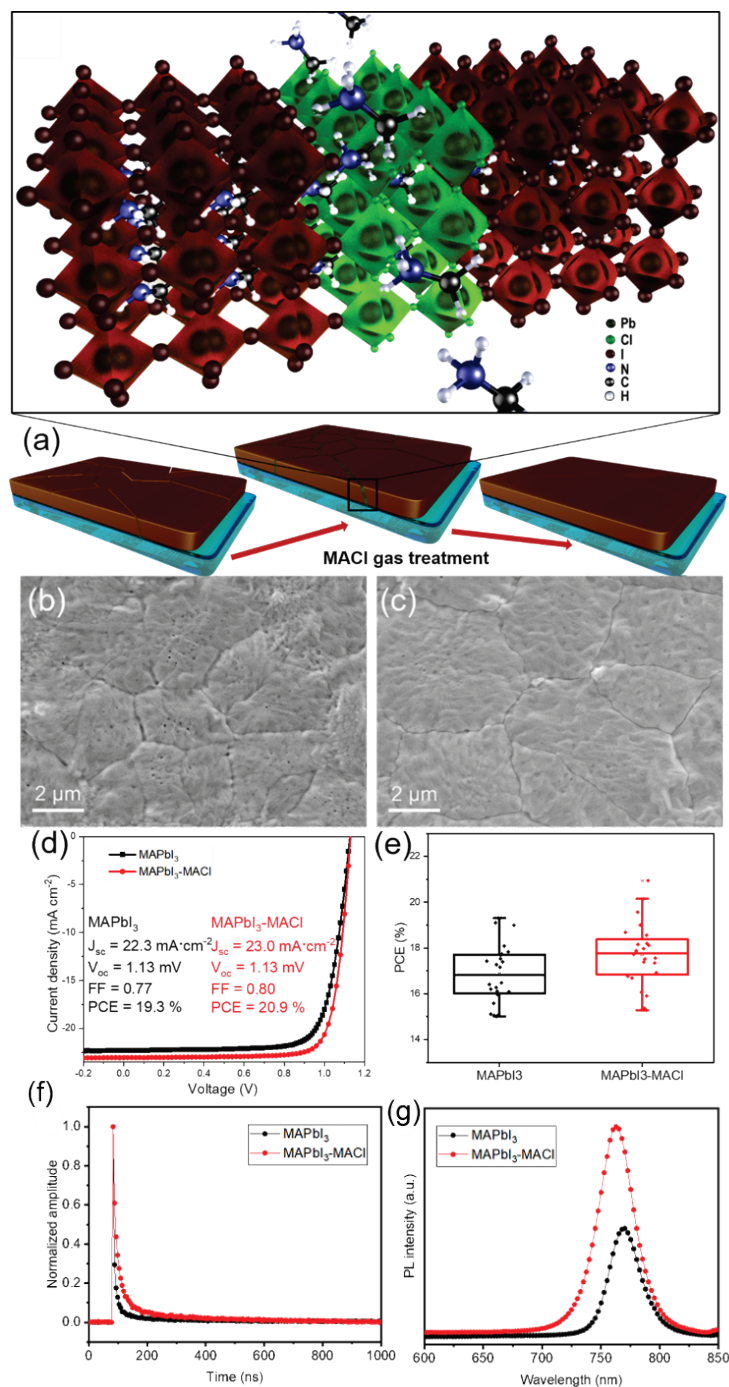


Figure 4.5 (a) The mechanism of MACl influence on MAPbI₃; SEM images of (b) the pristine MAPbI₃ thin film, and (c) the MAPbI₃ thin films treated with MACl vapor annealing process; (d) *J-V* curve of the champion device employing MAPbI₃ and MACl treated MAPbI₃ as the absorbing layer; (e) Box chart comparison between devices fabricated by MAPbI₃ and MACl treated MAPbI₃; (f) Time-resolved PL decay plots and (g) photoluminescence of MAPbI₃ films with and without MACl treatment.²³¹ Copyright

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Compared to the reference MAPbI₃ devices, the median average value of *PCE* increases by 8.3% when the perovskite film is treated by MACl gas. The *PCE* improvement is mainly dependent on the *J_{sc}* enhancement. As is shown in **Figure 4.6a**, *FF* and *V_{oc}* show no significant difference. The dark current plots (See **Figure 4.6b**) are also studied showing *J₀* values of 7.28×10^{-7} and 2.39×10^{-5} mA cm⁻² for the modified and control devices, respectively. The lower leakage density (*J₀*) of the modified device points to a better diode quality. As the maximum power point curve is shown in **Figure 4.6c**, the champion *PCE* of MAPbI₃ devices and MACl-gas treated devices are stable at 18.83% and 20.13%, respectively. A further test of the maximum power tracking curve at 85 °C and humidity of ~50% RH, and it can be seen that the devices made from MACl-treated films are not as stable at high temperatures as the control device (See **Figure 4.6d**). Further discussion, based on appropriate testing tools, is needed for a more detailed mechanistic study.

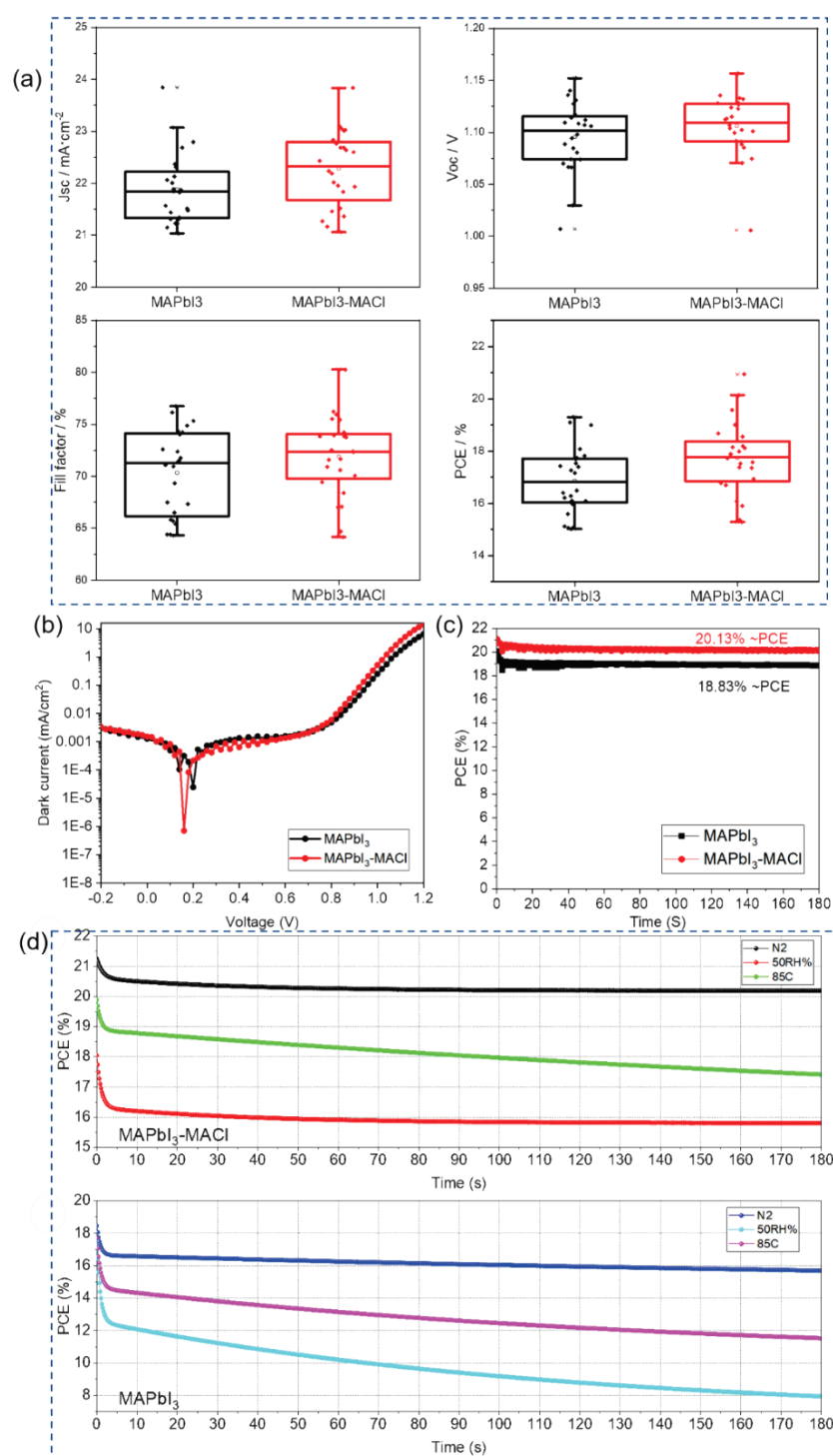


Figure 4.6 (a) Statistics of the PV parameters for 24 devices based on MAPbI₃ with and without MACl gas treatment; (b) The dark current curve and (c) the maximum power point curve of devices fabricated by MAPbI₃ and MACl treated MAPbI₃; (d) The maximum power point curve of devices measure in different conditions fabricated by MAPbI₃ and MACl treated MAPbI₃.²³¹ Copyright 2022, American Chemical Society

4.3 Conclusion

The role of MACl vapor pressure on the fabrication of chlorine-doped perovskite films is investigated, and the crystallization process of chloride-doped perovskites is studied using in situ 2D-GIXRD and in situ real-time X-ray diffraction. It is found that an external chloride source with a higher MACl vapor pressure is appropriate for the crystallization of perovskite crystals and also optimizes the quality of perovskite films. Compared to the external chloride source, the internal chloride source in our experiment is uniformly distributed in the perovskite film and forms more pinholes randomly during annealing. What's more, MACl vapor from internal chloride cannot provide enough momentum to support crystallization, and processed perovskite films are not densified. As demonstrated, MAPbI_{3-x}Cl_x-based PSCs champion devices present PCEs of 15.80%, 17.01%, and 18.69% under these three conditions, respectively. Considering the beneficial effects of a higher concentration of MACl vapor, films are fabricated based on MAPbI₃ using this treatment. SEM images show an evident improvement in morphology for the films processed with MACl vapor treatment. This difference could also be seen in device performance, where the PCE of the champion device improved from 19.3% to 20.9%. Thus, MACl vapor is presented as an easy and universal method to enhance the performance of PSCs.

5 Crystallization and stabilization of the FAPbI₃ single-phase perovskite

This work was published as: Zuo, W.; Fu, W.; Wang, K.; Das, C.; Byrannvand, M. M.; Wang, K. L.; Chaudhary, A.; Lim, J.; Li, M.; Saliba, M. Crystallization dynamics and stabilization of FAPbI₃ single-phase perovskite. *Energy Environ Sci* 2023, 17 (4), 1407–1415.

Zuo, W. fabricated and measured the films and devices, analysed the data, wrote the manuscript and contributed to the design and implementation of the research as well as the analysis of the results.

This work was also selected for the Cover of the corresponding issue (See **Figure 5.0**):

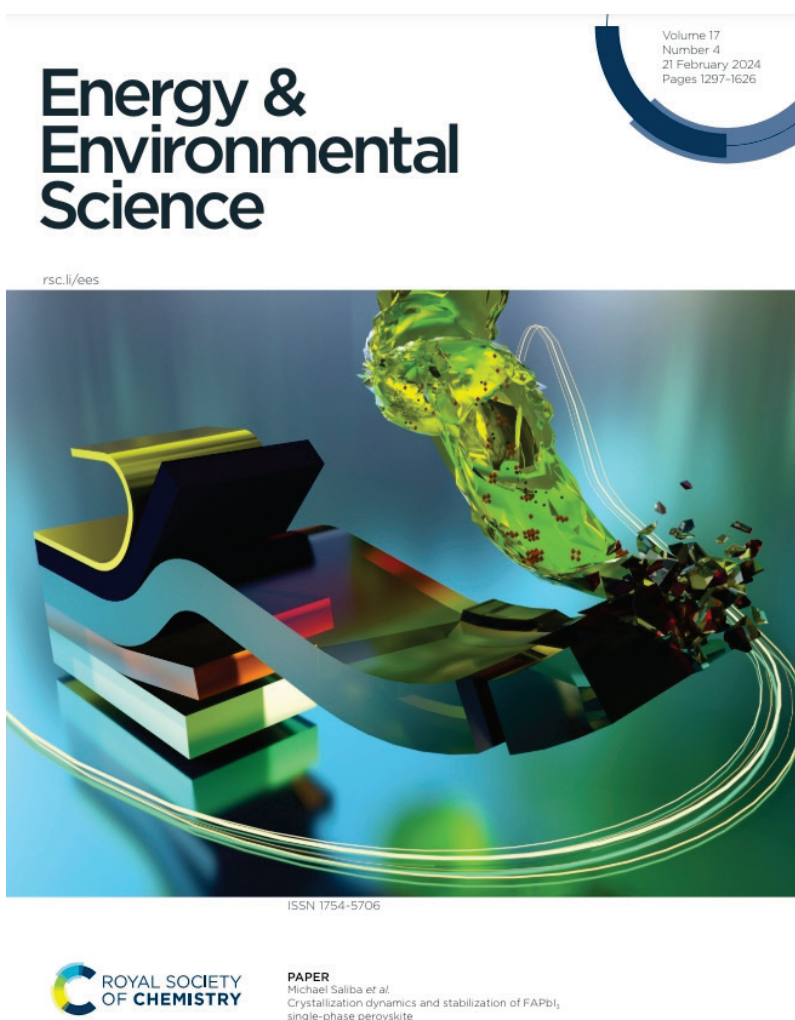


Figure 5.0 Front cover of the journal, designed and illustrated by Weiwei Zuo. Copyright 2024, Royal Society of Chemistry

The fundamental ABX₃ structure of lead-based perovskites is crucial for maintaining the crystal lattice stability and also enables exceptional optoelectronic properties. One main goal of the research community is to produce phase-stable (“ α -phase”) FAPbI₃. Here, this experiment used the LaMer model and an in-situ optical microscope to investigate the perovskite crystallization process. It is aimed to develop a stable single-phase FAPbI₃ perovskite by investigating the [PbX₆]⁴⁻ template and associated A-cations. Specifically, the addition of MACl is shown to facilitate the crystallization of FAPbI₃ without changing the [PbI₆]⁴⁻ skeleton. On the other hand, MABr acts as a passivation agent for the film, resulting in stable FAPbI₃ films with a pure α -phase when exposed to air with a humidity of 45%.

5.1 Introduction

PSCs have achieved an excellent power conversion efficiency (PCE) of 26.1%,⁴ which can be attributed to an exceptional light absorption coefficient,²³⁸ low Urbach energy,²³⁹ long carrier diffusion length,²⁴⁰ and defect tolerance²⁴¹. These materials have the generic formula “ABX₃”, where A is a monovalent organic cation, B is a divalent metal cation, and X is a halide anion. They consist of [BX₆]⁴⁻ octahedra that form a three-dimensional (3D) structure combined with A cations.^{9,242–244} These corner-sharing [BX₆]⁴⁻ octahedra leave voids, where the A cations reside. As the A-cations do not interact strongly with the metal-halide octahedra,^{245,246} rendering the [BX₆]⁴⁻ structure acts as a “skeleton” or scaffold.

While in the multi-component perovskite, the different ionic radii of I⁻, Br⁻, and Cl⁻ lead to different miscibility between the [PbX₆]⁴⁻ sublattices, especially between I⁻ and Cl⁻.²⁴⁷ The different ionic radii will also influence the bond length. For example, the Pb-Br bond length is 3.067 Å, whereas the Pb-I bond length is 3.279 Å.²⁴⁸ It will cause stacking faults in the lattice, leading to potential crystallographic defects.

However, the disadvantage caused by multi-phase or multi-component perovskite could be resolved through a controlled crystallization process of single-component perovskite film. For example, Peng et al. fabricated single-crystal MAPbBr₃²⁴⁹ and MAPbI₃²⁵⁰ solar cells with PCEs of 6.53% and 21.1% respectively. However, single crystal thin films^{249,251–255} are arduous to fabricate and may be impractical for industrial manufacturing. Hence, fabricating

perovskite films in a simple method that minimizes carrier defect compounding while maintaining the stability of the single-phase perovskite would be an important target.

Therefore, this work focuses on the growth of stable black single-phase FAPbI₃ perovskite. After studying the crystallization time of various perovskites, MACl is selected as a crystallization aid, which can be easily removed due to its low boiling point,²³¹ making it an ideal choice for single-phase perovskite growth. Furthermore, the post-treatment with MABr introduces MA⁺ ions to repair defects in FAPbI₃ films, while Br⁻ ions could enhance the film stability. Notably, the modified perovskite films by MABr exhibit higher stability compared to the MABr-free perovskite films. The corresponding solar cell exhibits a PCE of 23.0%.

5.2 Result and discussion

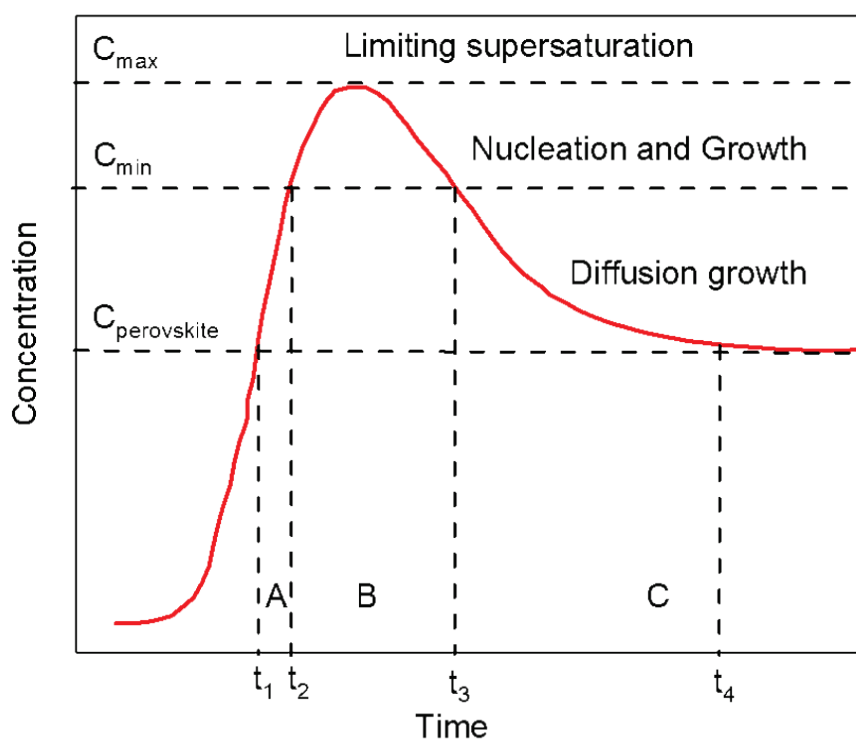


Figure 5.1 The LaMer model of a perovskite film crystallization process. Copyright 2024, Royal Society of Chemistry

The crystallization characteristics of perovskite films prepared by the solution method are similar to the crystallization model in solution.²⁵⁶ The formation of nuclei depends mainly on the solution concentration variation. Using spin coating as an example, the solution concentration changing in the pre-crystallization film period comes primarily from the spin coating speed and the ambient temperature.²⁵⁷ Although water affects perovskite, a vapor film that prevents the entry of water and oxygen forms above the perovskite film during the evaporation of the solvent for crystallization,²⁵⁸ and our work discusses the perovskite crystallization process based on this situation. The LaMer model^{259,260} of perovskite film crystallization is shown in **Figure 5.1**. As the concentration increases, in part A where homogenization is just completed, 0D perovskite monomers are dispersed in the solution. When the concentration reaches the minimum limit concentration ($C_{\min}$), nuclei appear. From t_2 onwards, in part B, new nuclei are formed and accompanied by rapid crystal growth. Part C completes the final growth of grains and is accompanied by Ostwald ripening²⁶¹, and finally, perovskite films are formed.

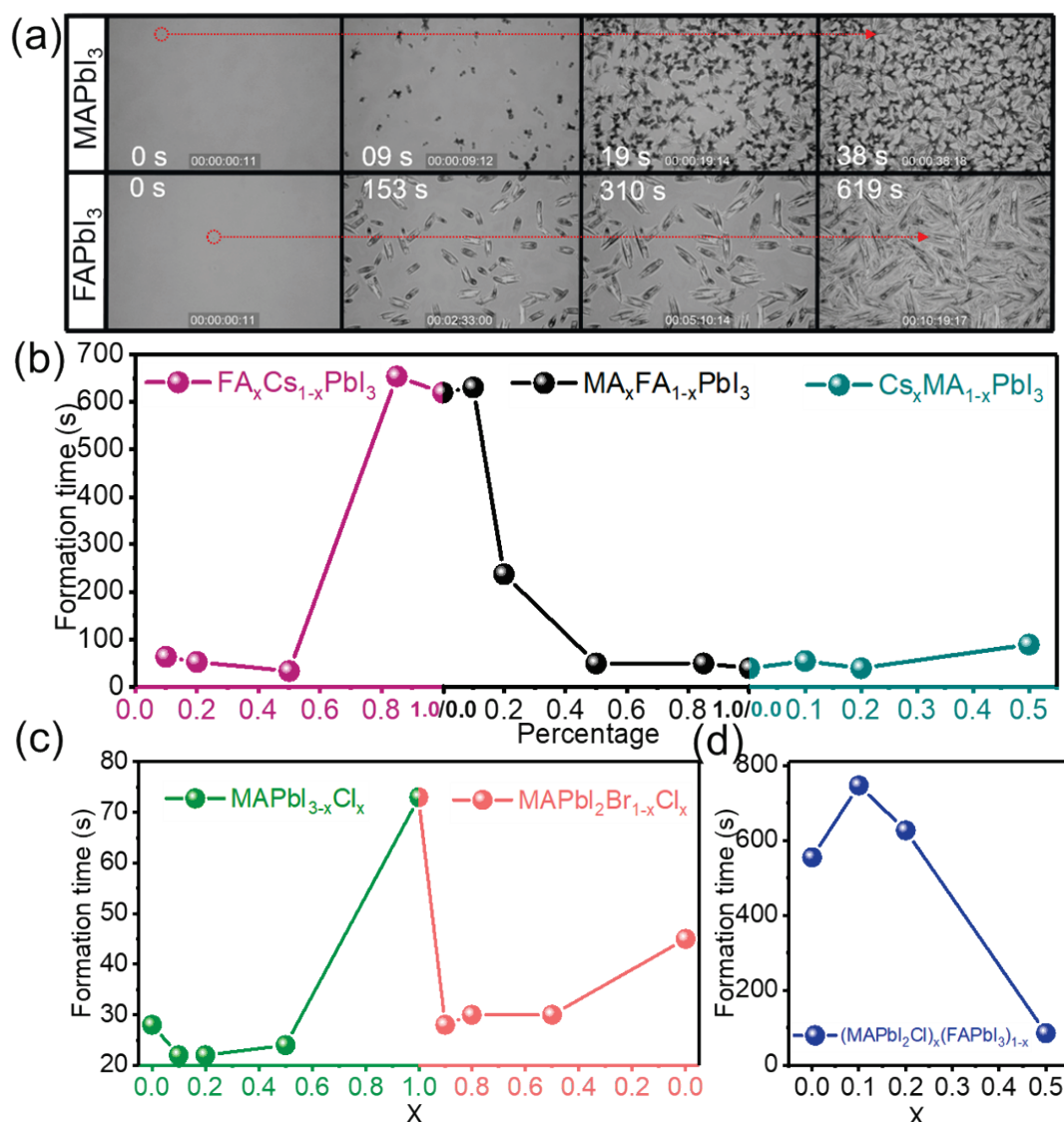


Figure 5.2 Crystal formation time of perovskites with different compositions. (a) In situ optical microscopy images of MAPbI₃ and FAPbI₃, Crystallization rate trend of perovskite with (b) different A-cations, (c) different halides, and (d) mixed A-cations and halides. Copyright 2024, Royal Society of Chemistry.

This experiment roughly counted the duration of $t_4 - t_3$ as the crystal formation time t_{form} (See **Figure 5.1**) indifferent perovskite crystallization processes by in situ optical microscopy (See **Figure 5.2**) (t_3 marks the moment when nuclei become visible. This occurrence is a result of varying growth times for microcrystalline particles in the film. When observing the film through an optical microscope, the first microcrystalline particle becomes apparent at t_3 , and this time is defined as the point at which nuclei appear. t_4 is the approximate time at which the change in crystal growth cannot be readily observed.). The cations currently used

in solar cell fabrication suitable for perovskite formation are Cs⁺, MA⁺, and FA⁺, and the cation size order is Cs⁺ (1.81 Å) < MA⁺ (2.70 Å) < FA⁺ (2.79 Å).²⁶² In order to discuss the effect of different radii of A cations on the crystallization rate, perovskite containing different cations was mixed in different proportions at the same concentration in this work to examine the t_{form} . To exclude the influence of factors other than the environment on the crystallization of perovskite, the perovskite films were fabricated by spin-coating at 45% humidity and 24.4-25.5 °C (indoor environmental conditions) without the application of anti-solvents and were observed under an in situ optical microscope in the same environment. As is shown in **Figure 5.2a**, the t_{form} increases very significantly during the gradual shift of the A cation from Cs⁺ to FA⁺ in FA_xCs_{1-x}PbI₃ from 33 s (Cs_{0.5}FA_{0.5}PbI₃) to more than 600 s with $x < 0.5$, where x is the mole ratio of FA⁺ to all the A-cations in the precursor. Likewise, the t_{form} is significantly decreased from 237 s (MA_{0.2}FA_{0.8}PbI₃) to 49 s (MA_{0.5}FA_{0.5}PbI₃) when the ratio of MA⁺ in MA_xFA_{1-x}PbI₃ exceeds 50%. And due to the fast crystallization rate of CsPbI₃, the short t_{form} could not be counted, which is consistent with the literature reports.²⁶³ As the size of the A-cation in perovskites increases from Cs⁺ to MA⁺ to FA⁺, the t_{form} increases (e.g. t_{form} of MAPbI₃ is 39 s and FAPbI₃ is 620 s), with the longest t_{form} observed for FA⁺ and the shortest t_{form} observed for Cs⁺.

In addition to discussing the effect of A cations on the crystallization rate of perovskite, the effect of the radius of halogen ions on the crystallization rate of perovskite is also discussed (See **Figure 5.2b**). The indoor environment at this time is 49% humidity and 23-25 °C. For MAPbI_{3-x}Cl_x, the t_{form} of the perovskite increases significantly when $x < 0.2$ as the Cl ratio increases from 22 s (MAPbI_{2.8}Cl_{0.2}) to 73 s (MAPbI₂Cl), where x is the mole ratio of Cl to all the halogens in the precursor. For MAPbI₂Br_{1-x}Cl_x, the t_{form} becomes shorter with higher Br content (e.g. t_{form} of MAPbI₂Cl is 73 s and MAPbI₂Br is 45 s). However, when the Cl content is further decreased beyond $x < 0.5$, the t_{form} increases from 22 s (MAPbI_{2.9}Cl_{0.1}) to 28 s (MAPbI₃), indicating that additional factors may affect t_{form} . For example, the number of ionic species and the number of ions in the perovskite precursors. Even so, when halogens are replaced in equal proportions, comparing MAPbI₃, MAPbI₂Br and MAPbI₂Cl, it is evident that the t_{form} follows t_{MAPbI_3} (28 s) < $t_{\text{MAPbI}_2\text{Br}}$ (45 s) < $t_{\text{MAPbI}_2\text{Cl}}$ (73 s), which indicates that the larger the radius of the halogen atom the shorter the crystallization time. This regular pattern changes when both the A cation and the halogen are varied, as in **Figure 5.2d**. It suggests when the ionic species become more complex (possibly because of complex

mixing), it may also affect t_{form} . However, this goes beyond the scope of the current work as it requires multivariable analysis. Nevertheless, it can still be seen that in $(\text{MAPbI}_2\text{Cl})_x(\text{FAPbI}_3)_{1-x}$, the addition of an appropriate concentration of MACl to FAPbI₃ results in a longer t_{form} for FAPbI₃ (e.g. t_{form} of $(\text{MAPbI}_2\text{Cl})_{0.1}(\text{FAPbI}_3)_{0.9}$ is 747 s compare to FAPbI₃ with 555 s). Based on our observations, in lead-based perovskite, it can be inferred that the t_{form} will be longer when as r_A increases or r_X decreases ($A=\text{Cs}, \text{MA}, \text{FA}$; $X=\text{Cl}, \text{Br}, \text{I}$).

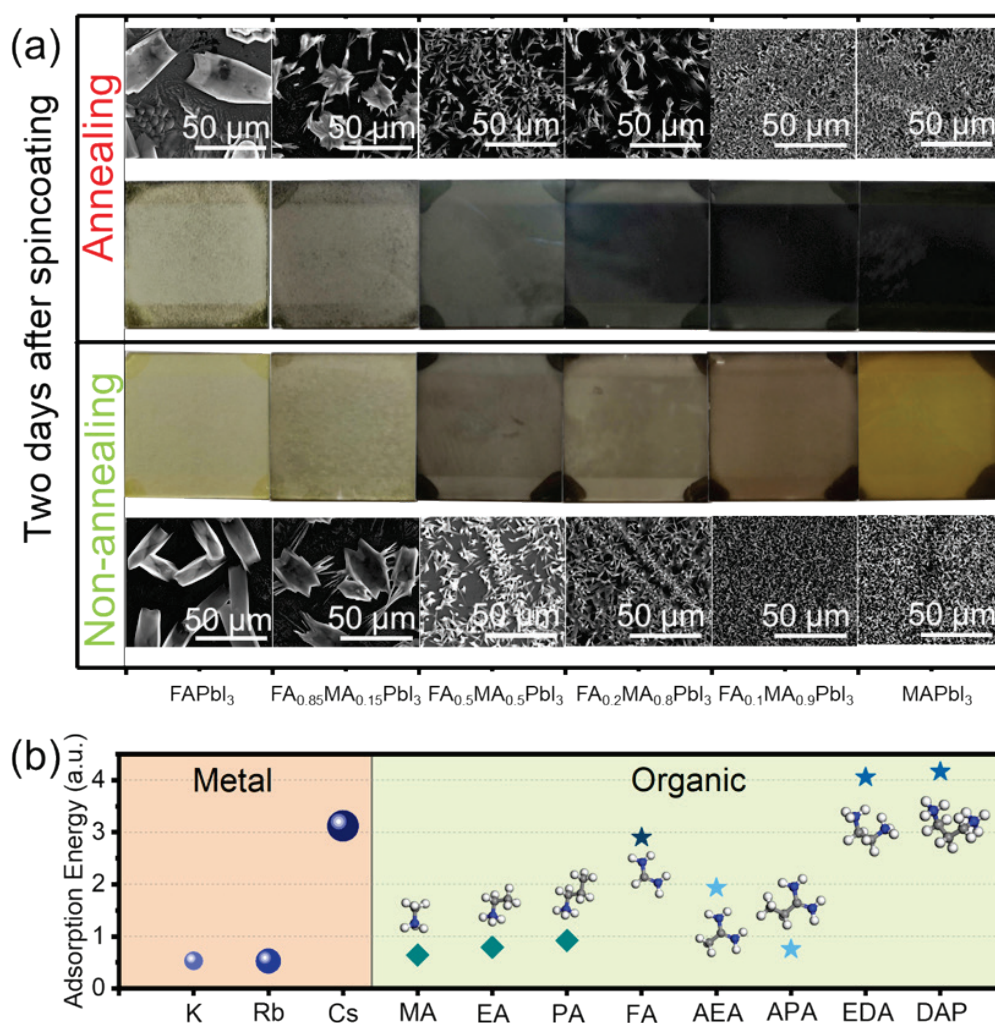


Figure 5.3 (a) Morphology of perovskite films with different A-cations with annealing at 100 °C (top) and without annealing (bottom). (b) Adsorption energies of different A-cations on PbI₂ (0 0 1) surfaces using the following cations: K, Rb, Cs, MA(CH₃NH₃), EA(CH₃CH₂NH₃), PA(CH₃CH₂CH₂NH₃), FA(CH(NH₂)₂), AEA(CH₃CH(NH₂)₂), APA(CH₃CH₂CH(NH₂)₂), EDA(NH₃CH₂CH₂NH₃), DAP(NH₃CH₂CH₂CH₂NH₃). The diamond shape denotes a small molecule with a single amino group; the star denotes a small molecule with two amino groups. Copyright 2024, Royal Society of Chemistry.

The A-cation strongly influences the perovskite crystallization, as it affects the t_{form} and determines whether a relatively stable intermediate phase can be obtained. To study the diffusion of A cations in the PbI₂ lattice, the crystallization properties of perovskite are investigated by spin-coating FA_xMA_{1-x}PbI₃ films without anti-solvent. The films were kept in N₂ atmosphere for three days to ensure that spontaneous crystallization had ended (See **Figure 5.3a**). One of the films was subsequently annealed at 100 °C for 1 h and the unannealed film was used as the control sample. The color of FAPbI₃ stays in the yellow phase after being placed in nitrogen for three days, and when the percentage of MA in perovskite gradually increased, the color of the film began to change from yellow to dark yellow, indicating that MA diffuses differently from FA.

Annealing does not only let the solvent vapor escape but also increases the thermal movement of the ions inside the film and thus increases the chance of their bonding to form perovskite grains. Upon annealing the films at 100 °C in nitrogen for three days, the films with higher MA content turn darker, indicating the formation of black MAPbI₃ perovskite. Pure FAPbI₃ can also form the black phase on its film surface after annealing at 100 °C.

Subsequently, the films are analyzed using scanning electron microscopy (SEM). The results in **Figure 5.3a** show that the grain shape of pure MAPbI₃ without annealing and anti-solvent treatment is needle-like, which is similar to the grain shape of PbI₂,²⁶⁴ suggesting that perovskite crystals are directed by the formed PbI₂ crystals, which then become perovskite upon addition of cation salts.

As the MA⁺ content in perovskite is decreased, an increase in grain size within the film is observed. Once the proportion of FA surpassed 50%, a significant change is noticed in grain shape, with a single display becoming more prominent. This is consistent with the observation in **Figure 5.2**, i.e. that the crystallization rate increases when the A cation ratio exceeds 50%.

This suggests that the steric effects of different anion sizes result in differences in the crystallization rate and that the differences in steric effects result in differences in the shape of the grains. To determine the mechanism, the initial adsorption of different A-cations is investigated using density functional theory (DFT) to understand the interactions between A-cations and PbI₂ (0 0 1) surfaces (See **Figure 5.3b**). In the calculations, the adsorption

energies (E_{ads}) of A-cations on the (0 0 1) surface of PbI₂ are calculated according to the following formula:²⁶⁵

$$E_{\text{ads}} = E_{\text{surface}} - E_{\text{PbI}_2(0\ 0\ 1)} - E_{\text{molecule}} \quad (5)$$

In the formula, E_{surface} represents the total energy of the relaxed (0 0 1) surface with A-cations. $E_{\text{PbI}_2(0\ 0\ 1)}$ corresponds to the energy of the clean and relaxed (0 0 1) surface of PbI₂. Our calculations indicate that the effect of spin polarization on the energy of A-cations is negligible. Therefore, the energy (E_{molecule}) of isolated A-cations is obtained without spin polarization. Based on our calculations, the adsorption energies E_{ads} of A-cations are negative, implying an exothermic process for A-cations adsorption. The following discussion focuses on the absolute values of the adsorption energies. The adsorption energy values of all the calculations are shown in **Table 5.1**.

Strong adsorption energy between molecules typically can increase steric effects and affect their movement. The adsorption energy of Cs on PbI₂ (0 0 1) surface is significantly higher compared to Rb and K. This aligns with the article Hu et al.²⁶⁶, which states that Cs reduces the defect density and charge complexation rate of the perovskite layer while Rb increases the carrier mobility. In the adsorption energy comparison in **Figure 5.3b** (Specific values are shown in **Table 5.1**), the adsorption energy of MA on PbI₂ (0 0 1) surface is significantly lower than that of Cs and FA in the A-cations commonly used to make 3D perovskite. Additionally, the evidence presented in **Figure 5.3b** corroborates the observation depicted in **Figure 5.3a**. To delve deeper into the study, it is analyzed how the longer alkyl chain affects the adsorption energy. **Figure 5.3b** clearly indicates that as we move from MA to Ethylammonium (EA) to Propylammonium (PA), adsorption energy is slightly increased from 0.64 to 0.79 to 0.92 eV/molecule. However, the size of the A cation also increases, making EA and PA unsuitable for 3D perovskite preparation.

On the other hand, when the alkyl chain is extended from FA to 1-aminoethaniminium (AEA) to 1-aminopropyliminium (APA) based on a similar amidine group, the adsorption energy decreases significantly from 2.90 to 1.93 to 0.76 eV/molecule. Interestingly, if the terminal group is turned into two amino groups and extends the alkyl chain, the adsorption energy increases when comparing FA with ethylenediaminium (EDA) and 1,3-Diaminopropanium (DAP) from 2.90 to 4.06 to 4.16 eV/molecule. These findings highlight the unique structure of FA, which makes it suited for stable single-phase perovskites.

Table 5.1 Energy parameters of the compounds

K-PbI ₂ (001)	E _{surface} ^a (eV)	E _{molecule} ^b (K) (eV)	E _{PbI₂(001)} ^c (eV)	E _{ads} ^d (eV/ molecule)
	-7812.55416	-791.1924123	-7020.831775	-0.529973372
Rb-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (Rb) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-7711.632619	-690.2741841	-7020.831181	-0.527253647
Cs-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (Cs) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-7797.6696	-774.5739346	-7019.973243	-3.12242318
MA-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (MA) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-7550.283042	-528.8319102	-7020.806812	-0.644320149
EA-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (EA) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-7740.666725	-719.0640037	-7020.811435	-0.791286108
PA-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (PA) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-7930.698615	-908.9995526	-7020.776379	-0.922683654
FA-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (FA) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-7812.145232	-789.2545921	-7019.991361	-2.899279163
AEA-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (AEA) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-8001.740487	-979.5397581	-7020.269344	-1.931384394
APA-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (APA) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-8191.230903	-1169.669312	-7020.806442	-0.755148682
EDA-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (EDA) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-8049.690018	-1025.64825	-7019.986427	-4.055341106
DAP-PbI ₂ (001)	E _{surface} (eV)	E _{molecule} (DAP) (eV)	E _{PbI₂(001)} (eV)	E _{int} (eV/ molecule)
	-8239.558908	-1215.305392	-7020.090472	-4.16304469

a: The total energy of the relaxed PbI₂ (0 0 1) surface.**b:** The energy of A-cations on the clean relaxed (0 0 1) surface of PbI₂.**c:** The energy of the clean relaxed (0 0 1) surface of PbI₂.**d:** The adsorption energy.

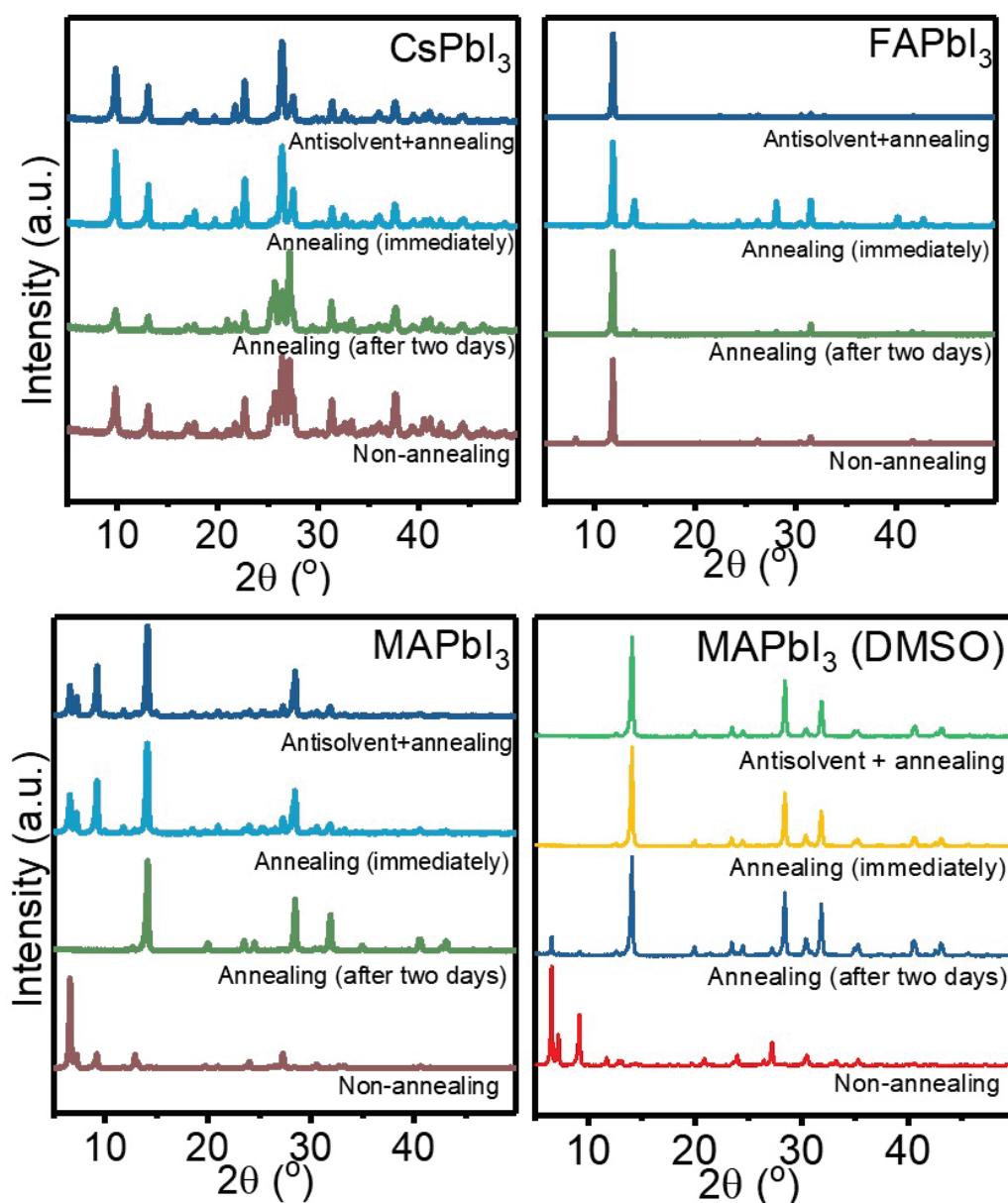


Figure 5.4 XRD Pattern of different A cations perovskite films by different treatments and MAPbI₃ films fabricated by pure DMSO. Copyright 2024, Royal Society of Chemistry

As described above, the crystallization of perovskite is related to the appropriate A-cation, which not only affects its crystallization rate but also determines whether a relatively stable intermediate phase can be obtained. Three factors influence the perovskite crystallization: time, anti-solvent, and annealing. These were compared to clarify this issue (**Figure 5.4**). Taking MAPbI₃ as an example, low diffraction angles (6.55°, 7.21°, and 9.17°) shown in

the XRD pattern suggest that films mainly have some different structures formed by the intercalation of DMF and MAI molecules into a layered PbI₂ phase and MAI-PbI₂-DMSO intermediate phase^{267–269} when the films placing in nitrogen without any treatment for two days. A pure perovskite phase could be obtained after annealing at 100 °C.

However, when the MAPbI₃ film is annealed immediately just after spin coating, the film still has an intermediate phase as can be seen from the XRD shown in **Figure 5.4**. This could not be eliminated even with anti-solvent. Considering that the intermediate phase is related to the precursor solvent, the DMSO: DMF solvent mixture is replaced with pure DMSO (**Figure 5.4**) to increase the purity of the intermediate phase. The XRD results show that the samples annealed immediately after spin-coating or annealed after spin-coating with anti-solvent yield a relatively pure perovskite phase, while the films annealed two days after spin-coating show residual MAI-PbI₂-DMSO intermediate phase.^{267–269}

This also demonstrates the possible involvement of DMSO in the formation of low-dimensional perovskite in the precursors, which can remain during post-treatment. The complex can be attributed to the peak at 6.48° corresponding to a larger distance between the crystallographic planes (~ 13 Å).²⁶⁸ For FAPbI₃, the film annealed two days after spin-coating gives a pure FAPbI₃ hexagonal phase, while immediate annealing causes the precipitation of PbI₂. Annealing after the dropwise addition of the anti-solvent also gives a pure FAPbI₃ hexagonal phase. The perovskite films with the smaller A-cation Cs do not show much difference under various conditions.

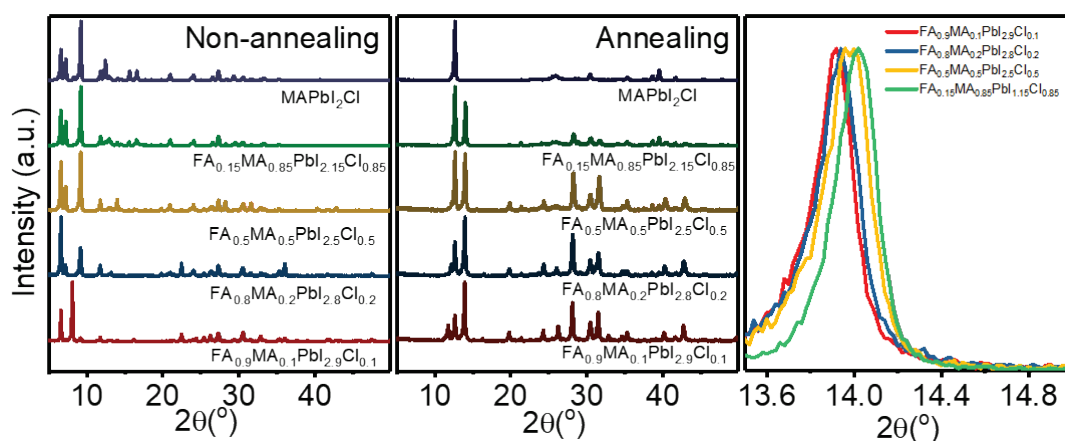


Figure 5.5 The effect of capping agent concentration on the crystallization of FAPbI₃ perovskite was explored by XRD and comparison of the perovskite phase. Copyright 2024, Royal Society of Chemistry

Based on observations, it appears that the MA cation is the most appropriate option for the intermediate phase. To confirm this, a study is conducted using FAPbI₃ perovskite as the primary material and introduced MA ions as the doping cation to investigate how to produce α -phase FAPbI₃ perovskite. With reference to our previous work²³¹, MACl was chosen as an additive to help FAPbI₃ form the perovskite phase (See **Figure 5.5**). Then a series of FA_{1-x}MA_xPbI_{3-x}Cl_x films are fabricated, and the phases of annealed and unannealed films are compared after two days in nitrogen. The XRD results clearly show a significant change in the composition of the interstitial phase in the films without intervention in the static nitrogen when the MA percentage reaches 50%, which agrees with the previous discussion. This indicates a strong correlation between the stable homogeneous intermediate phase and the ratio of capping additives. Meanwhile, after annealing at 150 °C²⁷⁰ FAPbI₃ hexagonal phase still exists in perovskite at $x = 0.1$, while the FAPbI₃ hexagonal phase disappears when $x = 0.2$, which proves that MACl as an additive can well help FAPbI₃ to form black perovskite phase. The PbI₂ in the film is mainly from the component residue after the volatilization of MACl, which is easily found in the XRD of the component with $x = 1$. **Figure 5.5** (right side) also shows that in the zoomed-in X-ray diffraction patterns between 13° to 15°, the peak of the black perovskite phase is shifted with the increase of MACl content in the precursor, which is consistent with the description of Bragg's law. Interestingly, at $x = 0.5$, an α -phase also formed spontaneously in perovskite that had been left for a long time and had not been annealed.

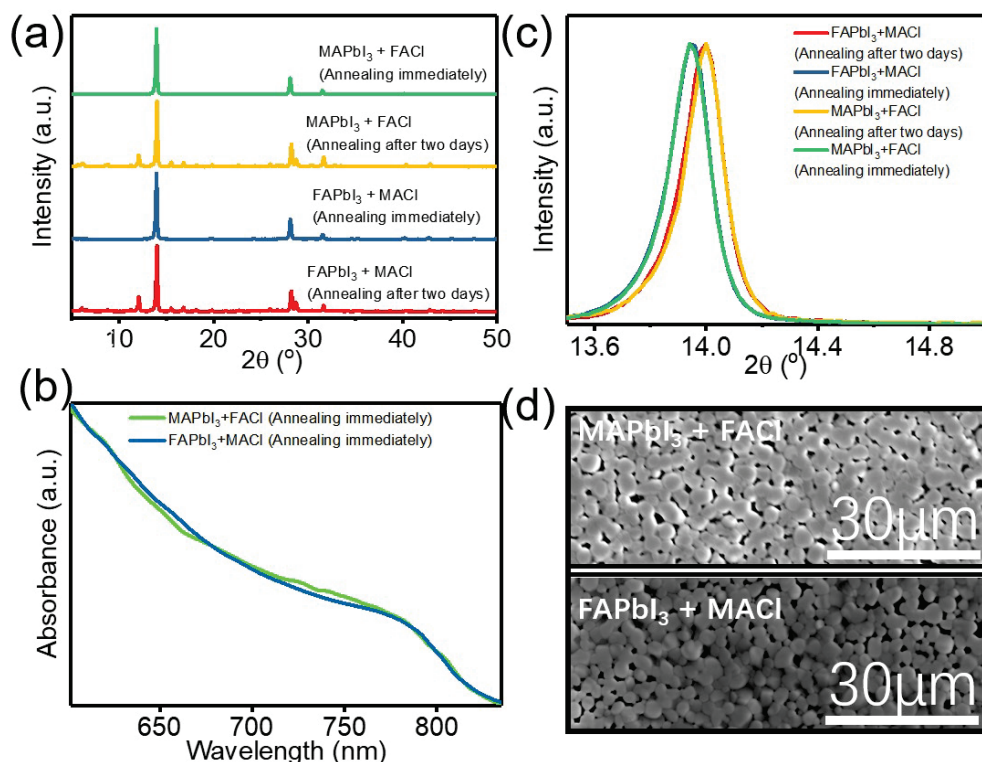


Figure 5.6 FAPbI₃ films fabricated by the same ratio of ions but different materials.

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Therefore, two perovskite precursor ratios, 50 mol% MACl in 1.5 M FAPbI₃ and 50 mol% FACl in 1.5 M MAPbI₃ are compared and analyzed the films using three characterization tools, XRD, UV-Vis, and SEM (See **Figure 5.6**). As **Figure 5.6a** shows, when perovskite films are stored in nitrogen for more than two days after spin coating and then annealed, both δ - and α -phases FAPbI₃^{271,272} will be present in the film. But when the film is spin-coated and immediately annealed, a pure α -phase perovskite is obtained. From XRD it can be seen that the perovskite formed by both formulations is of the same type, as also evidenced by the UV-vis absorption curves (**Figure 5.6b**). However, when the XRD pattern is zoomed in at 13° to 15° (**Figure 5.6c**), not only is it possible to determine that the crystalline surfaces of the crystals in the perovskite films produced by the two precursors are the same immediately annealed and two days later annealed, but the presence of the yellow phase also affects the crystalline surface of the perovskite according to Bragg's law. However, from the scanning electron microscopy observation, the film obtained from the 1.5 M precursor solution did not cover the substrate completely (**Figure 5.6d**), so it is considered to increase the concentration of the precursor solution to achieve full coverage

of the substrate. But still, the perovskite crystal phases presented in the SEM prove that the perovskite produced by the two precursors is of the same type. The above data also indicate that perovskite is always in ionic form from the precursor solution to crystal formation.

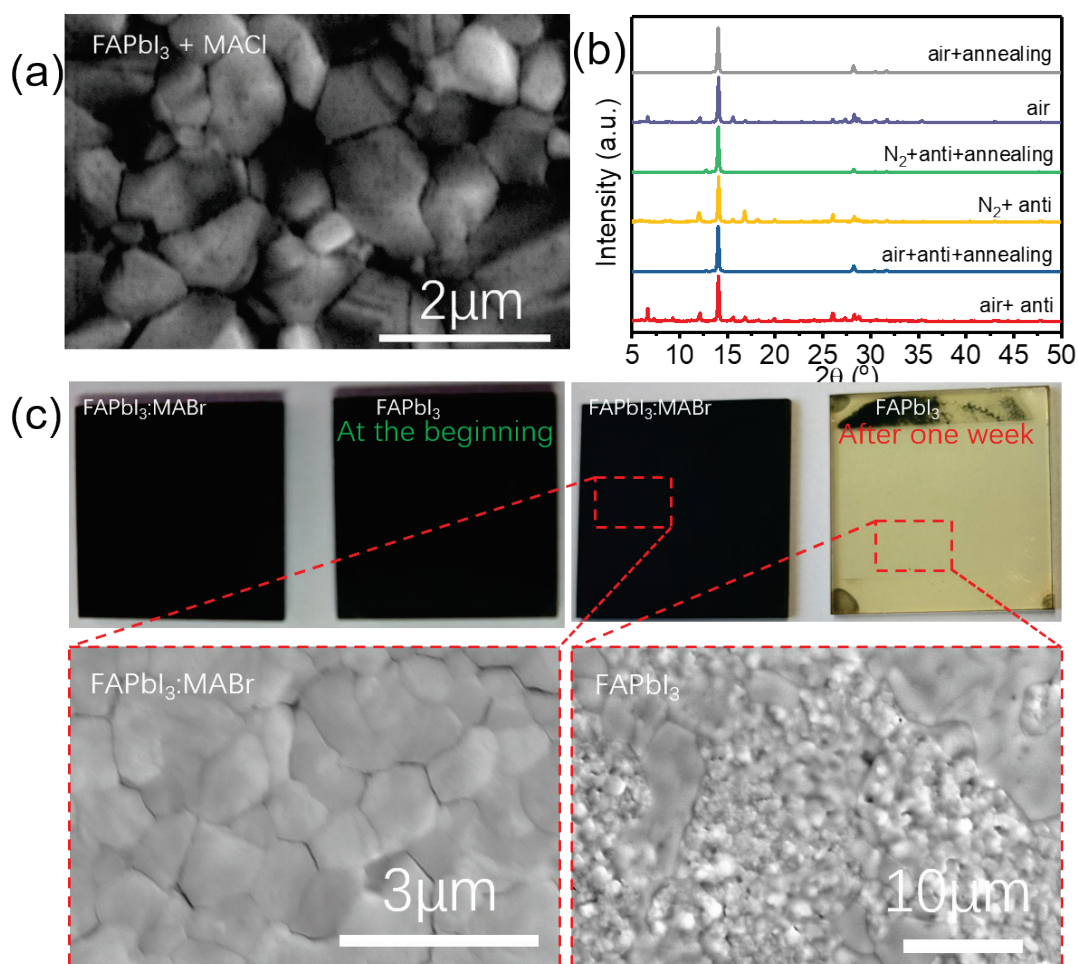


Figure 5.7 (a) Top view of FAPbI₃ film fabricated by 1.8 M precursor; (b) XRD pattern of FAPbI₃ films treated by different conditions; (c) the stability test of control and target film and their SEM top view after degradation. Copyright 2024, Royal Society of Chemistry

As described in the previous data, the doping concentration of A cations above 50% causes the change of perovskite grain shape, which is also reflected in **Figure 5.6d**, and according to the description of **Figure 5.5**, only MACl concentration greater than 20% can completely convert FAPbI₃ into α -phase perovskite, and this result agrees with the report of Jeon et al.²⁷³ Thus, the concentration of MACl in the precursor was specified as 35% in the later study of this work, and the concentration of the precursor solution is increased to 1.8 M, which concentration can make perovskite films full cover on the substrate (See **Figure 5.7a**).

In this study, the surface of perovskite grains should also be modified, so the passivation ions can be chosen, which also resist well against water erosion. To probe the effect of air humidity on FAPbI₃, experiments were also designed in this work to discuss it. Similarly, the effect of annealing and anti-solvent on perovskite films is considered, the wet films with and without anti-solvent during spin coating and the films annealed with anti-solvent were divided into two groups, one group was placed in air overnight and the other group was placed in dry nitrogen overnight, then all the films were uniformly annealed at 150 °C and then the resulting films were analyzed by XRD. As shown in **Figure 5.7b**, the films annealed in advance in all conditions, with or without the use of an anti-solvent, yielded perovskite in the pure phase after placing them in the air overnight. In contrast, the other conditions either contained PbI₂ or δ -phase. And annealing at 150 °C is necessary to form the α -phase FAPbI₃, so it is said that after the formation of the α -phase a capping agent is still needed to keep the perovskite stable, and the water or oxygen in the air acts as such a capping agent to some extent. But after all, water or oxygen will eventually decompose FAPbI₃, so it is only used to discuss the mechanism. But this experiment shows us from the side that capping agents are useful for stabilizing α -FAPbI₃, and the use of capping agents includes doping as well as interfacial modifications. These two methods need to act at the same location as the initial decomposition of perovskite by water and oxygen in the air.²⁷⁴⁻

²⁷⁶ In addition, MAI dopant is chosen to aid the crystallization of α -FAPbI₃, and will not exist in the film during the 150 °C annealing. Analogous to FA⁺ for [PbI₆]⁴⁻, MA⁺ is chosen to match the size of the [PbBr₆]⁴⁻ backbone and to add MABr as the passivation material for the FAPbI₃ layer. However, since the excess MABr may cause instability, IPA is used to clean off the excess MABr immediately afterwards. This avoids the cell instability caused by phase separation of MAPbBr₃ doped in FAPbI₃. Thus, MAPbBr₃ is expected can form a protective shell outside the FAPbI₃ grains to resist the moisture attack. FAPbI₃ films were used as control films and MABr passivated FAPbI₃ films were used as target films and exposed to air at 45% humidity for one week. **Figure 5.7c** shows that the control film turns yellow, but the target film stays black. As the SEM shows in **Figure 5.7c**, control films already degrade, but the target film still keeps big crystals in the film.

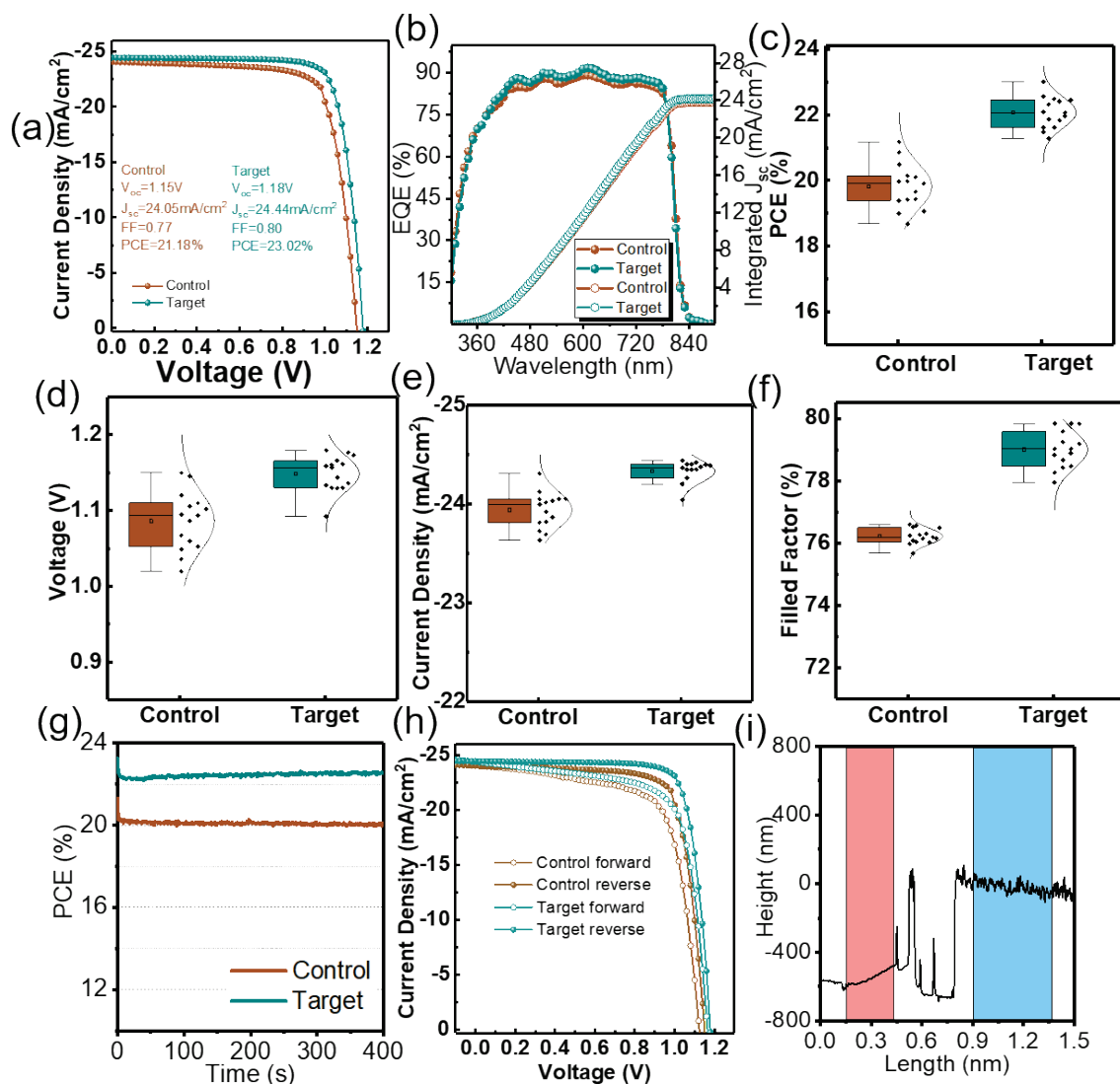


Fig. 5.8 (a) Current density-voltage (J-V) curves, (b) External quantum efficiency (EQE), (c) PCE, (d) V_{oc}, (e) J_{sc} and (f) FF box statistical chart, (g) Maximum power point tracking (MPPT) and (h) forward and reverse scan J-V curves of pure FAPbI₃ and MAPbBr₃ doped PSCs; (i) thickness measurement of perovskite film by profilometer. Copyright 2024,

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Based on this, solar cells are fabricated by employing the structure ITO/SnO₂/perovskite (~530 nm)/spiro-OMeTAD/Au to study the performance. Champion PCE of control devices is 21.2%, V_{oc} is 1.15 V, J_{sc} is 24.05 mA/cm², and FF is 0.77. Compared to the control devices, target devices using MABr get a higher performance, the champion PCE reached in 23.0%, V_{oc} is 1.18 V, J_{sc} is 24.44 mA/cm², and FF is 0.80. However, the hysteresis of the champion devices is similar (See **Figure 5.8h**). The EQEs of the cells are presented in

Figure 5.8b. Based on the EQE spectra, integral J_{sc} values of 23.90 and 24.31 mA/cm² are obtained for champion control and target solar cells, respectively, which are basically consistent with the experimental results. When MABr is used, the averaged PCE (15 cells) is improved from 19.8% to 22.0%, with an average V_{oc} , J_{sc} , and FF of 1.14 V, 24.33 mA/cm², and 78.99 as shown in **Figure 5.8c-f**. This result also confirms the superiority of the FAPbI₃:MABr film to the pure FAPbI₃ film. Besides, the stability of both solar cells is tested over 400 s. The MPPT curves showed that both the control cell and the target cell were stable, but the control cell had a slight downward trend while the target cell had a slight upward trend, which might be due to the reorganization of different crystalline phases of perovskite in the target cell.

5.3 Conclusion

The controllable crystallization rate of perovskite enables the production of high-quality films, resulting in high-performance solar cells. This work shows that FAPbI₃ has the longest crystallization formation time of all the pure phases of perovskite and can therefore be used to obtain films with low defect density. However, it tends to form the hexagonal non-perovskite polymorph, making it challenging for photovoltaic applications. After thoroughly discussing the perovskite crystallization process, MACl is selected as the crystallizing aid to modulate the intermediate phase getting a controllable crystallization rate to fabricate black-phase perovskite films. In addition, MABr is used to heal defects and passivate FAPbI₃ to obtain air-stable films. The PSC fabricated using this strategy presents an efficiency of 23.0%.

6 Heterogeneous nucleation for processing green perovskite solar cells

This work was published as: **Zuo, W.**; Byravnand, M. M.; Kodalle, T.; Zohdi, M.; Lim, J.; Carlsen, B.; Magorian Friedlmeier, T.; Kot, M.; Das, C.; Flege, J. I.; Zong, W.; Abate, A.; Sutter-Fella, C. M.; Li, M.; Saliba, M. *Coordination Chemistry as a Universal Strategy for a Controlled Perovskite Crystallization. Adv. Mater.* 35, 2302889 (2023).

Zuo, W. fabricated and measured all devices, analysed the data, wrote the manuscript and contributed to the design and implementation of the research as well as the analysis of the results.

The excellent performance of solution-processed PSCs has attracted enormous attention in the photovoltaics field. The most efficient PSCs require an anti-solvent to instigate nucleation and subsequent crystallisation of an oversaturated perovskite liquid precursor. However, even with the most potent anti-solvents, frequently traces of the original perovskite solvent remain in the finally formed film. This is especially true for the commonly used dimethyl sulfoxide (DMSO) solvent, which tends to complexation during crystallization and has a relatively high boiling point. Consequently, DMSO lingers in the final thin film affecting detrimentally the original film formation, crystal quality and long-term stability. It is now conventional wisdom that perovskite precursors dissolved in DMSO only are not suitable for high-quality film formation. Thus, practically every high-performance perovskite film uses large amounts of more volatile solvents, such as N, N-dimethylformamide (DMF) or γ -butyrolactone (GBL), which degas during crystallization. This is also the reason why MAPbI₃ perovskite precursors, which are dissolved in DMSO only and tend to degrade when volatile solvents are used, never reached high performances and are now not seen as a viable route for the highest performances. Here, dimethyl sulfide (DMS) is introduced as an efficient and green heterogeneous coordination solvent replacing the common toxic anti-solvents such as chlorobenzene (CB) or toluene. A better electron donor property of DMS compared to DMSO is demonstrated, resulting in excellent perovskite crystallization with reduced defect density despite the use of a pure DMSO

precursor. Finally, MAPbI₃ PSCs are processed in a completely green solvent system using DMS, water, DMSO and anisole as the only solvents. And 21.6% efficiency is achieved, which to the best of our knowledge, is the highest reported efficiency for fully green solvent-processed MAPbI₃ PSCs. This work provides important progress toward a better understanding of perovskite crystallization to achieve highly efficient PSCs and also puts MAPbI₃ back on the map.

6.1 Introduction

Organic-inorganic metal halide perovskite materials have shown excellent optoelectronic properties such as tunable bandgap, strong light absorption, and defect tolerance, which introduces them as the most suitable materials as promising light absorbers for solar cell applications.^{7,277,278} The PSCs' performances rapidly increased from 3.8%²⁷⁹ in 2009 approaching now 26.1%.^{4,213} This is why PSCs have become a strong competitor to already matured solar cell technologies such as crystalline silicon, cadmium telluride (CdTe), or copper indium gallium selenide (CIGS). The facile and cost-effective solution processability of PSCs is the key advantage for futuristic development. At the initial stages of PSC development, perovskite film comfort quality-related issues, such as nonuniformity, pinholes and rough surface, led to low-performance devices.²⁸⁰ Under laboratory conditions, so far, the anti-solvent-assisted one-step spin-coating method has shown excellent potential to provide high-quality perovskite films, leading to the highest PCE records.^{281–283} The role of the anti-solvent is the extraction of the solvent of perovskite precursor during the spin coating process to provide a stable intermediate phase (IP) before the annealing process, leading to homogeneous nucleation and crystallization of pinhole-free films.^{284–286} So far, chlorobenzene (CB) and toluene (Tol) have been frequently used as anti-solvents for depositing uniform perovskite films with full surface coverage. However, the toxicity of this anti-solvent and its impact on human health could be considered a significant challenge for further developing high-efficiency PSCs.²⁸⁷

On the other hand, γ -butyrolactone (GBL), N, N-dimethylformamide (DMF), and (DMSO) are the most common solvents for the preparation of perovskite precursor. However, even in the presence of an anti-solvent, adding a small amount of DMSO to GBL²⁶⁷ and DMF²⁸⁸ solvents is one strategy to achieve high-quality perovskite films by forming MAI·PbI₂·DMSO adducts. Nevertheless, both GBL and DMF are super toxic, while DMSO

could be considered a green solvent due to its relatively low hazard effects.^{287,289} Therefore, besides the toxicity of lead components, replacing the toxic anti-solvents or solvents of perovskite processing with greener counterparts is a requirement for future commercialization of perovskite photovoltaics. Initially, diethyl ether (DE) was introduced as a greener anti-solvent than CB and Tol. Due to the difference of only one C=O double bond, ethyl acetate (EA), which has a different polarity from DE, has become a common anti-solvent for researchers to use as a green solvent.²⁹⁰ Compared to other halogenated anti-solvents, the low boiling point of ethyl acetate (77 °C) and diethyl ether (35 °C), led the perovskite films to exhibit non-negligible surface roughness and abundant.^{285,287,291} Subsequently, hexane,²⁹¹ isopropyl alcohol²⁹² and petroleum ether²⁹³ are used in EA together to fix this problem. Although mixed anti-solvents can solve the issues arising from single anti-solvents to a certain extent, the mixing ratio always leads to uncorrectable errors due to artificial deviations, which is unsuitable for a fine industry such as the fabrication of solar cells. Because of this, it is essential to find a more excellent single anti-solvent.

In this study, the effect of dimethyl sulfide (DMS) as a coordination solvent on perovskite films based on the chemical structure of diethyl ether (DE) is discussed by replacing the oxygen atom in it with a sulfur atom of the same group as it. In the previous report, the Pb-S bond can provide a new channel for the transmission and extraction of charges.²⁹⁴ Due to the sulfur compounds can passivate the interfacial trap states and reduce charge recombination, this kind of single coordination solvent can have the function of the mixed-anti-solvent. At the same time, dimethyl sulfide (DMS), diethyl sulfide (DES) and dipropyl sulfide (DPS) with different boiling points based on different carbon-branched chain lengths of sulfur atoms are compared. At last, CH₃NH₃PbI₃ (MAPbI₃) is used to investigate the mechanism of the effect of DMS. As a classical model for experimental and theoretical simulations,^{295–298} MAPbI₃ can be well used to study the mechanism of effect for various optimization perovskite strategies. It is found that because of the difference in boiling point, DMS is a good solution to the problem of MA⁺ escaping from perovskite due to high annealing temperature. Then a high efficiency of 21.6% is obtained in a fully green solvent process MAPbI₃-based comparable PSC.

6.2 Results and discussion

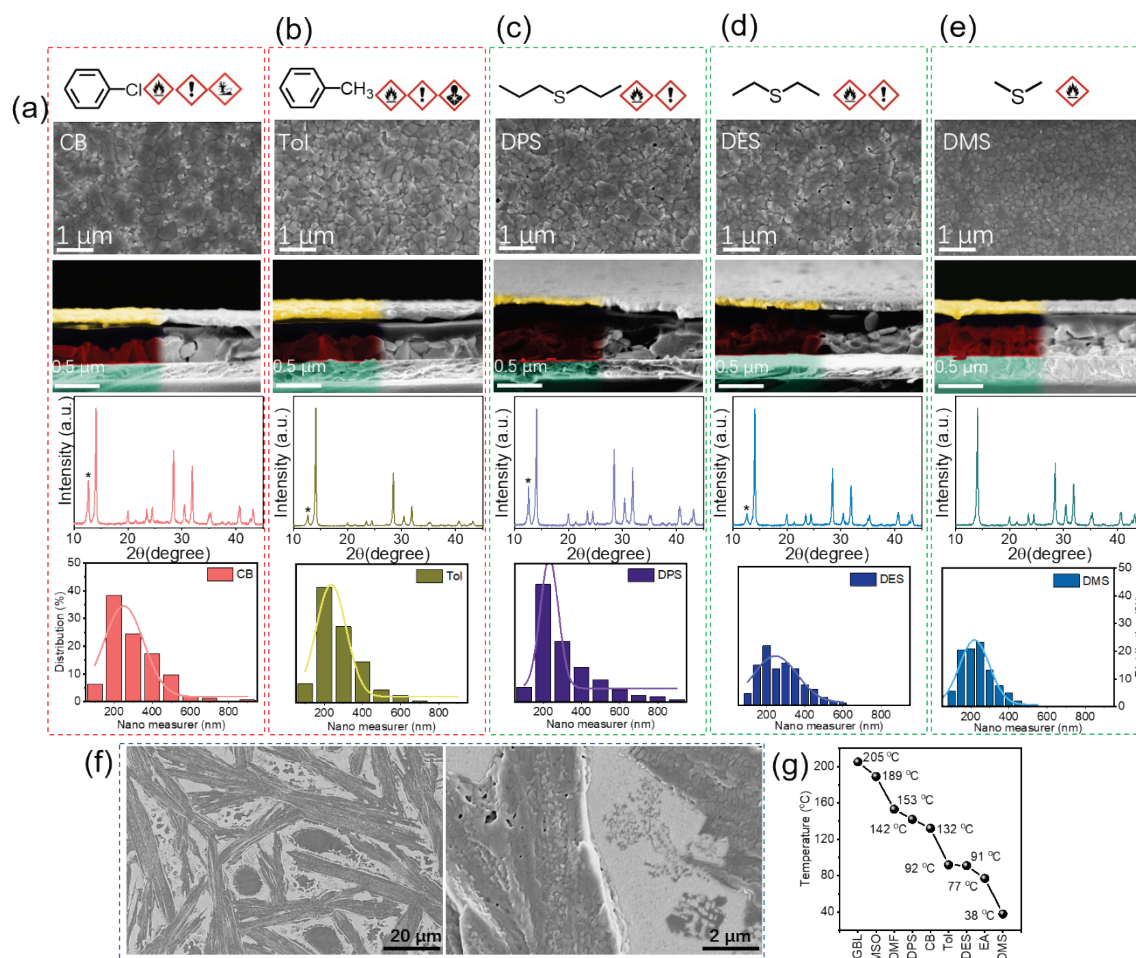


Figure 6.1 The morphology and crystallinity of MAPbI₃ films fabricated with different anti-solvents. Chemical structures and safety information of anti-solvents (a) CB, (b) Tol, (c) DPS, (d) DES and (e) DMS. Top-view and cross-sectional SEM images, XRD pattern and grain size statistics of films fabricated by corresponding anti-solvents. (f) Top-view of perovskite film without anti-solvent and (g) boiling point of different solvent.

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A high-resolution scanning electron microscope (SEM) is used to observe the changes in the morphology of the perovskite films. **Figure 6.1** presents the fabricated perovskite films by CB and Tol as common toxic anti-solvents, as well as, DPS, DES, and DMS as green anti-solvents. Compared to the fabricated film without an anti-solvent (See **Figure 6.1f**), films with anti-solvents are fully covered over the ITO/ SnO_x ETL substrates. However,

perovskite film was treated by DMS exhibits a denser surface morphology than films that use DES and DPS as the anti-solvent, and more homogeneous than films treated by all other anti-solvent. The grain size statistics are performed for different films based on SEM top view in **Figure 6.1** and the results showed no significant difference in grain size distribution between CB, Tol and DPS. However, since Tol, unlike CB, does not dissolve the intermediate phase, which in turn enables perovskite to crystallize slowly,^{299,300} the films fabricated using Tol as an anti-solvent have a relatively low PbI_2 content, which can be demonstrated by X-ray diffraction (XRD). In comparison, for the films made with DPS as an anti-solvent, a step-like morphology (See **Figure 6.1**) was observed due to the formation of secondary recrystallized grains, resulting in films that were not dense and had pinholes. The surface morphology of the films can also be confirmed in the cross-sectional SEM images. The morphology of perovskite films also depends on the miscibility of the anti-solvent and solvent,³⁰¹ as well as the temperature at which the anti-solvent is dropped.³⁰² The boiling points of DPS, DES and DMS are different (See **Figure 6.1f**), so they evaporate at different rates when added drops to the film, resulting in a difference in film temperature during spin coating, which causes a tendency for perovskite grains with DPS, DES and DMS to become smaller (See **Table 6.1**).

Table 6.1 SEM particle size analysis statistics

	Total ^a	Max. ^b /nm	Min. ^c /nm	Mean ^d /nm
CB	266	832.39	62.70	248.80
Tol	294	847.12	64.81	230.63
DPS	247	843.70	68.91	262.34
DES	354	590.56	68.80	251.27
DMS	562	520.64	47.17	213.18

a: Number of grains in the SEM top view image

b: Maximum grain size in SEM image

c: Minimum grain size in SEM image

This is clearly shown in the SEM images as well as in the grain size analysis statistics. As the number of alkyl carbons decreases, the grain size of perovskite films becomes

homogeneous, while XRD also demonstrates that the characteristic peaks of PbI_2 become weaker from DPS to DES until they disappear completely when DMS is used. This suggests that besides the role of sulfur on perovskite films, the size of the alkyl chain leading to the molecular size also plays a significant role.

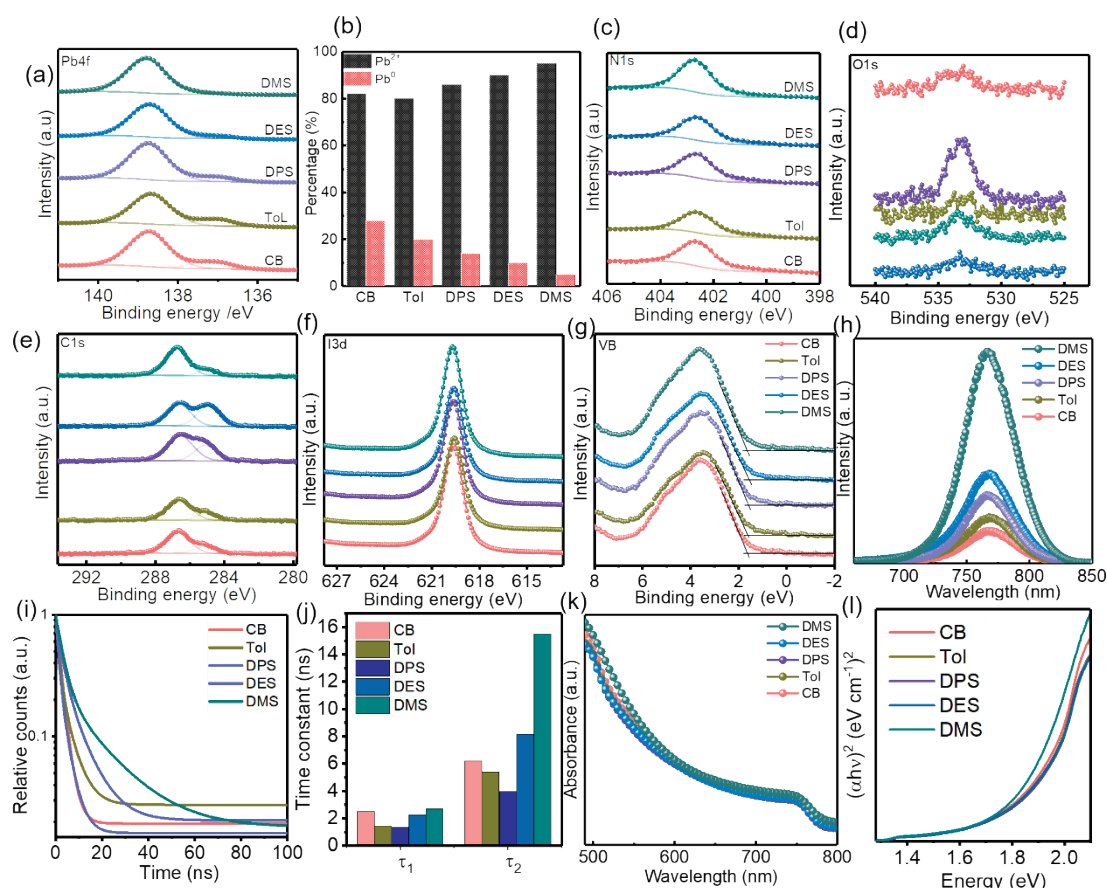


Figure 6.2 XPS/UPS measurement and PL/TRPL measurement of MAPbI₃ films by different anti-solvent treatments. Survey X-ray photoelectron spectroscopy (XPS) spectra of MAPbI₃ films treated by anti-solvent CB, Tol, DPS, DES and DMS, (a) Pb 4f, (b) Pb⁰ and Pb²⁺ density comparison bar chart, (c) N 1s, (d) O 1s, (e) C 1s and (f) I 3d spectra of MAPbI₃ films, (g) Ultraviolet photoelectron spectroscopy (UPS) of MAPbI₃ films; (h) photoluminescence (PL) intensity spectra and (i) time-resolved PL (TRPL) decay spectra of MAPbI₃ films with different anti-solvent treatment; (j) Carrier lifetimes for fast decay processes (τ_1) and slow decay processes (τ_2), (k) and (l) UV-Vis absorbance of perovskite films based on CB, Tol, DPS, DES and DMS treatment. Copyright 2023,

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The X-ray photoelectron spectroscopy (XPS) and ultraviolet photoelectron spectroscopy (UPS) are conducted to deeper understanding of perovskite film properties. The I3d5/2, N 1s, O 1s, C 1s and I 3d, and Pb4f5/2 (See **Figure 6.2a, c, e, f and g**) detailed core level spectra and valance band (VB) (See **Figure 6.2g**) are measured. The peak position of core level I3d5/2, C1s, and Pb4f5/2 spectra are at 619.8, 286.7 and 138.8 eV and the valance band maxima (VBM) at 1.5 eV corresponding to the MAPbI₃ perovskite. In all the samples, the peak positions and the VBM remain the same demonstrating similar electronic properties of the perovskite layers. Similarly, the UV-Vis absorption curves suggest identical band edges for all perovskite films (See **Figure 6.2k and l**). However, in the case of C1s and Pb4f5/2, the peak intensities at lower binding energy are changed by different anti-solvent treatments. In the case of C1s two peaks at 286.7 and 285.0 eV from the methylammonium and advantageous carbon respectively. While in the case of Pb4f a lower intense peak at around 137.0 eV appears which is from the Pb⁰ state. A close look at the Pb shows that the DMS (3%) and DES (8%) treated have lower concentrations of Pb⁰ compared to the DPS (13%), CB (19%), and Tol (21%) (See **Figure 6.2b**). The device prepared with DMS and DES has lower Pb⁰ states, which often work as nonradiative recombination centers³⁰³. This can be clearly observed in steady-state photoluminescence (SSPL) spectra of different perovskite films (See **Figure 6.2h**). The fabricated perovskite film with DMS with minimum Pb⁰ content shows the highest PL intensity among all anti-solvents, demonstrating the lowest nonradiative recombination. However, fabricated films with non-green solvents, i.e., CB and Tol, show the lowest peak intensities compared to green solvents due to higher defect state intensities, considering the XPS observation. The lifetime of carriers is also longer in DMS-based perovskite films (See **Figure 6.2i**), due to the lower interference of carrier transport by traps. The carrier lifetime statistics are shown in **Figure 6.2j**, τ_1 denotes the carrier lifetime of the fast decay process, which is related to the non-radiative recombination at the interface, and τ_2 denotes the carrier lifetime of the slow decay process, which is related to the radiative recombination inside the grain. The carrier lifetime of perovskite films treated with different types of anti-solvents is less than 10 ns, which indicates that the carrier annihilation of perovskite films is severe. When DMS was used, the carrier lifetime of the perovskite films reached the highest, with τ_1 increasing to 2.74 ns and τ_2 increasing to 15.52 ns, at which time the perovskite films had the least internal defects, further demonstrating the significant improvement of the perovskite film quality by the action of DMS.

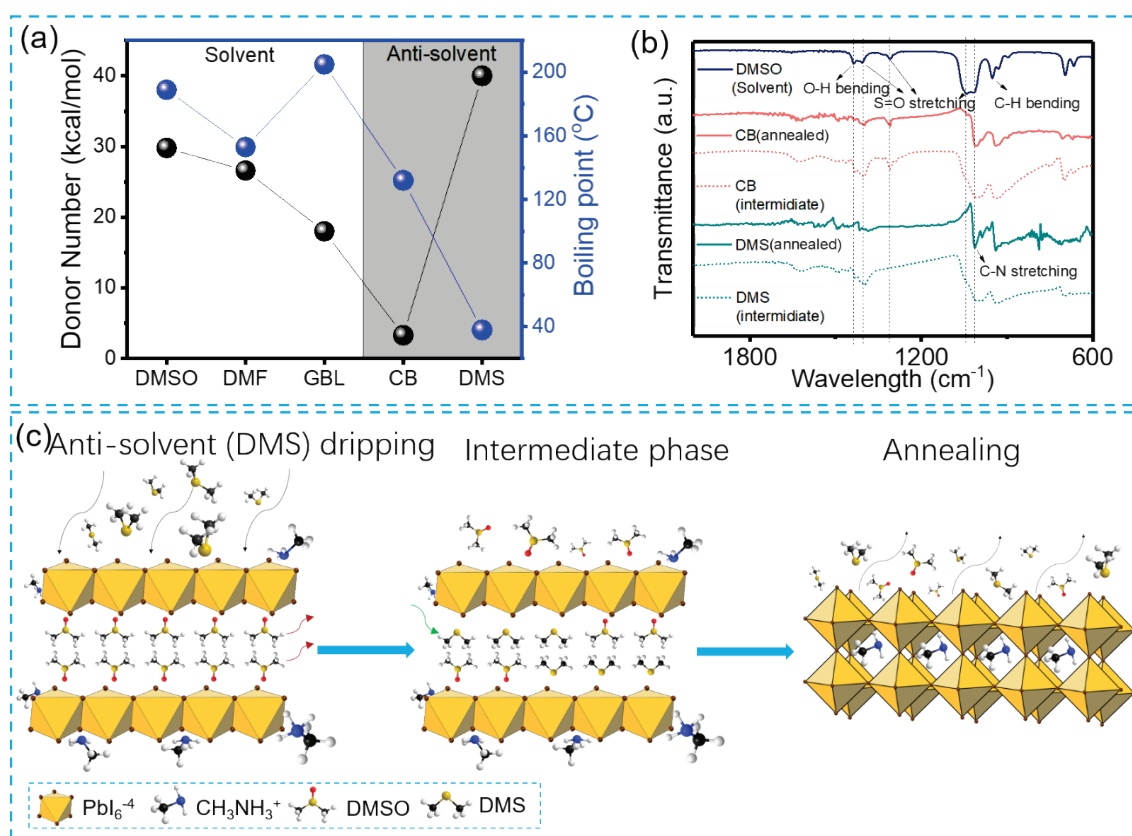


Figure 6.3 Mechanism of DMS working principle as a coordination solvent. (a) Gutmann donor numbers and boiling points of perovskite precursor solvents (DMSO, DMF and GBL) and anti-solvents CB and DMS. (b) The Fourier-transform infrared spectroscopy (FTIR) spectra of MAPbI₃ treated with CB and coordination solvent DMS anti-solvents before (intermediate phase) and after annealing. (c) Schematic illustration of the proposed performing mechanism of DMS in perovskite film fabrication. Copyright 2023, Wiley-VCH GmbH

The donor number (D_N) is a measure of the ability of a solvent to solvate cations and Lewis acids.^{304,305} **Figure 6.3a** shows the D_N s and boiling points of the common solvents DMSO, DMF and GBL for perovskite precursor, as well as DMS and CB as green and toxic solvents, respectively (all the detail parameters include donor Number, Dielectric constant, boiling point and dipole moment are shown in **Table 6.2**).

Table 6.2 Basic parameters of different solvents

	Donor Number (kcal/mol)	Vapor pressure at 20 °C (kPa)	Dielectric constant	Boiling point (°C)	Dipole moment (D)
DMSO	29.8	0.05	46.68	189	4.1
DMF	26.6	0.38	36.71	153	3.79
GBL	18	0.2	40.96	205	4.27
DE	19.2	58.9	4.33	35	1.30
EA	17.1	9.69	6.40	77	1.88
CB	3.3	1.17	5.62	132	1.54
Tol	0.1	2.93	2.38	92	0.31
DMS	40	53.7	6.3	38	1.5

Normally, high D_N solvents coordinate more strongly with the Pb^{2+} center, varying the concentration of high- D_N additives in precursor solutions tunes the strength of lead-solvent interactions.³⁰⁶ It is clear that DMS has the highest D_N but the lowest boiling point among these solvents. It is commonly known that $MAPbI_3$ readily decomposes to CH_3I , NH_3 , and PbI_2 at 100 °C and was also degraded by prolonged exposure (>1 h) to 80 °C,³⁰⁷ but the boiling point of DMSO is 189 °C (See **Figure 6.3a**), which leaves a contradiction between the degradation temperature of perovskite films and the temperature required for evaporation of DMSO from perovskite films. To explore the mechanism of DMS effect on perovskite films, Fourier-transform infrared spectroscopy (FTIR) is used to analyze the intermediate phases produced by the interaction of CB and DMS on perovskite, respectively, and subsequently the black phase after annealing at 80 °C for 40 min, and used the FTIR spectrum of pure DMSO as a reference. The pure DMSO shows the stretching peaks of S=O at 1400 cm^{-1} , 1309 cm^{-1} and 1043 cm^{-1} , respectively. In the spectra of the perovskite intermediate phase obtained using the anti-solvent of CB and DMS, the S=O stretching peak at 1043 cm^{-1} is covered by the stretching peak signal of C-N at 1010 cm^{-1} , which then

appears as a companion peak in the spectra, and in the annealed perovskite phase, this stretching peak disappears. The S=O stretching peak at 1400 cm^{-1} disappears only in the spectra of the perovskite phase with DMS as the coordination solvent, however, the stretching peak of S=O at 1309 cm^{-1} disappears in the spectra of the perovskite intermediate phase and the annealed black phase when DMS is used as the coordination solvent. This indicates that due to the high D_N of DMS and the mutual solubility with DMSO³⁰⁸, when DMS is used as a coordination solvent, DMS washes off part of DMSO and replaces the position of DMSO to continue binding PbI_2 to form the intermediate phase, and due to the low boiling point of DMS, better perovskite films can be formed at a lower annealing temperature. Based on previous observations, the possible mechanism of action of DMS as a coordination solvent for efficient crystallization of MAPbI_3 perovskite films is schematically illustrated (See **Figure 6.3c**).

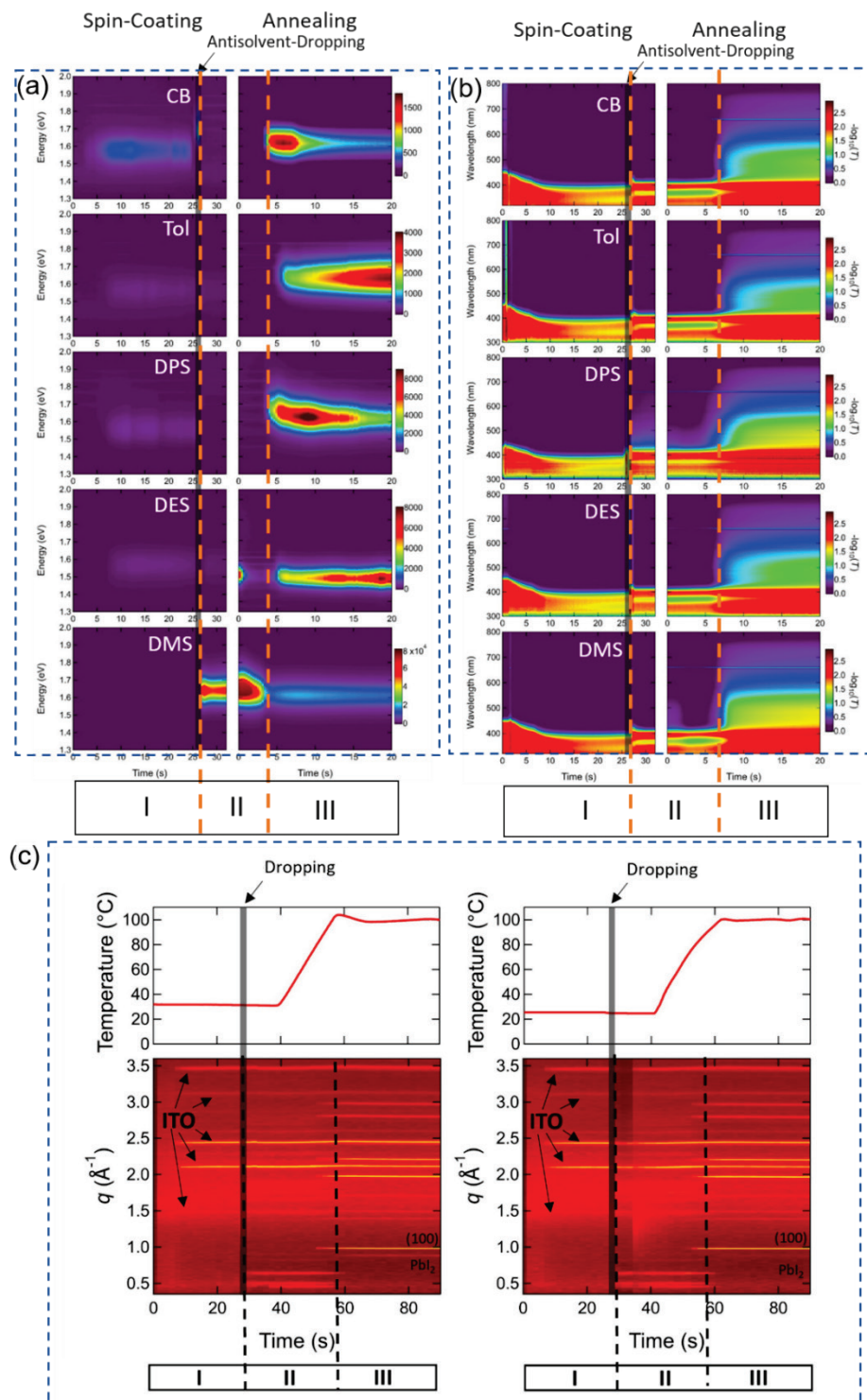


Figure 6.4 In-situ measurements of the evolution of (a) the photoluminescence signal during spin-coating (left column) and annealing (right column) of a MAPbI_3 sample deposited using CB, Tol as anti-solvent and DPS, DES, DMS as coordination solvent, (b) in situ UV-Vis of all samples and (c) in situ GIWAXS of the CB and the DES sample.

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To get a more detailed understanding of the perovskite crystallization, the in-situ film characterization is provided. **Figure 6.4a** shows the evolution of the PL measured in-situ during spin-coating and annealing at 100 °C using CB, Tol, DPS, DES and DMS, respectively. To elucidate the formation dynamics at the structural level, in-situ grazing incidence wide-angle X-ray scattering (GIWAXS) patterns of the CB sample are shown in **Figure 6.4c (left)**. Due to the very low viscosity of DMS, it is not easy to measure in-situ GIWAXS on that corresponding sample. Instead, **Figure 6.4c (right)** shows the GIWAXS evolution in the case of DES. Combining the PL and the GIWAXS data, the film formation is divided into three growth stages that will be discussed in the following. Comparing the PL-data of all five cases suggests that the perovskite growth follows similar dynamics via these three stages for all solvents.

In Stage I, after the initial solution ejection during the first ~5 s of the spin-coating process, there is a weak PL signal emerging centered around 1.58 eV. This signal is also present in the case of DMS but not visible in **Figure 6.4a** due to the different intensity-scaling of both panels (see color bars on the right side of **Figure 6.4a**) – a differently scaled version of the same plot showing the existence of the weak PL-signal at 1.58 eV is shown. GIWAXS during the first stage only shows peaks associated with the ITO substrates which cannot explain the PL emission (see **Figure 6.4c**). Interestingly, the PL emission peak position coincides with the bandgap of MAPbI₃ but MAPbI₃ crystallization or formation is not expected and has not been reported before at this early stage and at room temperature. Instead, during spin coating, before dripping the solvent, and depending on the exact precursor composition and solvent choice, solvate phases may form quickly.³⁰⁹ it is shown before, that small volume inclusions of the perovskite phase may not show up in the diffraction pattern but PL is very sensitive to these bright emitting phases.³¹⁰ Other in-situ PL studies during perovskite formation found blue-shifted PL emission at the early crystallization stages explained by excitonic confinement in low dimensional perovskite structures.³¹¹ Here, it can only be speculated that the weak PL signal in Stage I may be attributed to an amorphous or intermediate, MAPbI₃-containing phase.

Upon the solvent drop, the beginning of Stage II, the initial PL-signal vanishes, and distinctive diffraction peaks appear (e.g. at 0.46 Å⁻¹, 0.51 Å⁻¹, and 0.65 Å⁻¹) that can be attributed to the orthogonal crystalline solvent-complex, Pb₃I₈·2(CH₃)₂SO·2CH₃NH₃ (MAPbI₃·DMSO solvent complex).⁴⁰⁻⁴² In addition, dropping DMS or CB induces the

occurrence of a very intense PL-peak centered at 1.64 and 1.7 eV, respectively, but is not present in the other cases at room temperature. Two possible explanations are considered to explain the PL emission energy higher than the bandgap energy. Dripping of DMS or CB likely leads to an exchange mechanism of DMS or CB with DMSO leading to the formation of a luminescent intermediate phase containing 2D-layered nanosheets or other low dimensional perovskite clusters.³¹¹ In the case of CB, and in analogy to a previous report, the short PL "flash" in the case of CB is interpreted as immediate MAPbI₃ nanocrystal formation. The PL is blue-shifted compared to the bulk bandgap due to quantum confinement effects. The emergence of PL in the cases of DMS and CB but not for DPS and DES possibly indicates a different interaction with DMSO. For DMS and CB, DMSO is likely removed efficiently and locally during the dripping event while it is not as efficiently extracted with DPS and DES. In the case of CB, the quantum-confined perovskite nuclei formation appears metastable at this stage as large amounts of DMSO are still present in the wet film, hence the PL signal disappears very quickly. As the signal remains constant in the case of DMS, it appears that DMS is more efficiently extracting the DMSO, creating more stable nuclei. In addition, the short-lived PL signal appearing right after CB dripping is more blue-shifted than the PL peak appearing after DMS dripping which indicates that nuclei are smaller in the case of CB.³¹²

During annealing, i.e., in Stage III, the PL-emission of the DMS sample slightly redshifts before the appearance of the MAPbI₃-related PL-peak centered around 1.61 eV (see **Figure 6.4a**). The PL intensity drop and re-appearance of the MAPbI₃-related PL in the DMS case is likely caused by a second nucleation and growth event upon MAPbI₃·DMSO dissociation and DMSO evaporation when reaching the final annealing temperature. With increasing annealing duration, this signal fades due to temperature-induced PL-quenching³¹³ as well as degradation due to the constant laser-illumination. These dynamics are in good agreement with the GIWAXS data, which shows that the intermediate phase (in all investigated cases) undergoes thermal dissociation and subsequent solvent evaporation leading to the growth of MAPbI₃ and PbI₂ crystals (see **Figures 6.4c**).

Judging by the in-situ PL and GIWAXS data, the choice of the solvent and consequently the solvent-solvent interaction as well as the subsequent coordination chemistry do affect the formation dynamics of the final MAPbI₃ crystals during annealing. However, there are indications that DMS dripping is the most efficient in extracting DMSO, leads to larger

initial nuclei, a faster and more defined MAPbI₃ absorbance edge as measured by in-situ transmission (see highlighted area in **Figure 6.4b**), and a significantly more intense PL-signal, indicating higher crystal quality.

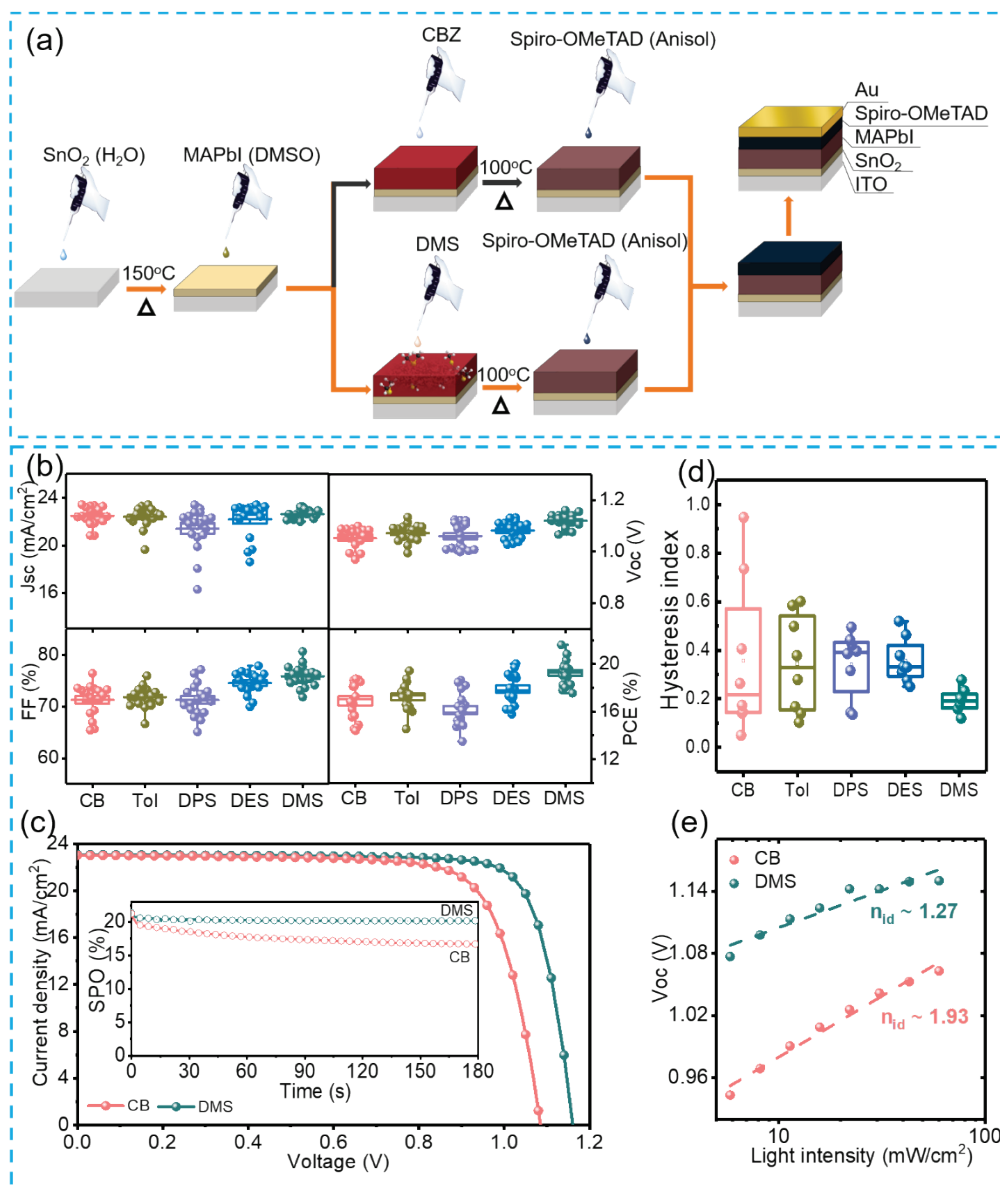


Figure 6.5 PSCs fabrication process and characterization for devices. (a) Schematic diagram of PSCs fabrication by fully green and non-green solvent process, PSC (b) statistical photovoltaic parameters and (c) hysteresis index (HI) extracted from the J-V measurement based on different anti-solvent treatment perovskite films. Comparison of (d) champion J-V curves and (e) ideality factor determined from the light intensity dependence of V_{oc} of PSCs with CB anti-solvent and DMS coordination solvent treatment.

In addition to discussing the mechanism of different anti-solvent effects on perovskite film crystallization, PSCs are fabricated based on these anti-solvents to discuss their effects on cell performance. **Figure 6.5a** shows the fabrication process of PSCs. First, the cleaned ITO is spin-coated with colloidal SnO₂ nanoparticles (diluted with H₂O), annealed at 150 °C and cooled to room temperature, then spin-coated with perovskite films (precursor solution using DMSO as solvent) and used CB and DMS (also extent to Tol, DPS and DES) as the anti-solvent. Previously, our group introduced anisol (ANS) not only as a green anti-solvent but also as the solvent for Spiro-OMeTAD instead of CB.³¹⁴ Therefore, after annealing the perovskite film at 100 °C, ANS is used to formulate the precursor solution for the spiro-OMeTAD hole transport layer, and the final layer of gold electrodes was vaporized onto the hole transport layer to complete the solar cell fabrication. The solvents used in the precursor solutions for the fabrication of each layer in the solar cell are green solvents. The statistical photovoltaic data of 18 devices are included in the box chart for different anti-solvents presenting the reproducibility of all the device parameters (See **Figure 6.5b**). The average values of J_{sc} for the devices fabricated using CB as an anti-solvent were essentially convergence with those of Tol and DMS, but there was a significant increase from DPS to DES to DMS, which was positively correlated with the performance of the perovskite films fabricated by DPS, DES and DMS, and the same trend appears in the statistical plot of the V_{oc} averages. Moreover, due to the molecular properties of DMS, the average FF (Fill factor) value of solar cells fabricated using DMS as a coordination solvent is significantly increased. And then, the solar cells using DMS also demonstrate the best photovoltaic conversion efficiency (PCE). To estimate the magnitude of hysteresis, the hysteresis index (HI) was calculated using the previously reported equation (1).³¹⁵

$$HI = \frac{|PCE_{reverse} - PCE_{forward}|}{PCE_{reverse}} \quad (1)$$

The statistical results of the hysteresis are shown in **Figure 6.5c**, solar cells using CB as the anti-solvent have a larger case oscillation, but the average value is similar to that of the solar cells using DMS as the coordination solvent, which is caused by the fact that the perovskite film based on CB is not as good as that of DMS. Whereas the Tol-based PSCs have larger hysteresis, similarly, the DPS, DES and DMS-based solar cells exhibit increasingly smaller hysteresis, indicating that the solar cells made based on DMS are more stable. The champion device is based on DMS with V_{oc} of 1.161 V, J_{sc} of 23.08 mA/cm², FF reaching 0.81, and PCE of 21.6%. In comparison, the CB-based champion solar cell has a lower V_{oc} of 1.085 V,

a similar J_{sc} of 23.02 mA/cm^2 , and a slightly lower FF than the DMS-based solar cell of 0.76, while the PCE is 19.1%. The J-V curves are shown in **Figure 6.5d**, and the stabilized maximum power output (SPO) of the champion devices was determined by holding the voltage at the maximum power point (0.99 V and 0.86 V). The SPO verifies for 180 s shows that their PCEs are stable at 20.17% and 16.71%, respectively.

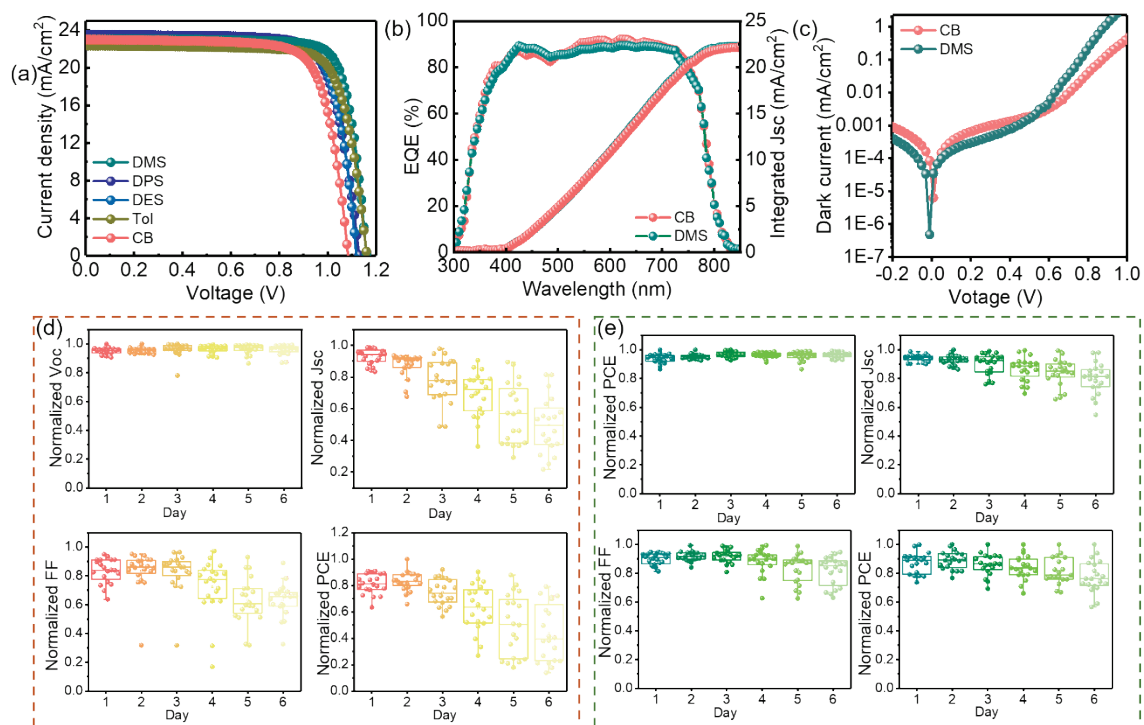


Figure 6.6 (a) Champion devices J-V curves based on the treatment of CB, Tol, DPS, DES and DMS; (b) EQE spectra and (c) dark current comparison of champion devices based on CB and DMS treatment; Performance tracking of PSCs fabricated based on CB (d) and (e) DMS. Copyright 2023, Wiley-VCH GmbH

J-V curves of all the champion cells fabricated based on CB, Tol, DPS, DES and DMS are shown in **Figure 6.6a**, and detailed performance parameter is shown in **Table 6.3**. The external quantum efficiency (EQE) of champion solar cells (See **Figure 6.6b**) was demonstrated to cover the whole visible spectrum and the integrated current densities are $\sim 22.29 \text{ mA/cm}^2$ and $\sim 22.27 \text{ mA/cm}^2$ of the champion solar cells based on CB and DMS, which were inconsistent with the measured J_{sc} . **Figure 6.6c** shows the dark J-V curves. Solar cells fabricated based on DMS have lower dark currents at negative and positive bias voltages. The lower leakage dark current suggests improved charge transport and decreased

recombination loss in the devices. As shown in **Figure 6.5e**, the V_{oc} increases monotonically with logarithmic light intensity, indicating that trap-assisted Shockley-Read-Hall recombination dominates in these devices. The DMS-based cell has a smaller ideality factor (1.27) than the CB-based cell (1.93). Then a one-week data tracking measurement was performed on 20 devices, and the results showed that the DMS-based solar cells were more stable, while the CB-based solar cells started to show more significant performance degradation on the third day (See **Figures 6.6d and 6.6e**). In addition, the reports of MAPbI₃-based solar cells with efficiencies exceeding 20% in recent years are counted. It shows that the performance of the all-green solvent solar cell fabricated in this experiment is close to the world efficiency record.

Table 6.3 Comparison of performance value between solar cells fabricated based on different anti-solvent

	V_{oc} (V)	J_{sc} (mA/cm ²)	FF (-)	PCE (%)
DPS	1.099	23.42	0.72	18.6
DES	1.119	23.06	0.79	20.4
DMS	1.161	23.08	0.81	21.6
Tol	1.162	22.36	0.78	20.3
CB	1.085	23.02	0.76	19.1

6.3 Conclusion

This study reports a green coordination solvent DMS and investigated its working mechanisms by comparing it to the common anti-solvents CB and Tol, and the similar anti-solvents DPS and DES based on different alkyl chains. The results show that DMS, a coordination solvent for perovskite films, can replace DMSO and form the intermediate phase in a short time due to its high Gutmann donor number and mutual solubility with DMSO. This leads to a higher quality perovskite layer than the other similar anti-solvents by resolving the contradiction between the annealing and degradation temperature of perovskite. All-green solvent PSCs are fabricated based on MAPbI₃, resulting in a record

efficiency of 21.6%. This work is an important step toward the commercialization of toxic-free, high-performance PSCs.

7 Summary and outlook

Perovskites are promising, next-generation materials for solar cells. They are frequently deposited from a solution-based process resulting in a thin-film formation that starts with nucleation and subsequent crystallization. Such crystallized thin films typically suffer from defects during growth due to the difficulty of controlling the film formation process. At the same time, as defect-free crystals as possible are desired. Hence, studying the crystal growth process and identifying possible defects in the finally formed film has become an active area of research.

Moreover, the bandgap of perovskites can be adjusted by interchanging the different components making it attractive for general optoelectronic applications. Nevertheless, the effect of these complex perovskite compositions with respect to phase transition as a function of temperature requires careful consideration in the process of obtaining stable and perfect perovskite materials.

Compared to growing single crystals, the environment in which thin film crystallization occurs is more complex and more susceptible to external influences. For example, a fast nucleation followed by a slow crystallization rate resulted in a more complete coverage and more defect-free materials. To regulate the nucleation and crystallization rate of perovskite, different strategies were proposed, i.e. mainly classified here as homogeneous or heterogeneous nucleating agents and homogeneous or heterogeneous crystallizing agents. However, heterogeneous dopants, if not used for passivation, can often turn into impurities in the film and thus affect non-radiative recombination.

Therefore, it is posited in this thesis that avoiding external agents during the growth of perovskite crystals facilitates the fabrication of more defect-free perovskite films.

The thesis described the use of a homogeneous dopant, MAI, to aid the crystal growth of perovskite films. MAI has a low boiling point and although it is also involved in the production of perovskite, when the annealing temperature reaches 100 °C, MAI leaves the film without affecting the composition of the main perovskite. However, experimental results show that when the dopant is removed from the film, it also forms larger grains, but leaves more pinholes and does not even allow for full film coverage.

This thesis therefore designed experiments to apply MACl as a vapor to perovskite films during annealing, resulting in perovskite films with larger grains. Although large grains were obtained, the crystallization rate of the different components of perovskite still needed to be explored. Thus, the focus was on the crystallization rate. It was found that the crystallization formation time was longer when the ionic radius r_A (of the cation component) was increased or r_X (the ionic radius of the halide component) was decreased using FAPbI₃ as a control. To ensure pure phase growth of FAPbI₃, the experiments in this part still used MACl as a dopant due, as concluded earlier, to the evaporation of MACl which can lead to pinholes and defects.

The next experiments used MABr to passivate the FAPbI₃ films. On the one hand, MA⁺ heals the pinholes left by the evaporation of MACl, on the other hand, Br⁻ distorts the lattice at the edge of the PbI₂ backbone thus protecting the stability of the FAPbI₃ grains. The experimental results showed that the pure FAPbI₃ films quickly turned into the yellow phase, while the MABr-passivated films remained in the black phase after one week. However, this method of pure phase crystallization uses homogeneous doping, and the precursor solvents are all mixed solvents, where there is also a risk of lattice mismatch leading to defects during the crystallization process due to the different boiling points of the two solvents.

The next experimental design therefore used DMSO as the only precursor solvent and the newly introduced DMS coordination solvent as a nucleating agent. The advantage of DMS is that it can replace DMSO during the crystallization forming intermediates briefly before the nucleation of the perovskite film, which then crystallizes into the finished thin film. The XRD data shows that there are no peaks of PbI₂ in the films using DMS as a coordination solvent, indicating that there is no MAI free from the PbI₂ backbone and therefore fewer defects in the films. The coordination solvent DMS was used to fabricate MAPbI₃-based perovskite solar cells (dissolved in pure DMSO). This resulted in >21% performance, among the highest for this system, showing the soundness of this approach.

However, all these solvent systems and additives add complexity to the overall thin-film formation. Therefore, one alternative approach could be solid-solid reactions without solvents. One common example is vacuum evaporation, albeit it may require more development to deposit high-quality thin films at a low price towards commercialization.

Hence, at this stage, the deposition of perovskite thin films, using solution-based methods, will be an ongoing challenge in the quest for inexpensive, large-area perovskite solar cells.

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Publications

1. **Zuo, W.**; Fu, W.; Wang, K.; Das, C.; Byranvand, M. M.; Wang, K. L.; Chaudhary, A.; Lim, J.; Li, M.; Saliba, M. Crystallization Dynamics and Stabilization of FAPbI₃ Single-Phase Perovskite. *Energy Environ Sci* **2023**, 17 (4), 1407–1415. (*Zuo, W. fabricated all devices, wrote the manuscript and contributed to the design and implementation of the research as well as the analysis of the results*)
2. **Zuo, W.**; Byranvand, M. M.; Kodalle, T.; Zohdi, M.; Lim, J.; Carlsen, B.; Magorian Friedlmeier, T.; Kot, M.; Das, C.; Flege, J. I.; Zong, W.; Abate, A.; Sutter-Fella, C. M.; Li, M.; Saliba, M. Coordination Chemistry as a Universal Strategy for a Controlled Perovskite Crystallization. *Advanced Materials* **2023**, 35 (39). (*Zuo, W. fabricated all devices, wrote the manuscript and contributed to the design and implementation of the research as well as the analysis of the results*)
3. Li, G.; Su, Z.; Canil, L.; Hughes, D.; Aldamasy, M. H.; Dagar, J.; Trofimov, S.; Wang, L.; **Zuo, W.**; Jerónimo-Rendon, J. J. Highly Efficient Pin Perovskite Solar Cells That Endure Temperature Variations. *Science (1979)* **2023**, 379 (6630), 399-403. (*Zuo, W. performed the device performance measurements and contributed to suggestions for the manuscript*)
4. Liu, H.; Zhang, Z.; **Zuo, W.**; Roy, R.; Li, M.; Byranvand, M. M.; Saliba, M. Pure Tin Halide Perovskite Solar Cells: Focusing on Preparation and Strategies. *Adv Energy Mater* **2023**, 13 (3), 2202209. (*Zuo, W. contributed to the writing and suggestions for the manuscript*)
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Erklärung

Ich versichere hiermit, dass ich die vorliegende Arbeit selbstständig verfasst und keine anderen als die angegebenen Hilfsmittel verwendet habe.

Stuttgart, 01. März 2025

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